



Mathematics for Computer Science Engineers

HYPOTHESIS and INFERENCE

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Introduction

A sample of 50 micro drills had an average lifetime of $\bar{X} = 12.68$ holes drilled and a standard deviation of $s = 6.83$.

Main question is: Is the population mean lifetime $\mu > 11$?

We address this question by examining the value of the sample mean $\bar{X}$.

We see that $\bar{X} > 11$, but because of the uncertainty in X , this does not guarantee that $\mu > 11$.

We would like to know just how certain we can be that $\mu > 11$.

The confidence interval does not directly tell us how confident we can be that $\mu > 11$.

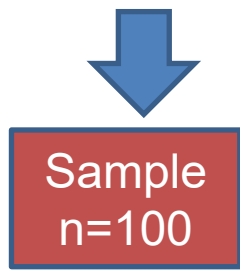
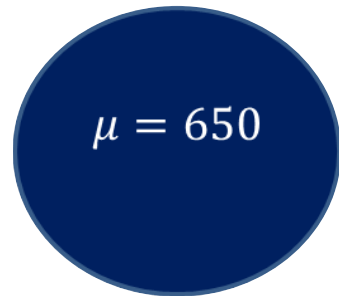
The statement " $\mu > 11$ " is a hypothesis about the population mean μ .

To determine just how certain we can be that a hypothesis is true, we must perform a hypothesis test.

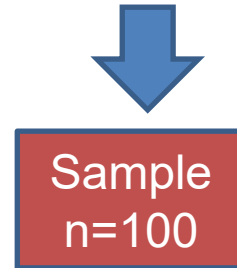
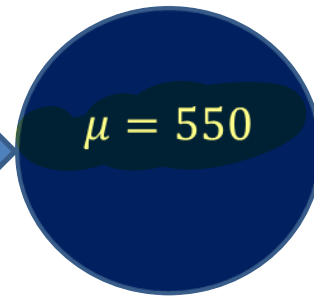
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Introduction

**Bore well water
(TDS)**



**water after
filtration**



Is it plausible
that this sample,
with its mean of
550, could have
come from a
population
whose mean is
650 or more?

- It turns out that hypothesis tests are closely related to confidence intervals.
- In general, whenever a confidence interval can be computed, a hypothesis test can also be performed, and vice versa.
- **A Hypothesis is an assumption about population Parameter.**

A hypothesis test produces a number between 0 and 1 that measures the degree of certainty we may have in the truth of a hypothesis about a quantity such as a population mean or proportion.

- Represented by H_0
- The Null Hypothesis says that the effect indicated by the sample is due only to random variation between the sample and the population.
- In performing a hypothesis test, we essentially put the Null Hypothesis on trial.
- In scientific research, the null hypothesis is the claim that the effect being studied does not exist.

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Alternate Hypothesis

- Represented by H_1
- The Alternate Hypothesis says that the effect indicated by the sample is real, in that it accurately represents the whole population.
- In statistical hypothesis testing, the Alternative Hypothesis is one of the proposed proposition in the hypothesis test.

Hypothesis Testing



Null Hypothesis: H_0



- State the Hypothesized value of the parameter before Sampling
- The assumption we wish to test.



Alternate Hypothesis: H_1



- All possible alternatives other than the null Hypothesis

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Introduction

Example:

Do men and women having different average salaries after graduating University?

Solution: $H_0: \mu_M = \mu_F$ $H_1: \mu_M \neq \mu_F$

Example:

A coin was flipped 50 times, resulting in 40 Heads and 10 Tails.

$H_0: p = 0.5$ $H_1: p \neq 0.5$

Example:

A new type of battery will be installed in heart pacemakers if it can be shown to have a mean lifetime is greater than eight years.

Solution: $H_0: \mu \leq 8$ $H_1: \mu > 8$

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Introduction

- A parameter is the characteristic of the population Like its mean or variance
- The parameter must be identified before the analysis.

I Assume the average weight
of this class is 58 Kg



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Introduction

- The best way to determine whether a statistical hypothesis is true would be to examine the entire population.
- Since that is often impractical, researchers typically examine a random sample from the population.
- If sample data are not consistent with the statistical hypothesis, the hypothesis is rejected.

Steps in Performing a Hypothesis Test

1. Define H_0 and H_1 .
2. Assume H_0 to be true.
3. Compute a **test statistic**. A test statistic is a statistic that is used to assess the strength of the evidence against H_0 .
4. Compute the P -value of the test statistic. The P -value is the probability, assuming H_0 to be true, that the test statistic would have a value whose disagreement with H_0 is as great as or greater than that actually observed. The P -value is also called the **observed significance level**.
5. State a conclusion about the strength of the evidence against H_0 .

Note:

In any hypothesis test, we are calculating conditional probabilities based on the assumption that the null hypothesis is true.

P-value: **measures the plausibility of H_0** . The smaller the P-value, the stronger the evidence is against H_0 .

If the P -value is sufficiently small, we may be willing to abandon our assumption that H_0 is true and believe H_1 instead.
This is referred to as rejecting the null hypothesis.

Note:

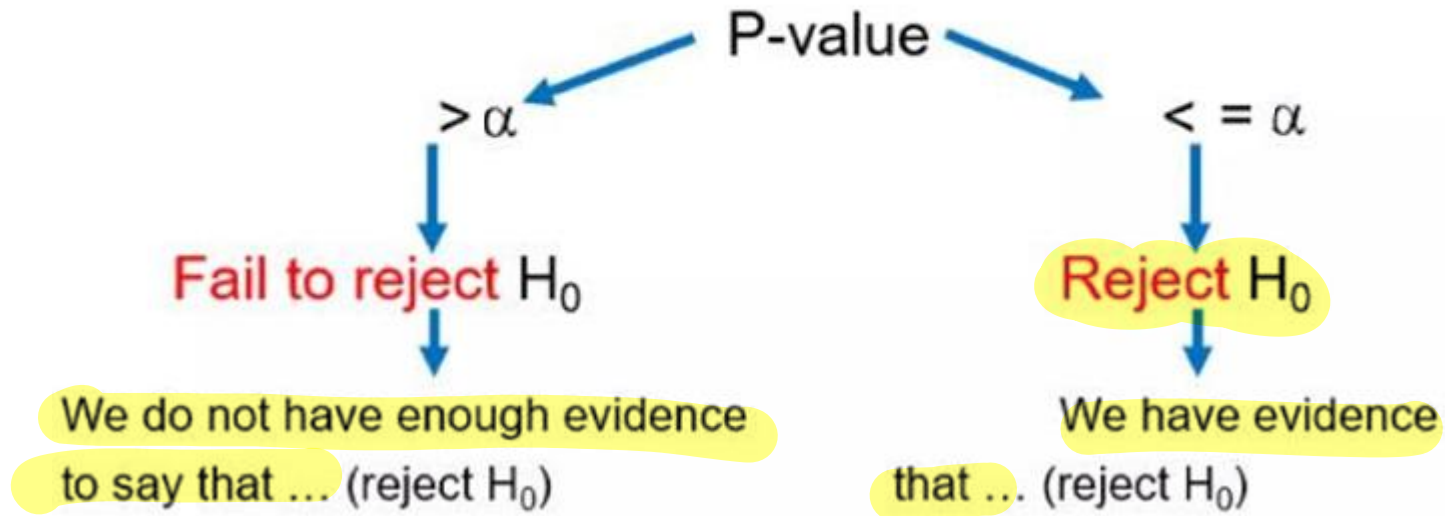
It is natural to ask how small the P-value should be in order to reject H_0 .
Some people use the “5% rule”; they reject H_0 if $P \leq 0.05$.

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Conclusions from hypothesis

- IF $p\text{-value} < \alpha \rightarrow \text{Reject } H_0$
- IF $p\text{-value} > \alpha \rightarrow \text{Fail to Reject } H_0$

α -is the level of significance



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Type of Hypothesis Tests:

Sl No	Type of Hypothesis Test	Alternate Hypothesis	Null Hypothesis
1	Left Tailed Test 	$<$	$\geq$
2	Right Tailed Test 	$>$	$\leq$
3	Two Tailed Test 	$\neq$	$=$

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Note:

Hypothesis Test	Population S.D. known	Population S.D. unknown
$n < 30$ Small sample drawn from Normal population	Use z- test	Use t-test
$n \geq 30$ Large sample	Use z- test	Use z- test Or t- test

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Topics Covered



❖ 6.1 - Large-Sample Tests for a Population Mean

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❖ Large-Sample Tests for a Population Mean

We want to test a claim about the population mean μ .

- Null hypothesis:

$$H_0 : \mu = \mu_0$$

- Alternative hypothesis:

- Two-tailed: $H_1 : \mu \neq \mu_0$
- Right-tailed: $H_1 : \mu > \mu_0$
- Left-tailed: $H_1 : \mu < \mu_0$

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❖ Large-Sample Tests for a Population Mean

- If population standard deviation σ is known:

Test Statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

- If σ is unknown but $n \geq 30$:

$$Z = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

(where s is the sample standard deviation).

- IF $p\text{-value} < \alpha \rightarrow \text{Reject } H_0$
- IF $p\text{-value} > \alpha \rightarrow \text{Fail to Reject } H_0$

α -is the level of significance

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Large-Sample Tests for a Population Mean

- Compute the P -value.
- The P -value is an area under the normal curve, depends on the alternate hypothesis

Alternate Hypothesis	P -value
$H_1: \mu > \mu_0$	Area to the right of z
$H_1: \mu < \mu_0$	Area to the left of z
$H_1: \mu = \mu_0$	Sum of the areas in the tails cut off by z and $-z$

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Example

The article “Wear in Boundary Lubrication” (S. Hsu, R. Munro, and M. Shen, *Journal of Engineering Tribology*, 2002: 427– 441) discusses several experiments involving various lubricants. In one experiment, **45 steel balls** lubricated with purified paraffin were subjected to a 40 kg load at 600 rpm for 60 minutes . **The average wear, measured by the reduction in diameter, was $673.2\text{ }\mu\text{m}$, and the standard deviation was $14.9\text{ }\mu\text{m}$.** Assume that the **specification for a lubricant is that the mean wear be less than $675\text{ }\mu\text{m}$.** Find the P -value for testing $H_0: \mu \geq 675$ versus $H_1: \mu < 675$.

$$\bar{x} = 673.2\text{ }\mu\text{m} \text{ (sample mean)}$$

$$\mu_0 = 675\text{ }\mu\text{m} \text{ (population mean under the null hypothesis)}$$

$$s = 14.9\text{ }\mu\text{m} \text{ (sample standard deviation)}$$

$$n = 45 \text{ (sample size)}$$

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Example - Continued

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{673.2 - 675}{14.9/\sqrt{45}}$$
$$z = \frac{673.2 - 675}{14.9/\sqrt{45}} = \frac{-1.8}{14.9/6.708}$$

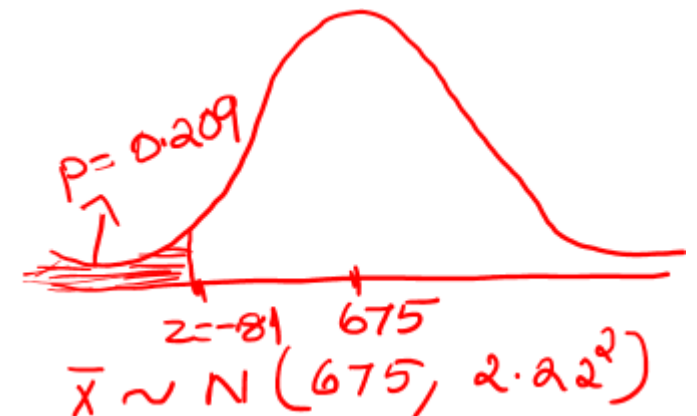
Now, we compute the z-value:

$$z = \frac{-1.8}{2.221} \approx -0.81$$

Using a standard normal table or calculator:

$$P(Z < -0.81) \approx 0.2090$$

Thus, the P -value is approximately 0.209.

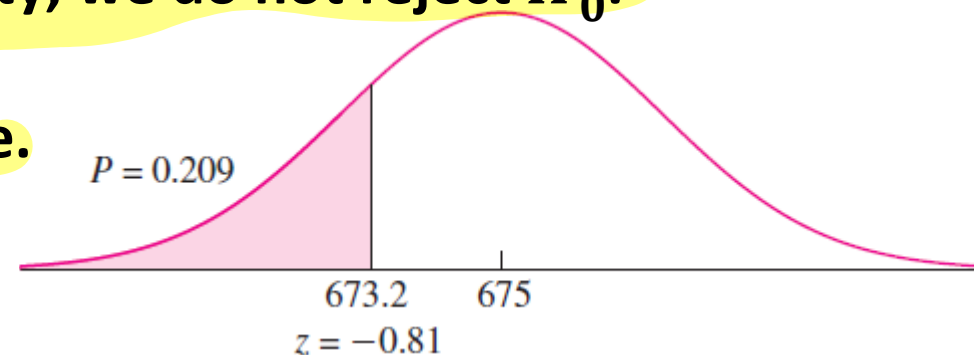


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Example - Continued

Solution continued:

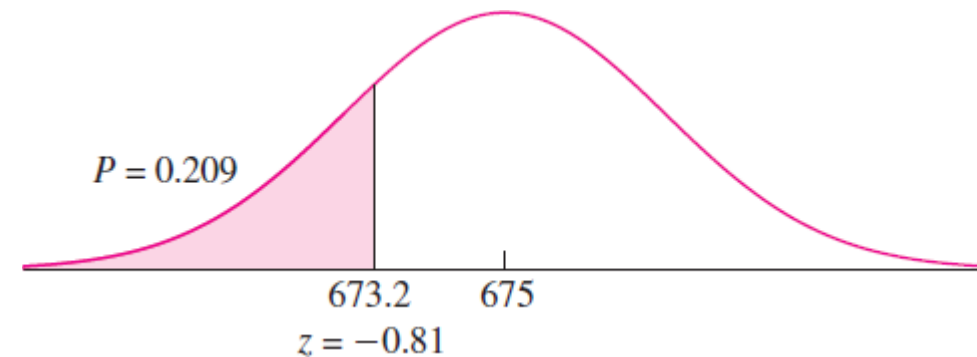
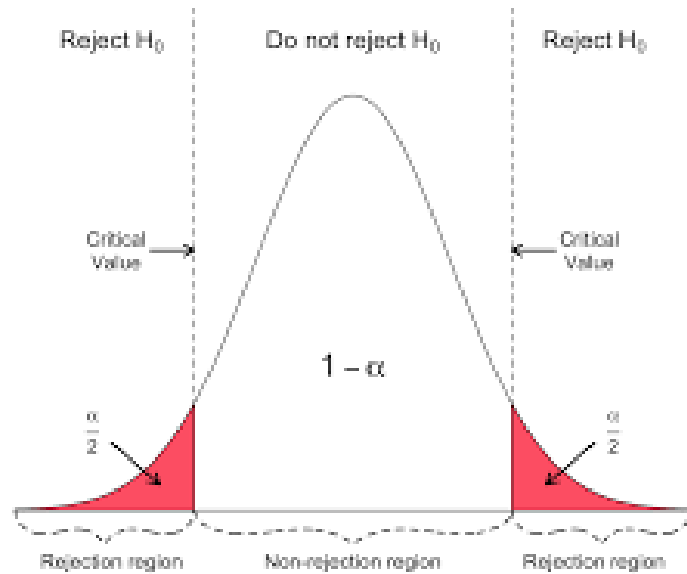
- We got P – value is 0.209 .
- Therefore if H_0 is true, there is a 20.9% chance to observe a sample whose disagreement with H_0 is as least as great as that which was actually observed.
- Since 0.209 is not a very small probability, we do not reject H_0 .
- Instead, we conclude that H_0 is plausible.



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Example - Continued

Rejection based on the critical value



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Example



A scale is to be calibrated by weighing a 1000 g test weight 60 times. The 60 scale readings have mean 1000.6 g and standard deviation 2 g.

- Find the P -value for testing

$$H_0 : \mu = 1000 \text{ versus } H_1 : \mu \neq 1000.$$

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Example - Continued



Solution:

$H_0 : \mu = 1000$ versus $H_1 : \mu \neq 1000$.
We assume H_0 is true

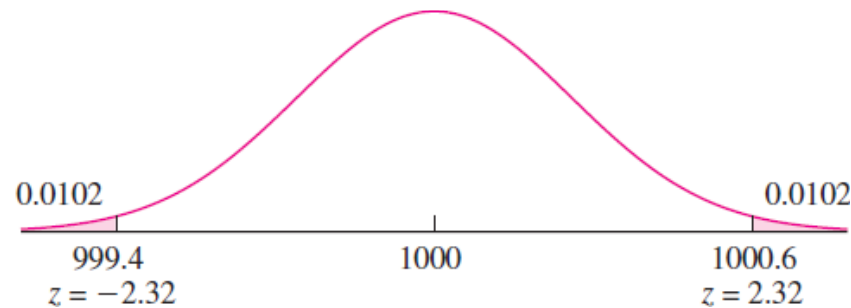
$$z = \frac{1000.6 - 1000}{0.258} = 2.32$$

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Example - Continued

Solution continued:

- The P -value is the sum of the areas in both of these tails, which is 0.0204.
- Therefore, if H_0 is true, the probability of a result as extreme as or more extreme than that observed is only 0.0204.
- The evidence against H_0 is pretty strong. It would be prudent to reject H_0 and to recalibrate the scale.



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Large-Sample Tests for a Population Mean

Example:

- A sample of 400 male students is found to have a mean height 67.47 inches. Can it be reasonably regarded as a sample from a large population with mean height 67.39 inches and standard deviation 1.30 inches?

→ Population Parameters

μ & σ

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Large-Sample Tests for a Population Mean

Solution:

$$H_0: \mu = 67.39 \text{ inches}$$

$$H_1: \mu \neq 67.39 \text{ inches}$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} = \frac{67.47 - 67.39}{1.30 / \sqrt{400}} = 1.231$$

Now, calculate the P -value:

$$P(Z > 1.23) = 1 - 0.8907 = 0.1093$$

Two-tailed test
 $P(Z > z)$

Since it's a two-tailed test, the total P -value is:

$$P = 2 \times 0.1093 = 0.2186$$

Thus, the P -value is approximately 0.219.

Since the P -value is greater than 0.05, we fail to reject the null hypothesis. This means it can reasonably be regarded as a sample from the population with mean height 67.39 inches.

🌟 Step 1: Recall what the *P-value* represents

In hypothesis testing, the **p-value** measures how extreme your sample statistic (like Z) is, **under the null hypothesis** H_0 .

Depending on the **alternative hypothesis** H_1 , you will look at:

- **Right tail** $\rightarrow P(Z > z_{\text{obs}})$
- **Left tail** $\rightarrow P(Z < z_{\text{obs}})$
- **Both tails** $\rightarrow 2 \times P(Z > |z_{\text{obs}}|)$

🔍 Step 2: Use the direction of the *alternative hypothesis* H_1

Type of test	Alternative hypothesis H_1	Tail to use	Compute p-value as...
Right-tailed test	$H_1 : \mu > \mu_0$	Upper (right) tail	$p = P(Z > z_{\text{obs}})$
Left-tailed test	$H_1 : \mu < \mu_0$	Lower (left) tail	$p = P(Z < z_{\text{obs}})$
Two-tailed test	$H_1 : \mu \neq \mu_0$	Both tails	$(p = 2 \times P(Z > z_{\text{obs}}))$

Example 1 — Right-tailed

- $H_0 : \mu = 50, H_1 : \mu > 50$
- Suppose $z_{\text{obs}} = 1.8$
- $p = P(Z > 1.8) = 0.0359$

You're testing if the mean is **greater**, so you use the **upper tail**.

Example 2 — Left-tailed

- $H_0 : \mu = 50, H_1 : \mu < 50$
- Suppose $z_{\text{obs}} = -1.8$
- $p = P(Z < -1.8) = 0.0359$

You're testing if the mean is **smaller**, so you use the **lower tail**.

Example 3 — Two-tailed

- $H_0 : \mu = 50, H_1 : \mu \neq 50$
- Suppose $z_{\text{obs}} = -2.1$
- $p = 2 \times P(Z > 2.1) = 2 \times 0.0179 = 0.0358$

You're testing for **any difference**, so you look at **both extremes**.

🔑 Summary Shortcut

If your test asks whether...	Use
"Greater than" $\rightarrow H_1 : \mu > \mu_0$	$P(Z > z)$
"Less than" $\rightarrow H_1 : \mu < \mu_0$	$P(Z < z)$
"Different from" $\rightarrow H_1 : \mu \neq \mu_0$	$(2 \times P(Z > z))$

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Large-Sample Tests for a Population Mean

Example:

- A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis that the coin is unbiased?

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Large-Sample Tests for a Population Mean

Example:

- A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis that the coin is unbiased?

Solution:

$$H_0: p = \frac{1}{2}$$

$$H_1: p \neq \frac{1}{2}$$

$$z = \frac{x - np}{\sqrt{npq}} = 1.6$$

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Large-Sample Tests for a Population Mean

For $z = 1.6$:

$$P(Z \leq 1.6) \approx 0.9452$$

2. Since this is a two-tailed test, the P -value is:

$$P = 2 \times (1 - 0.9452) = 2 \times 0.0548 = 0.1096$$

Thus, the P -value for $z = 1.6$ is approximately 0.1096.

$P > 0.05$, Accept H_0
that coin is unbiased

or Fail to reject H_0 , Hence we can conclude

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Large-Sample Tests for a Population Mean

Example:

- A trucking firm is suspicious of the claim that the average lifetime of certain tires is at least 28,000 miles.
- To check the claim, the firm puts 40 of these tires on its trucks and gets a mean lifetime of 27,463 with a standard deviation 1,348 miles.

Find the P value for testing

$$H_0: \mu \geq 28,000 \text{ miles}$$

$$H_1: \mu < 28,000 \text{ miles?}$$

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Large-Sample Tests for a Population Mean

Solution:

$$H_0: \mu \geq 28,000 \text{ miles}$$

$$H_1: \mu < 28,000 \text{ miles}$$

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{27,463 - 28,000}{1,348/\sqrt{40}} = -2.52 < -2.33$$

P – Value is 0.0059

Since P- Value is a very small probability we need to reject H_0

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Large-Sample Tests for a Population Mean

Summary

Let $X_1, \dots, X_n$ be a *large* (e.g., $n > 30$) sample from a population with mean μ and standard deviation σ .

To test a null hypothesis of the form $H_0: \mu \leq \mu_0$, $H_0: \mu \geq \mu_0$, or $H_0: \mu = \mu_0$:

- Compute the z -score: $z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$.

If σ is unknown it may be approximated with s .

- Compute the P -value. The P -value is an area under the normal curve, which depends on the alternate hypothesis as follows:

Alternate Hypothesis

$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

$$H_1: \mu \neq \mu_0$$

P -value

Area to the right of z

Area to the left of z

Sum of the areas in the tails cut off by z and $-z$

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MCQS

- . Which is an example of a null hypothesis?
- a) "The new drug has no effect on blood pressure"
 - b) "The new drug lowers blood pressure"
 - c) "The average score is higher than 70"
 - d) "There is a difference between men and women"

✓ **Answer: a)**

A two-tailed test is used when:

- a) We are only interested if the parameter is greater than a value
- b) We are only interested if the parameter is less than a value
- c) We are interested in deviations in both directions from the null value
- d) The sample size is greater than 30

✓ **Answer: c)**

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References:

- https://en.wikipedia.org/wiki/Alternative_hypothesis
- https://en.wikipedia.org/wiki/Null_hypothesis
- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.



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HYPOTHESIS and INFERENCE

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UNIT- 2 HYPOTHESIS and INFERENCE

Drawing conclusions from the results of Hypothesis Tests
(p-valued approach, Rejection Region approach)

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Topics Covered

- ❖ **Drawing Conclusions from the Results of Hypothesis Tests**
- ❖ **P Value and Acceptance and Rejection region Approach**
- ❖ **Statistical significance.**

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Drawing Conclusions from the Results of Hypothesis Tests

- The only two conclusions that can be reached in a hypothesis test are that
 - H_0 is false or that H_0 is plausible.
 - One can never conclude that H_0 is true.
 - How do we know when to reject H_0 ?
- The smaller the P -value, the less plausible H_0 becomes.
- A common rule of thumb is to draw the line at 5%. According to this rule of thumb, if $P \leq 0.05$, H_0 is rejected; otherwise H_0 is not rejected.

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Drawing Conclusions from the Results of Hypothesis Tests

- The smaller the P -value, the more certain we can be that H_0 is false.
- The larger the P -value, the more plausible H_0 becomes, but we can never be certain that H_0 is true.
- There is no sharp dividing line between conclusive evidence against H_0
- So while this rule of thumb is convenient, it has no real scientific justification.

p-value Approach

p-value = Probability of obtaining results as extreme as observed, assuming H_0 is true.

- **Steps:**

1. **Compute test statistic (Z, t, etc.).**
2. **Find p-value.**
3. **Compare with α .**

- **Rule:**

- **If $p \leq \alpha \rightarrow$ Reject H_0**
- **If $p > \alpha \rightarrow$ Fail to Reject H_0**

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Drawing Conclusions from the Results of Hypothesis Tests

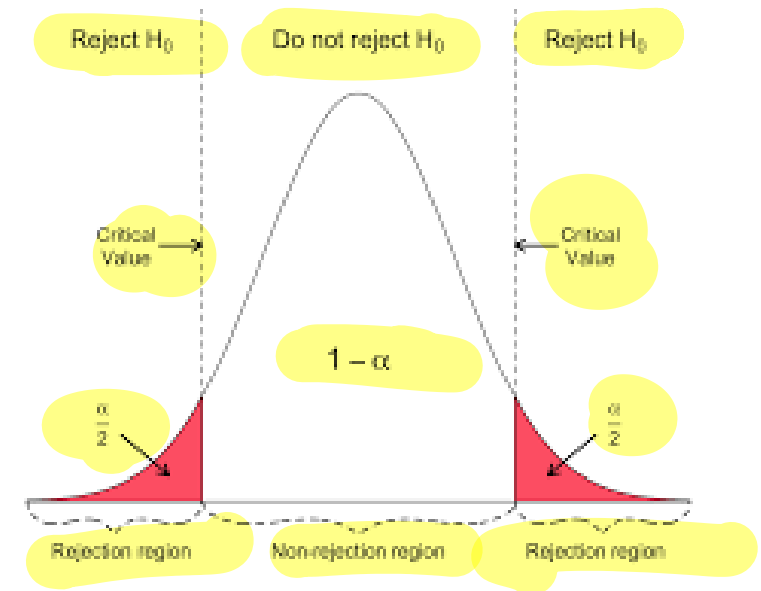
Rejection Region approach for Hypothesis Test

Critical Point & Rejection Region

- A critical point is a value of the test statistic that produces a P-value exactly equal to α .
- The region on the side of the critical point that leads to rejection is called the rejection region.
- The critical point itself is also in the rejection region.

Rejection region Approach

- Based on rejection and acceptance regions of the test statistic.
- Steps:
 1. Choose significance level (α).
 2. Find critical value(s) from statistical tables.
 3. Define acceptance & rejection regions.
 4. Compare test statistic with critical value(s).
- Rule:
 - If test statistic $\in$ rejection region $\rightarrow$ Reject H_0
 - Else $\rightarrow$ Fail to Reject H_0



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Drawing Conclusions from the Results of Hypothesis Tests

Comparison of Approaches

p-value Approach:

- Gives exact strength of evidence against H_0 .
- Common in modern software.

Rejection region Approach:

- Relies on cutoff values of test statistic.
- Traditional/manual method.

Note: Both lead to the same decision.

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Drawing Conclusions from the Results of Hypothesis Tests

Statistical Significance:

- Whenever the **P -value is less than a particular threshold, the result is said to be “statistically significant” at that level.**
- So, for example:
 - if $P \leq 0.05$, the result is statistically significant at the 5% level
 - if $P \leq 0.01$, the result is statistically significant at the 1% level, and so on.
- If a result is statistically significant at the $100\alpha\%$ level, we can also say that the null hypothesis is “rejected at level $100\alpha\%$.”

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Drawing Conclusions from the Results of Hypothesis Tests

- The null hypothesis is rejected at the $100\alpha\%$ level.
- When reporting the result of a hypothesis test, report the P –value, rather than just comparing it to 5% or 1%.
- Let α be any value between 0 and 1. Then, if $P \leq \alpha$,
- The result of the test is said to be statistically significant at the $100\alpha\%$ level.

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Drawing Conclusions from the Results of Hypothesis Tests

Note:

The **P -value** Is Not the Probability That H_0 Is True.

The p -value is the probability (assuming H_0 is true) of obtaining data as extreme as, or more extreme than, what was observed.

A researcher aims to evaluate whether a new teaching method improves exam scores. The null hypothesis is:

$$H_0 : \mu = 70 \quad (\text{the average score is } 70)$$

and the alternative hypothesis is:

$$H_a : \mu > 70 \quad (\text{the average score is greater than } 70)$$

g → wrong

The researcher collects a sample of $n = 25$ students, computes the sample mean to be $\bar{x} = 74$, and assumes the population standard deviation is $\sigma = 10$.

Interpreting p -value in this example

In this example, the p -value is the probability (assuming H_0 is true) of obtaining a test statistic as extreme as 2.0 or more in the *right tail* of the standard normal distribution:

$$p\text{-value} = P(Z \geq 2.0) \approx 0.0228.$$

This means: if the true population mean were 70, the probability of seeing a sample mean of 74 or more (i.e. seeing $Z \geq 2.0$) is about 0.0228 (2.28%). Because 0.0228 is less than the significance level 0.05, we reject H_0 .

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Drawing Conclusions from the Results of Hypothesis Tests

The null hypothesis, on the other hand, either is true or is not true.
The truth or falsehood of H_0 cannot be changed by repeating the experiment.
It is therefore not correct to discuss the “probability” that H_0 is true.

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Drawing Conclusions from the Results of Hypothesis Tests

Example:

- A hypothesis test is performed of the null hypothesis $H_0: \mu = 0$.
- The P – value turns out to be 0.03.
- Is the result statistically significant at the 10% level? The 5% level? The 1% level?
- Is the null hypothesis rejected at the 10% level? The 5% level? The 1% level?

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Drawing Conclusions from the Results of Hypothesis Tests

Solution:

- The result is statistically significant at any level greater than or equal to 3%.
- Thus it is statistically significant at the 10% and 5% levels, but not at the 1% level.
- Similarly, we can reject the null hypothesis at any level greater than or equal to 3%
- So H_0 is rejected at the 10% and 5% levels, but not at the 1% level.

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Drawing Conclusions from the Results of Hypothesis Tests

Choose H_0 to Answer the Right Question:

- When performing a hypothesis test, it is important to choose H_0 and H_1 appropriately so that the result of the test can be useful in forming a conclusion.

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Drawing Conclusions from the Results of Hypothesis Tests

Choose H_0 to Answer the Right Question:

Example:

- Specifications for a water pipe call for a mean breaking strength μ of more than 2000 lb per linear foot.
- Engineers will perform a hypothesis test to decide whether or not to use a certain kind of pipe.
- They will select a random sample of 1 ft sections of pipe, measure their breaking strengths, and perform a hypothesis test.
- The pipe will not be used unless the engineers can conclude that $\mu > 2000$.
- Assume they test:
 - $H_0: \mu \leq 2000$ versus $H_1: \mu > 2000$.
- Will the engineers decide to use the pipe if H_0 is rejected? What if H_0 is not rejected?

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Drawing Conclusions from the Results of Hypothesis Tests

Choose H_0 to Answer the Right Question:

Solution:

- If H_0 is rejected, the engineers will conclude that $\mu > 2000$, and they will use the pipe.
- If H_0 is not rejected, the engineers will conclude that μ might be less than or equal to 2000, and they will not use the pipe.

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Drawing Conclusions from the Results of Hypothesis Tests

Choose H_0 to Answer the Right Question:

Example:

- In the previous example, Assume the engineers test:
- $H_0: \mu \geq 2000$ versus $H_1: \mu < 2000$.
- Will the engineers decide to use the pipe if H_0 is rejected?
- What if H_0 is not rejected?

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Drawing Conclusions from the Results of Hypothesis Tests

Choose H_0 to Answer the Right Question:

Solution:

- If H_0 is rejected, the engineers will conclude that $\mu < 2000$, and they will not use the pipe.
- If H_0 is not rejected, the engineers will conclude that μ *might* be greater than or equal to 2000, but that it also might not be. So again, they won't use the pipe.

Mathematics for Computer Science Engineers

Drawing Conclusions from the Results of Hypothesis Tests

Statistical Significance Is Not the Same as Practical Significance:

- The **P -value does not measure practical significance.**
- When a result has a small P -value, we say that it is “statistically significant.”
- In common usage, the word significant means “important.” It is therefore tempting to think that statistically significant results must always be important. This is not the case.
- Sometimes statistically significant results do not have any scientific or practical importance.

Mathematics for Computer Science Engineers

Drawing Conclusions from the Results of Hypothesis Tests

Statistical Significance Is Not the Same as Practical Significance:

- What it does measure is the degree of confidence we can have that the true value is really different from the value specified by the null hypothesis.
- When the P -value is small, then we can be confident that the true value is really different.
- This does not necessarily imply that the difference is large enough to be of practical importance.

Mathematics for Computer Science Engineers

MCQS/true or false

True or false:

If $P = 0.02$, then

- a. The result is statistically significant at the 5% level.
- b. The result is statistically significant at the 1% level.
- c. The null hypothesis is rejected at the 5% level.
- d. The null hypothesis is rejected at the 1% level.

Ans:

- (a) True. The result is statistically significant at any level greater than or equal to 2%.
- (b) False. $P > 0.01$, so the result is not statistically significant at the 1% level.
- (c) True. The null hypothesis is rejected at any level greater than or equal to 2%.
- (d) False. $P > 0.01$, so the null hypothesis is not rejected at the 1% level.

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MCQS/true or false

2) George performed a hypothesis test. Luis checked George's work by redoing the calculations. Both George and Luis agree that the result was statistically significant the 5% level, but they got different P-values. George got a P-value of 0.20 and Luis got a P-value of 0.02.

- a) Is it possible that George's work is correct? Explain.
- b. Is it possible that Luis's work is correct? Explain.

Ans:

- (a) No. If the P-value is 0.20, then the result is not statistically significant at the 5% level.
- (b) Yes. If the P-value is 0.02, then the result is statistically significant at the 5% level.

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References

- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.



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Mathematics for Computer Science Engineers

HYPOTHESIS and INFERENCE

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Mathematics for Computer Science Engineers

Relationship between Hypothesis and
Confidence Mean
Tests for a Population Proportion

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The Relationship Between Hypothesis Tests and Confidence Intervals:

- The values contained within a two-sided level $100(1 - \alpha)\%$ confidence interval for a population mean μ are precisely those values for which the P -value of a two-tailed hypothesis test will be greater than α .
- A one-sided level $100(1 - \alpha)\%$ confidence interval consists of all the values for which the P -value in a one-tailed test would be greater than α .

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Topics Covered



❖ 6.3 - Large-Sample Tests for a Population Proportion

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Tests for a Population Proportion

More than 85% of the students across the campuses participated in the revision session taken on 7/10/2020



Sample of 120 students



Found 75 attended



Can we accept the claim?

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Tests for a Population Proportion

- The **Sample Size Must Be Large**
- The test just described requires that the sample proportion be approximately normally distributed.
- This assumption will be justified whenever both $np_0 > 10$ and $n(1 - p_0) > 10$, where p_0 is the population proportion specified in the null distribution.
- Then the **z -score** can be used as the test statistic, making this a z test.

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Tests for a Population Proportion

- Let X be the number of successes in n independent Bernoulli trials, each with success probability p ; in other words, let $X \sim \text{Bin}(n, p)$.
- Null distribution of $\hat{p}$, $\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$
- p is approximated by using p_0 value.
- To test a null hypothesis of the form
 $H_0: p \leq p_0, H_0: p \geq p_0, \text{ or } H_0: p = p_0$

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Tests for a Population Proportion

- Assuming that both np_0 and $n(1 - p_0)$ are greater than 10:
- Compute the z-score:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

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Tests for a Population Proportion

- Compute the P –value.
- The P –value is an area under the normal curve, which depends on the alternate hypothesis as follows:

Alternate Hypothesis	P value
$H_1: p > p_0$	Area to the right of z
$H_1: p < p_0$	Area to the left of z
$H_1: p = p_0$	Sum of the areas in the tails cut off by z and $-z$

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Tests for a Population Proportion



Example:

- The article “Refinement of Gravimetric Geoid Using GPS and Leveling Data” (W. Thurston, *Journal of Surveying Engineering*, 2000:27–56) presents a method for measuring orthometric heights above sea level.
- For a sample of 1225 baselines, 926 gave results that were within the class C spirit leveling tolerance limits.
- Can we conclude that this method produces results within the tolerance limits more than 75% of the time?

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Tests for a Population Proportion

Solution

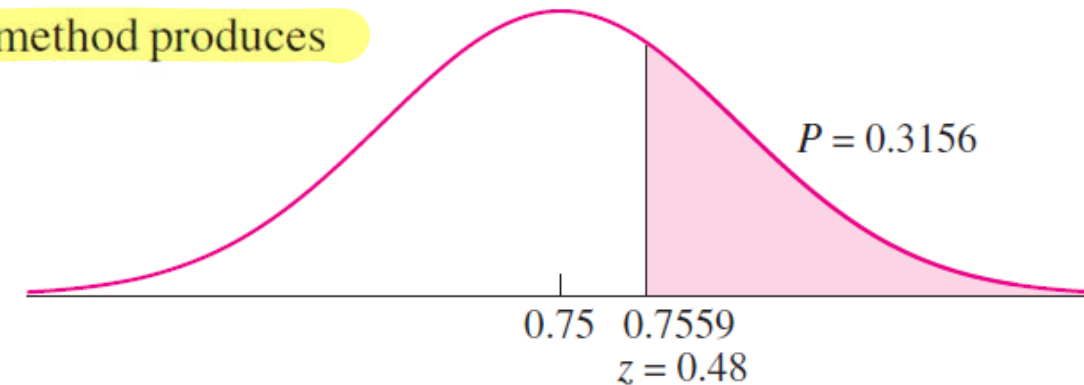
Let p denote the probability that the method produces a result within the tolerance limits. The null and alternate hypotheses are

$$H_0: p \leq 0.75 \quad \text{versus} \quad H_1: p > 0.75$$

The sample proportion is $\hat{p} = 926/1225 = 0.7559$. Under the null hypothesis, $\hat{p}$ is normally distributed with mean 0.75 and standard deviation $\sqrt{(0.75)(1 - 0.75)/1225} = 0.0124$. The z -score is

$$z = \frac{0.7559 - 0.7500}{0.0124} = 0.48$$

The P -value is 0.3156 (see Figure 6.6). We cannot conclude that the method produces good results more than 75% of the time.



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Tests for a Population Proportion

Example:

A commonly prescribed drug for relieving nervous tension is believed to be only 60% effective. Experimental results with a new drug administered to a random sample of 100 adults who were suffering from nervous tension show that 70 received relief.

Is this sufficient evidence to conclude that the new drug is superior to the one commonly prescribed? Use a 0.05 level of significance.

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Tests for a Population Proportion

Solution:

$$H_0: p \leq 0.6$$

$$H_1: p > 0.6$$

$$z = \frac{70 - 60}{\sqrt{100 \times 0.6 \times 0.4}} = 2.04$$

$$p(z > 2.04) = 0.0207 < 0.05$$

So reject H_0 and conclude that the new drug is superior.

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Tests for a Population Proportion

Relationship with Confidence Intervals for a Proportion:

- A level $100(1 - \alpha)\%$ confidence interval for a population mean μ contains those values for a parameter for which the P -value of a hypothesis test will be greater than α .
- For the confidence intervals for a proportion presented earlier and the hypothesis test presented here, this statement is only approximately true.
- The reason for this is that the methods presented earlier are slight modifications (that are much easier to compute) of a more complicated confidence interval method for which the statement is exactly true.

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Tests for a Population Proportion

Summary

Let X be the number of successes in n independent Bernoulli trials, each with success probability p ; in other words, let $X \sim \text{Bin}(n, p)$.

To test a null hypothesis of the form $H_0: p \leq p_0$, $H_0: p \geq p_0$, or $H_0: p = p_0$, assuming that both np_0 and $n(1 - p_0)$ are greater than 10:

- Compute the z -score: $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$.
- Compute the P -value. The P -value is an area under the normal curve, which depends on the alternate hypothesis as follows:

Alternate Hypothesis

$$H_1: p > p_0$$

$$H_1: p < p_0$$

$$H_1: p \neq p_0$$

P -value

Area to the right of z

Area to the left of z

Sum of the areas in the tails cut off by z and $-z$

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MCQS

Which statement correctly describes the relationship between Confidence Interval (CI) and Hypothesis Testing?

- a) If the CI contains the hypothesized parameter, we reject H_0
- b) b) If the CI excludes the hypothesized parameter, we reject H_0
- c) CI and hypothesis test are unrelated
- d) CI always gives wider results than a test

Answer b)

A company claims that 60% of its customers are satisfied. You take a sample of 200 customers and find that 120 are satisfied. Which hypothesis test setup is correct for testing the claim?

- a) $H_0 : p = 0.6, H_1 : p \neq 0.6$
- b) $H_0 : p \neq 0.6, H_1 : p = 0.6$
- c) $H_0 : \hat{p} = 0.6, H_1 : \hat{p} \neq 0.6$
- d) $H_0 : p > 0.6, H_1 : p < 0.6$

Answer) a

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References

- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.
- <https://www.geeksforgeeks.org/real-life-applications-of-hypothesis-testing/>



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Small-Sample Tests for a Population mean

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Mathematics for Computer Science Engineers

Topics Covered



❖ 6.4 - Small-Sample Tests for a Population mean

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Small-Sample Tests for a Population Mean

A small sample test for the population mean is typically conducted when the sample size ≤ 30 , and the population standard deviation is unknown. In such cases, a **t-test** is used instead of a **z-test**

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Small-Sample Tests for a Population Mean



Examples

1. Spacer collars for a transmission countershaft have a thickness specification of 38.98–39.02 mm. The process that manufactures the collars is supposed to be calibrated so that the mean thickness is 39.00 mm, which is in the center of the specification window.

A sample of six collars is drawn and measured for thickness. The six thicknesses are 39.030, 38.997, 39.012, 39.008, 39.019, and 39.002. Assume that the population of thicknesses of the collars is approximately normal. Can we conclude that the process needs recalibration?

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Small-Sample Tests for a Population Mean

Solution

$$H_0: \mu = 39.00 \quad \text{versus} \quad H_1: \mu \neq 39.00$$

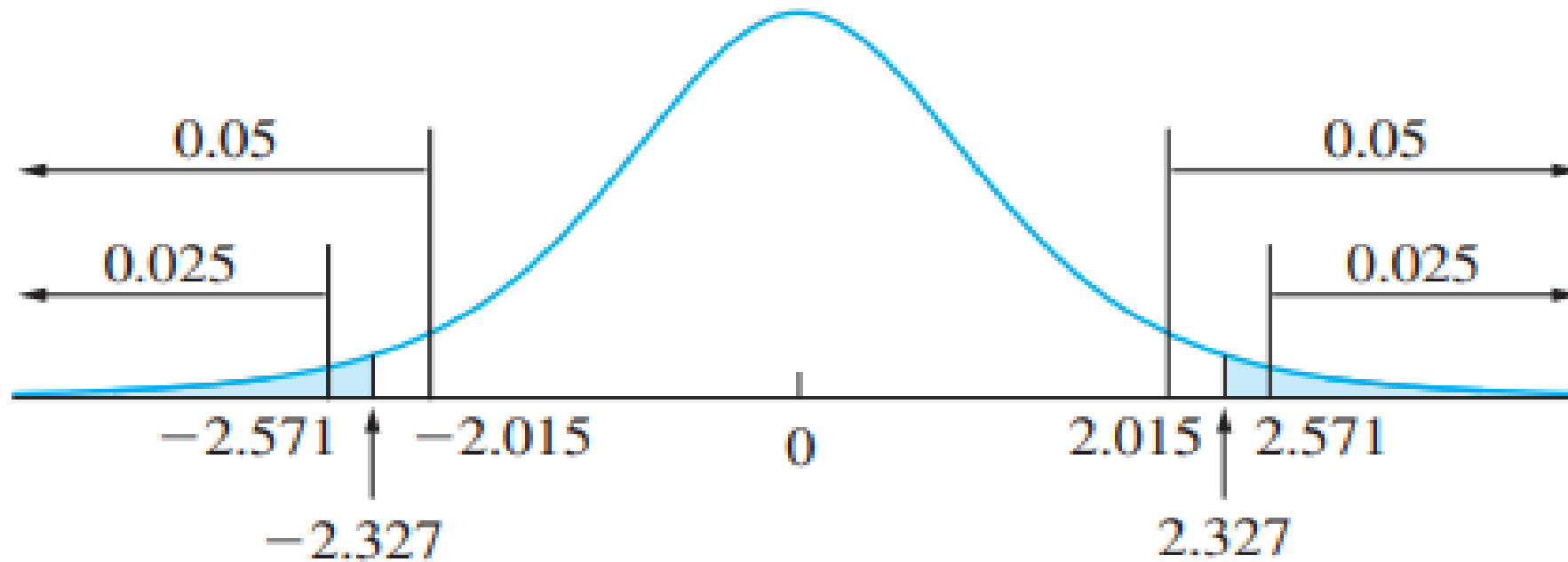
$$t = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

In this example the observed values of the sample mean and standard deviation are $\bar{X} = 39.01133$ and $s = 0.011928$. The sample size is $n = 6$. The null hypothesis specifies that $\mu = 39$. The value of the test statistic is therefore

$$t = \frac{39.01133 - 39.00}{0.011928 / \sqrt{6}} = 2.327$$

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Small-Sample Tests for a Population Mean



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Small-Sample Tests for a Population Mean

The P -value is the probability of observing a value of the test statistic whose disagreement with H_0 is as great as or greater than that actually observed. Since H_0 specifies that $\mu = 39.00$, this is a two-tailed test, so values both above and below 39.00 disagree with H_0 . Therefore the P -value is the sum of the areas under the curve corresponding to $t > 2.327$ and $t < -2.327$.

Figure 6.7 illustrates the null distribution and indicates the location of the test statistic. From the t table (Table A.3 in Appendix A) the row corresponding to 5 degrees of freedom indicates that the value $t = \pm 2.015$ cuts off an area of 0.05 in each tail, for a total of 0.10, and that the value $t = \pm 2.571$ cuts off an area of 0.025 in each tail, for a total of 0.05. Thus the P -value is between 0.05 and 0.10. While we cannot conclusively state that the process is out of calibration, it doesn't look too good. It would be prudent to recalibrate.

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Small-Sample Tests for a Population Mean



Example Solution:

1. Before a substance can be deemed safe for landfilling, its chemical properties must be characterized. The article “Landfilling Ash/Sludge Mixtures” (J. Benoit, T. Eighmy, and B. Crannell, *Journal of Geotechnical and Geoenvironmental Engineering*, 1999: 877–888) reports that in a sample of six replicates of sludge from a New Hampshire wastewater treatment plant, the mean pH was 6.68 with a standard deviation of 0.20. Can we conclude that the mean pH is less than 7.0?

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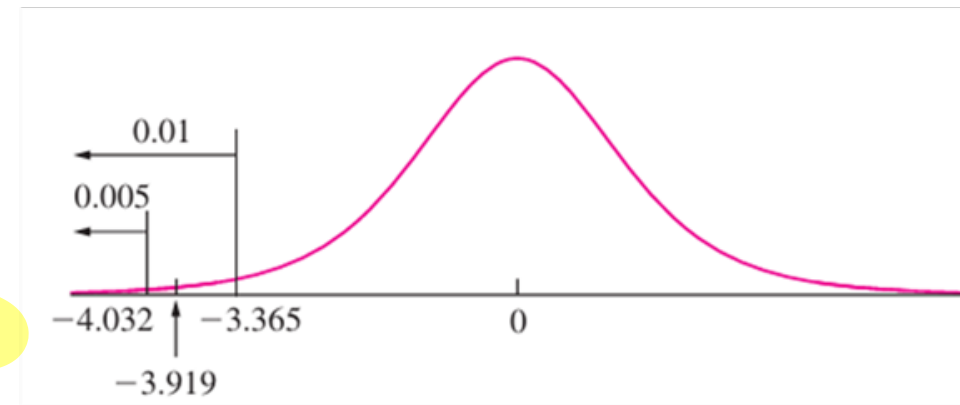
Small-Sample Tests for a Population Mean

Solution:

$$H_0: \mu \geq 7.0$$

$$H_1: \mu < 7.0$$

$$t = \frac{6.68 - 7.00}{0.20/\sqrt{6}} = -3.919$$



Consulting the t table, we find that the value $t = -3.365$ cuts off an area of 0.01 in the left-hand tail, and the value $t = -4.033$ cuts off an area of 0.005

We conclude that the P-value is between 0.005 and 0.01. There is strong evidence that the mean pH is less than 7.0.

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Small-Sample Tests for a Population Mean



More Example

2. A certain manufactured product is supposed to contain 23% potassium by weight. A sample of 10 specimens of this product had an average percentage of 23.2 with a standard deviation of 0.2. If the mean percentage is found to differ from 23, the manufacturing process will be recalibrated.
- State the appropriate null and alternate hypotheses.
 - Compute the P -value.
 - Should the process be recalibrated? Explain.

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Small-Sample Tests for a Population Mean



Solution:

(a) The null and alternate hypotheses are
 $H_0: \mu = 23$ versus $H_1: \mu \neq 23$.

(b) $\bar{X} = 23.2$, $s = 0.2$, $n = 10$. There are $10 - 1 = 9$ degrees of freedom.

$$t = (23.2 - 23) / (0.2 / \sqrt{10}) = 3.162.$$

Since the alternate hypothesis is of the form $\mu \neq \mu_0$, the P-value is the sum of the areas to the right of $t = 3.162$ and to the left of $t = -3.162$.

From the t table, $0.01 < P < 0.02$. A computer package gives $P = 0.0115$.

(c) Yes. Because the P-value is small, we can conclude that the mean percentage differs from 23, so the machine should be recalibrated.

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Small-Sample Tests for a Population Mean

Summary

Let $X_1, \dots, X_n$ be a sample from a *normal* population with mean μ and standard deviation σ , where σ is unknown.

To test a null hypothesis of the form $H_0: \mu \leq \mu_0$, $H_0: \mu \geq \mu_0$, or $H_0: \mu = \mu_0$:

- Compute the test statistic $t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$.
- Compute the *P*-value. The *P*-value is an area under the Student's *t* curve with $n - 1$ degrees of freedom, which depends on the alternate hypothesis as follows:

Alternate Hypothesis

$$H_1: \mu > \mu_0$$

$$H_1: \mu < \mu_0$$

$$H_1: \mu \neq \mu_0$$

P-value

Area to the right of t

Area to the left of t

Sum of the areas in the tails cut off by t and $-t$

- If σ is known, the test statistic is $z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$, and a *z* test should be performed.

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MCQS

For a two-tailed test at 5% significance using a sample of size 15, you calculate $t = 2.3$ (df = 14). The corresponding two-tailed **p-value** ≈ 0.037 . What is your conclusion?

- a) Fail to reject H_0
- b) Reject H_0
- c) Accept H_0
- d) Cannot decide

Answer b)

You sample 12 items from a population with unknown standard deviation. Which test should you use to test the population mean?

- a) Z-test
- b) T-test for mean (small sample, σ unknown)
- c) Chi-square test
- d) F-test

Answer b)

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References



- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.



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HYPOTHESIS and INFERENCE
Distribution Free Tests

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Distribution Free Tests.

Nonparametric methods :

- rank-based methods are used when we have no idea about the population distribution from which the data is sampled.
- Used for small sample sizes.
- Used when the data are measured on an ordinal scale and only their ranks are meaningful.
- Data are skewed or heavily tailed, or you detect outliers.
- Variances between groups are not equal and transformation either not possible or undesirable.

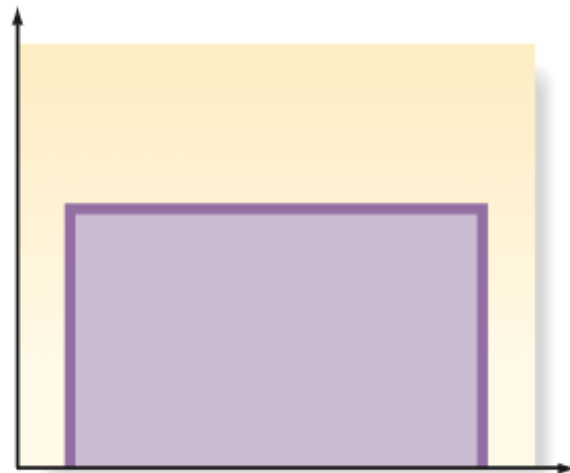
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Non-Parametric test procedure

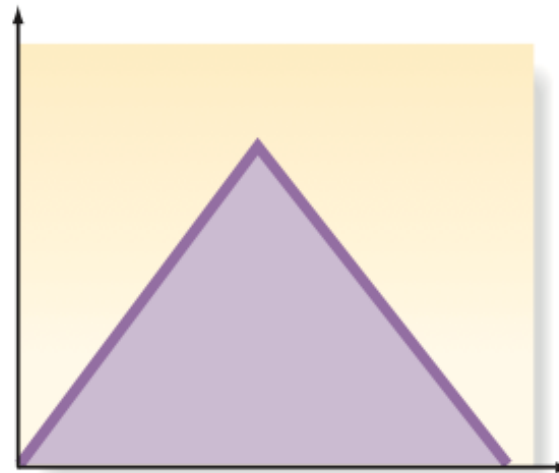
- Do not involve population parameters
 - Example: probability distributions, independence
- Data measured on any scale
 - Ratio or interval
 - Ordinal
 - Example: good-better-best
 - Nominal
 - Example: male-female
- Example: Wilcoxon rank sum test

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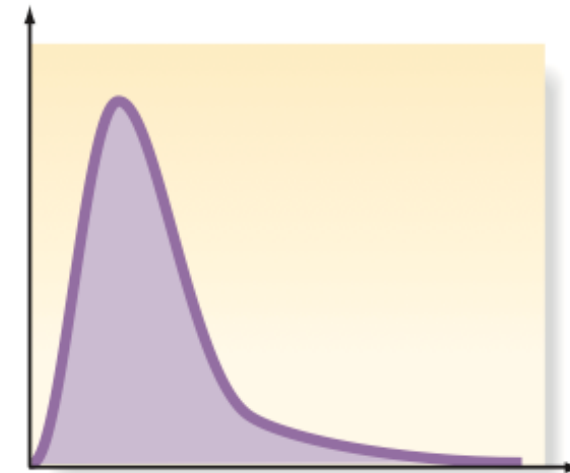
Nonnormal Distributions - t -Statistic is Invalid



a. Flat distribution



b. Peaked distribution



c. Skewed distribution

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Distribution Free Tests.

- The samples are not required to come from any specific distribution.
- While distribution free tests do require assumptions for their validity, these assumptions are somewhat less restrictive than the assumptions needed for the t test.
- Distribution-free tests are sometimes called nonparametric tests.
- We discuss two distribution-free tests in this section. The first, called the Wilcoxon signed-rank test, is a test for a population mean.
The second, called the Wilcoxon rank-sum test, or the Mann–Whitney test.

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Advantages and disadvantages of distribution free test

Advantages

- Used with all scales
- Easier to compute
 - Developed originally before wide computer use
- Make fewer assumptions
- Need not involve population parameters
- Results may be as exact as parametric procedures

Disadvantages

- May waste information
 - If data permit using parametric procedures
 - Example: converting data from ratio to ordinal scale
- Difficult to compute by hand for large samples
- Tables not widely available

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Distribution Free Tests: Wilcoxon Signed-Rank Test



The nickel content (in parts per thousand by weight) is measured for six welds, giving the following results:

9.3, 0.9, 9.0, 21.7, 11.5, and 13.9.

Let μ represent the mean nickel content for this type of weld.

It is desired to test whether the average nickel content is **less than 12**.

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Distribution Free Tests

- It is desired to test $H_0: \mu \geq 12$ versus $H_1: \mu < 12$.
- The Student's t test is not appropriate, because there are two outliers, 0.9 and 21.7, which indicate that the population is not normal.
- The Wilcoxon signed-rank test can be used in this situation.

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Distribution Free Tests

The Wilcoxon Signed-Rank Test:

- To compute the rank-sum statistic, we begin by subtracting 12 from each sample observation to obtain differences. The difference closest to 0, ignoring sign, is assigned a rank of 1.
- The difference next closest to 0, again ignoring sign, is assigned a rank of 2, and so on.
- Finally, the ranks corresponding to negative differences are given negative signs. The following table shows the results.
- Denote the sum of the positive ranks S_+ and the sum of the absolute values of the negative ranks S_- .
- Either S_+ or S_- may be used as a test statistic;
- we shall use S_+

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Distribution Free Tests

The Wilcoxon Signed-Rank Test:

X	X-12	Rank
11.5	-0.5	-1
13.9	1.9	2
9.3	-2.7	-3
9.0	-3.0	-4
21.7	9.7	5
0.9	-11.1	-6

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Distribution Free Tests

In this example

$$S_+ = 2+5 = 7, \text{ and } S_- = 1+3+4+6 = 14.$$

Note that since the sample size is 6,
 $(S_+) + (S_-) = 1 + 2 + 3 + 4 + 5 + 6 = 21.$

For any sample, it is the case that
 $(S_+) + (S_-) = 1+2+\cdots+n = n(n+1)/2.$

In some cases, where there are many more positive ranks than negative ranks, it is easiest to first compute S_- by summing the negative ranks and then computing
 $S_+ = n(n+1)/2 - (S_-).$

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Distribution Free Tests

how S_+ can be used as a test statistic.

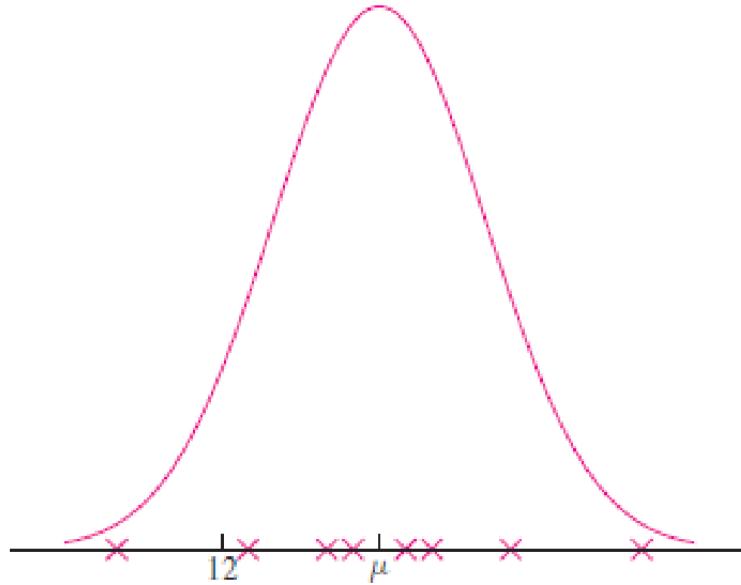


FIGURE 6.17 The true median is greater than 12. Sample observations are more likely to be above 12 than below 12. Furthermore, the observations above 12 will tend to have larger differences from 12 than the observations below 12. Therefore S_+ is likely to be large.

In Figure 6.17, $\mu > 12$. For this distribution, positive differences are more probable than negative differences and tend to be larger in magnitude as well.

Therefore it is likely that the positive ranks will be greater both in number and in magnitude than the negative ranks, so S_+ is likely to be large.

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Distribution Free Tests

In Figure 6.18, $\mu < 12$, and the situation is reversed.

positive ranks are likely to be fewer in number and smaller in magnitude, so S_+ is likely to be small.

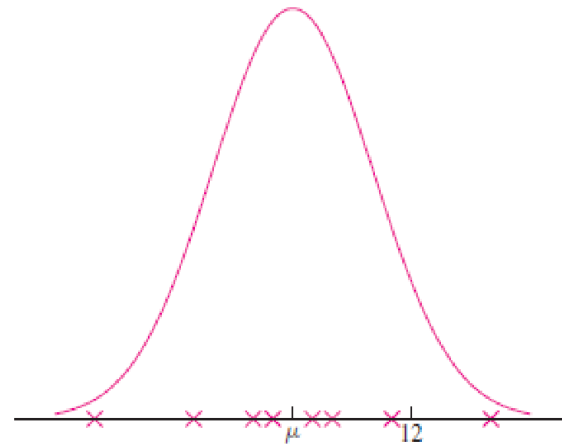


FIGURE 6.18 The true median is less than 12. Sample observations are more likely to be below 12 than above 12. Furthermore, the observations below 12 will tend to have larger differences from 12 than the observations above 12. Therefore S_+ is likely to be small.

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Distribution Free Tests

In general,

large values of S^+ will provide evidence against a null hypothesis of the form $H_0 : \mu \leq \mu_0$,

while small values of S^+ will provide evidence against a null hypothesis of the form $H_0 : \mu \geq \mu_0$.

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Distribution Free Tests

- The Null hypothesis is $H_0 : \mu \geq 12$, so a small value of S^+ will provide evidence against H_0 .
- We observe $S^+ = 7$. The P -value is the probability of observing a value of S^+ that is less than or equal to 7 when H_0 is true.
- For sample size $n = 6$, we find that the probability of observing a value of 4 or less is 0.1094.
- The probability of observing a value of 7 or less must be greater than this, so we conclude that $P > 0.1094$, and thus do not reject H_0 .

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Distribution Free Tests

Ties:

- Sometimes two or more of the quantities to be ranked have exactly the same value. Such quantities are said to be tied. The standard method for dealing with ties is to assign to each tied observation the average of the ranks they would have received if they had differed slightly.
- For example, the quantities 3, 4, 4, 5, 7 would receive the ranks 1, 2.5, 2.5, 4, 5
- The quantities 12, 15, 16, 16, 16, 20 would receive the ranks 1, 2, 4, 4, 4, 6.

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Distribution Free Tests

Differences of Zero:

- If the mean under H_0 is μ_0 , and one of the observations is equal to μ_0 , then its difference is 0, which is neither positive nor negative.
- An observation that is equal to μ_0 cannot receive a signed rank. The appropriate procedure is to drop such observations from the sample altogether, and to consider the sample size to be reduced by the number of these observations.

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Distribution Free Tests

The nickel content for six welds was measured to be 9.3, 0.9, 9.0, 21.7, 11.5, and 13.9. Use these data to test $H_0 : \mu \leq 5$ versus $H_1 : \mu > 5$.

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Distribution Free Tests

Solution

x	$x - 5$	Signed Rank
9.0	4.0	1
0.9	-4.1	-2
9.3	4.3	3
11.5	6.5	4
13.9	8.9	5
21.7	16.7	6

The observed value of the test statistic is $S+ = 19$.

Since the null hypothesis is of the form $\mu \leq \mu_0$, large values of $S+$ provide evidence against H_0 .

Therefore the P -value is the area in the right-hand tail of the null distribution, corresponding to values greater than or equal to 19.

Consulting Table A.5 shows that the P -value is 0.0469

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Distribution Free Tests

The nickel content for six welds was measured to be 9.3, 0.9, 9.0, 21.7, 11.5, and 13.9.
Use these data to test $H_0 : \mu = 16$ versus $H_1 : \mu \neq 16$.

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Distribution Free Tests

Solution

x	$x - 16$	Signed Rank
13.9	-2.1	-1
11.5	-4.5	-2
21.7	5.7	3
9.3	-6.7	-4
9.0	-7.0	-5
0.9	-15.1	-6

- null hypothesis is of the form $H_0 : \mu = \mu_0$, this is a two-tailed test.
- The observed value of the test statistic is $S^+ = 3$.
- Consulting Table A.5, we find that the area in the left-hand tail, corresponding to values less than or equal to 3, is 0.0781.
- The P -value is twice this amount, since it is the sum of areas in two equal tails.
- Thus the P -value is $2(0.0781) = 0.1562$.

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Distribution Free Tests

Use the data in previous Example to test $H_0 : \mu = 9$ versus $H_1 : \mu \neq 9$.

Solution

x	$x - 9$	Signed Rank
9.0	0.0	—
9.3	0.3	1
11.5	2.5	2
13.9	4.9	3
0.9	-8.1	-4
21.7	12.7	5

The value of the test statistic is $S^+ = 11$.

The sample size for the purposes of the test is 5, since the value 9.0 is not ranked.

Entering Table A.5 with sample size 5, we find that if $S^+ = 12$, the P -value would be $2(0.1562) = 0.3124$.

We conclude that for $S^+ = 11$, $P > 0.3124$.

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Distribution Free Tests

Large-Sample Approximation

When the sample size n is large, the test statistic S_+ is approximately normally distributed. A rule of thumb is that the normal approximation is good if $n > 20$. It can be shown by advanced methods that under H_0 , S_+ has

$$\text{mean} = n(n + 1)/4$$

$$\text{and variance} = n(n + 1)(2n + 1)/24$$

The Wilcoxon signed-rank test is performed by computing the z-score of S_+ , and then using the normal table to find the P-value.

The z-score is

$$z = \frac{S_+ - n(n + 1)/4}{\sqrt{n(n + 1)(2n + 1)/24}}$$

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Distribution Free Tests

The article “Exact Evaluation of Batch-Ordering Inventory Policies in Two-Echelon Supply Chains with Periodic Review” (G. Chacon, Operations Research, 2001: 79–98) presents an evaluation of a reorder point policy, which is a rule for determining when to restock an inventory. Costs for 32 scenarios are estimated. Let μ represent the mean cost. Test $H_0 : \mu \geq 70$ versus $H_1 : \mu < 70$. The data is presented in Table

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Distribution Free Tests

X	X	X
79.26	22.39	10.08
80.79	118.39	7.28
82.07	118.46	6.87
82.14	20.32	6.23
57.19	16.69	4.57
55.86	16.50	4.09
42.08	15.95	140.09
41.78	15.16	140.77
100.01	14.22	
100.36	11.64	
30.46	11.48	
30.27	11.28	

Given data

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Distribution Free Tests

x	$x - 70$	Signed Rank	x	$x - 70$	Signed Rank	x	$x - 70$	Signed Rank
79.26	9.26	1	30.27	-39.73	-12	11.48	-58.52	-23
80.79	10.79	2	22.39	-47.61	-13	11.28	-58.72	-24
82.07	12.07	3	118.39	48.39	14	10.08	-59.92	-25
82.14	12.14	4	118.46	48.46	15	7.28	-62.72	-26
57.19	-12.81	-5	20.32	-49.68	-16	6.87	-63.13	-27
55.86	-14.14	-6	16.69	-53.31	-17	6.23	-63.77	-28
42.08	-27.92	-7	16.50	-53.50	-18	4.57	-65.43	-29
41.78	-28.22	-8	15.95	-54.05	-19	4.09	-65.91	-30
100.01	30.01	9	15.16	-54.84	-20	140.09	70.09	31
100.36	30.36	10	14.22	-55.78	-21	140.77	70.77	32
30.46	-39.54	-11	11.64	-58.36	-22			

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Distribution Free Tests

Solution

The sample size is $n = 32$, so the mean is $n(n + 1)/4 = 264$ and the variance is $n(n + 1)(2n + 1)/24 = 2860$. The sum of the positive ranks is $S_+ = 121$. We compute

$$z = \frac{121 - 264}{\sqrt{2860}} = -2.67$$

Since the null hypothesis is of the form $H_0: \mu \geq \mu_0$, small values of S_+ provide evidence against H_0 . Thus the P -value is the area under the normal curve to the left of $z = -2.67$. This area, and thus the P -value, is 0.0038.

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MCQS

Which of the following is a key assumption of the Wilcoxon Signed-Rank Test?

- a) Data must be normally distributed
- b) Data must be **paired and at least ordinal**
- c) Samples must be independent
- d) Population variance must be known

Answer b)

In the Wilcoxon Signed-Rank Test, what do you do with zero differences (ties)?

- a) Count them as positive differences
- b) Count them as negative differences
- c) Exclude them from ranking
- d) Add them to the smallest rank

b) Answer C)

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References



- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.



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HYPOTHESIS and INFERENCE
Distribution Free Tests

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Distribution Free Tests

The Wilcoxon Rank-Sum Test:

- The Wilcoxon Rank–Sum Test (also known as the Mann–Whitney U Test) is a non-parametric test.
- Compares two independent samples to check if they come from the same population or if one tends to have larger values than the other.
- Two assumptions are necessary.
- First the populations must be continuous.
- Second, their probability density functions must be identical in shape and size; the only possible difference between them being their location.

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Distribution Free Tests

The Wilcoxon Rank-Sum Test:

- Let $X_1, \dots, X_m$ be a random sample from one population and let $Y_1, \dots, Y_n$ be a random sample from the other.
- We adopt the notational convention that when the sample sizes are unequal, the smaller sample will be denoted $X_1, \dots, X_m$.
- Thus the sample sizes are m and n , with $m \leq n$.
- Denote the population means by μ_X and μ_Y , respectively.

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Distribution Free Tests

The Wilcoxon Rank-Sum Test:

- The test is performed by ordering the $m + n$ values obtained by combining the two samples, and assigning ranks $1, 2, \dots, m + n$ to them.
- The test statistic, denoted by W , is the sum of the ranks corresponding to $X_1, \dots, X_m$.

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Distribution Free Tests

The Wilcoxon Rank-Sum Test:

- Since the populations are identical with the possible exception of location, it follows that if $\mu_X < \mu_Y$, the values in the X sample will tend to be smaller than those in the Y sample.
- So the rank sum W will tend to be smaller as well.
- By similar reasoning, if $\mu_X > \mu_Y$, W will tend to be larger.

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Distribution Free Tests

The Wilcoxon Rank-Sum Test:

Example:

- Resistances, in m, are measured for five wires of one type and six wires of another type. The results are as follows:

X : 36 28 29 20 38

Y : 34 41 35 47 49 46

- Use the Wilcoxon rank-sum test to test $H_0: \mu_X \geq \mu_Y$ versus $H_1: \mu_X < \mu_Y$.

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Distribution Free Tests

The Wilcoxon Rank-Sum Test:

Solution:

We order the 11 values and assign the ranks.

	Rank	Value
20	1	X
28	2	X
29	3	X
34	4	Y
35	5	Y
36	6	X

	Rank	Value
38	7	X
41	8	Y
46	9	Y
47	10	Y
49	11	Y

⇒ W

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Distribution Free Tests

The Wilcoxon Rank-Sum Test: Solution:

$$W = 1 + 2 + 3 + 6 + 7 = 19.$$

- To determine the P-value, we consult Table A.6 (in Appendix A).
- We note that small values of W provide evidence against $H_0: \mu_X \geq \mu_Y$, so the P value is the area in the left-hand tail of the null distribution. Entering the table with $m = 5$ and $n = 6$ we find that the area to the left of $W = 19$ is 0.0260. This is the P -value

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Distribution Free Tests

Large-Sample Approximation:

- When both sample sizes m and n are greater than 8, it can be shown by advanced methods that the null distribution of the test statistic W is approximately normal with mean $m(m + n + 1)/2$ and variance $mn(m + n + 1)/12$.

- z - score is

$$z = \frac{W - m(m + n + 1)/2}{\sqrt{mn(m + n + 1)/12}}$$

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Distribution Free Tests

The article “Cost Analysis Between SABER and Design Bid Build Contracting Methods” (E. Henry and H. Brothers, Journal of Construction Engineering and Management, 2001:359–366) presents data on construction costs for 10 jobs bid by the traditional method (denoted X) and 19 jobs bid by an experimental system (denoted Y). The data, in units of dollars per square meter, and their ranks, are presented in . Test $H_0 : \mu_X \leq \mu_Y$ versus $H_1 : \mu_X > \mu_Y$.

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Distribution Free Tests

TABLE 6.2 Data for Example 6.20

Value	Rank	Sample	Value	Rank	Sample
57	1	X	613	16	X
95	2	Y	622	17	Y
101	3	Y	708	18	X
118	4	Y	726	19	Y
149	5	Y	843	20	Y
196	6	Y	908	21	Y
200	7	Y	926	22	X
233	8	Y	943	23	Y
243	9	Y	1048	24	Y
341	10	Y	1165	25	X
419	11	Y	1293	26	X
457	12	X	1593	27	X
584	13	X	1952	28	X
592	14	Y	2424	29	Y
594	15	Y			

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Distribution Free Tests

Solution

The sum of the X ranks is $W = 1 + 12 + 13 + 16 + 18 + 22 + 25 + 26 + 27 + 28 = 188$.
The sample sizes are $m = 10$ and $n = 19$. We use the normal approximation and compute

$$z = \frac{188 - 10(10 + 19 + 1)/2}{\sqrt{10(19)(10 + 19 + 1)/12}}$$
$$= 1.74$$

Large values of W provide evidence against the null hypothesis. Therefore the P -value is the area under the normal curve to the right of $z = 1.74$. From the z table we find that the P -value is 0.0409.

The Wilcoxon Rank-Sum Test is used to:

- a) Compare means of two paired samples
- b) Compare medians of two independent samples when data are not normally distributed.
- c) Compare variances of two samples
- d) Test correlation between two variables

Answer b)

How are ranks assigned in the Wilcoxon Rank-Sum Test?

- a) Rank values within each group separately
- b) Rank values after combining both groups
- c) Only rank the larger group
- d) Only rank the smaller group

Answer b)



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HYPOTHESIS and INFERENCE

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UNIT-3 HYPOTHESIS and INFERENCE Chi-squared Test

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- Chi square test is used to **measure independence of two categorical variables**. i.e if one variable has any affect on another
- Ex: Does the gender of the person determine which chocolate they like?
- It is also **used to measure goodness of fit**. i.e if the **observed and expected values match**.
- For example, your model expected women winning lottery is 0.05 more than male, but is that really the case. Does the real data really match up with your prediction?
- Chi square is used when data is categorical (i.e can be classified into groups (yes/no)(red/blue/green))

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Chi-squared Test



Tender for collecting toll for a newly opened bridge

Day	Mon day	Tues day	Wedn esday	Thur sday	Frid ay	Saturd ay	Sun day
No	50	20	90	130	200	170	220

Day	Mon day	Tues day	Wedn esday	Thur sday	Frid ay	Saturd ay	Sun day
No	50	50	100	130	200	150	200

You want to know if the day of week affects

The number of people sending tenders for the bridge?

You accordingly set up your H_0 and H_1

Ex: H_0 probability of all day same

If H_0 rejected then the day affects number of tenders else not

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Chi-squared Test



- A generalization of the Bernoulli trial is the multinomial trial
- Which is an experiment that can result in any one of k outcomes, where $k \geq 2$.
- The probabilities of the k outcomes are denoted $p_1, \dots, p_k$.

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Chi-squared Test for best fit of model



- The null hypothesis has the form

$H_0 : p_1 = p_{01}, p_2 = p_{02}, \dots, p_k = p_{0k}$. (these are your expected probabilities)

For example:

A gambler wants to test a die to see whether it deviates from fairness.

Let p_i be the probability that the number i comes up. The null hypothesis will state that the die is fair.

The null hypothesis is $H_0 : p_1 = p_{01}, p_2 = p_{02}, \dots, p_6 = p_{06} = 1/6$.

- The gambler rolls the die 600 times and The results obtained are called the observed values.
- To test the null hypothesis, we construct a second column, labeled “Expected.” This column contains the expected values.
- The expected value for a given outcome is the mean number of trials that would result in that outcome if H_0 were true.

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Chi-squared Test

- The idea behind the hypothesis test is that if H_0 is true, then the observed and expected values are likely to be close to each other.
- If H_0 is false that implies that there is some hidden factor affecting the outcome (i.e the dice is not fair)
- Therefore we will construct a test statistic that measures the closeness of the observed to the expected values.

- The statistic is called the chi-square statistic. To define it, let k be the number of outcomes ($k = 6$ in the die example),
- Let O_i and E_i be the observed and expected numbers of trials, respectively, that result in outcome i .
- The chi-square statistic is

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

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Chi-squared Test

- When the expected values are all sufficiently large, a good approximation is available.
- It is called the chi-square distribution with $k - 1$ degrees of freedom, denoted χ^2_{k-1}
- A table for the chi-square distribution is available
- We look up the p-val associated for $k-1$ degree of freedom and given significance level (alpha) [this is critical value]
- If the calculated χ^2 is greater than critical value $\rightarrow$ reject null hypothesis
- Else $\rightarrow$ fail to reject null hypothesis

OR

You can look up p-val associated with chi-square closed to your calculated chi-square and if it is less than significance level $\rightarrow$ reject null hypothesis

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Chi-squared Test

Example:

Consider a die is thrown 600 times with following results.

Number turned up	1	2	3	4	5	6
Frequency	115	97	91	101	110	86

Is the die unbiased at 10% significance level?

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Chi-squared Test

Example:

Expected value=600/6=100 (unbiased die means equal outcome of all)

Hypothesis

H0: Die is unbiased

H1: Die is biased

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Catogory	Observed	Expected
1	115	100
2	97	100
3	91	100
4	101	100
5	110	100
6	86	100
Tot	600	600

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Chi-squared Test

$$\begin{aligned}\chi^2 &= \frac{(115 - 100)^2}{100} + \dots + \frac{(86 - 100)^2}{100} \\ &= 2.25 + \dots + 1.96 \\ &= 6.12\end{aligned}$$

Degrees of freedom = $n - 1 = 6 - 1 = 5$

χ^2 at 10% for df 5 is 9.236

Since $6.12 < 9.236$

We fail to reject H_0 , Conclude that die is unbiased

	P					
DF	0.995	0.975	0.2	0.1	0.05	0.025
1	.0004	.00016	1.642	2.706	3.841	5.024
2	0.01	0.0506	3.219	4.605	5.991	7.378
3	0.0717	0.216	4.642	6.251	7.815	9.348
4	0.207	0.484	5.989	7.779	9.488	11.143
5	0.412	0.831	7.289	9.236	11.07	12.833
6	0.676	1.237	8.558	10.645	12.592	14.449
7	0.989	1.69	9.803	12.017	14.067	16.013
8	1.344	2.18	11.03	13.362	15.507	17.535
9	1.735	2.7	12.242	14.684	16.919	19.023
10	2.156	3.247	13.442	15.987	18.307	20.483

Chi-squared Test

The Chi-Square test is used to:

- a) Compare means of two groups
- b) Compare medians of two groups
- c) Test the association between categorical variables or goodness-of-fit
- d) Test correlation between two quantitative variables

Answer b)

In a Chi-Square test, what happens to the test statistic if the difference between observed and expected frequencies increases?

- a) It decreases
- b) Remains the same
- c) It increases
- d) Becomes negative

Answer C)



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HYPOTHESIS and INFERENCE

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UNIT-3 HYPOTHESIS and INFERENCE Chi-squared Test

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The Chi-Square Test for Independence:

- In some cases, both row and column totals are random. In either case, we can test the null hypothesis that the probabilities of the column outcomes are the same for each row outcome, and the test is exactly the same in both cases.

📊 3. The data structure

You summarize the observed data in a **contingency table** ($r \times c$ table):

Beverage	Youth	Adult	Senior	Row Total
Tea	30	25	20	75
Coffee	40	35	30	105
Juice	30	40	50	120
Column Total	100	100	100	300

📊 4. Compute the expected counts

If the distributions are *homogeneous*, then the expected count in each cell is:

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

Example:

Expected number of *Youth who prefer Tea*:

$$E = \frac{(75)(100)}{300} = 25$$

Do this for each cell.

⚙️ 5. Compute the test statistic

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Where:

- O_{ij} = observed frequency in row i , column j
- E_{ij} = expected frequency under H_0



6. Degrees of freedom

$$df = (r - 1)(c - 1)$$

where r = number of rows (categories),
and c = number of columns (populations/groups).

Example: $df = (3 - 1)(3 - 1) = 4$

7. Find the p-value

Once you compute your χ^2 statistic, use the **Chi-square distribution** with the appropriate df to find the **p-value**:

$$p = P(\chi^2 > \chi_{\text{obs}}^2)$$

If $p < \alpha$ (commonly 0.05), reject H_0 :

→ conclude that the distributions are **not homogeneous**.

8. Summary of Steps

Step	Description	Formula / Notes
1	Set up hypotheses	H_0 : distributions are same
2	Make contingency table	observed data
3	Compute expected counts	$E_{ij} = \frac{(\text{row total})(\text{col total})}{\text{grand total}}$
4	Compute test statistic	$\chi^2 = \sum \frac{(O-E)^2}{E}$
5	Compute degrees of freedom	$df = (r - 1)(c - 1)$
6	Get p-value	from χ^2 table or software
7	Draw conclusion	Compare p with α



8. Summary of Steps

Step	Description	Formula / Notes
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2	Make contingency table	observed data
3	Compute expected counts	$E_{ij} = \frac{(\text{row total})(\text{col total})}{\text{grand total}}$
4	Compute test statistic	$\chi^2 = \sum \frac{(O-E)^2}{E}$
5	Compute degrees of freedom	$df = (r - 1)(c - 1)$
6	Get p-value	from χ^2 table or software
7	Draw conclusion	Compare p with α

The Chi-Square Test for Homogeneity:

- Sometimes several multinomial trials are conducted, each with the same set of possible outcomes.
- The null hypothesis is that the probabilities of the outcomes are the same for each experiment.
- We calculate the expected values for each cell using the formula
- $$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand total}}$$
- Then calculate overall chi-square (applying formula over all cells observed and expected)

- Then calculate overall chi-square (applying formula over all cells observed and expected)

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

- Then calculate degree of freedom using formula

$$df = (r-1) \times (c-1)$$

where r is the number of rows and c is the number of columns.

- Then look up critical χ^2 val and compare with calculated to determine whether to reject null hypothesis or not

The Chi-Square Test for Homogeneity:

Example:

- Use the following data to test the null hypothesis that the proportions of pins that are too thin, OK, or too thick are the same for all the machines.

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Chi-squared Test

The Chi-Square Test for Homogeneity:

Example: specification. A pin may meet the specification, or it may be too thin or too thick. Pins are sampled from each machine, and the number of pins in each category is counted. Table below presents the results. Use the data in Table to test the null hypothesis that the proportions of pins that are too thin, OK, or too thick are the same for all the machines.

	Too thin	OK	Too thick
Machine 1	10	102	8
Machine 2	34	161	5
Machine 3	12	79	9
Machine 4	10	60	10

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Chi-squared Test

Null Hypothesis (H_0): **The proportions of pins that are too thin, OK, or too thick are the same across all machines.**

Alternative Hypothesis (H_1): **The proportions of pins that are too thin, OK, or too thick differ for at least one machine.**

Expected Value Formula

For each cell in the table, the **expected value** is calculated as:

$$E = (\text{Row Total}) \times (\text{Column Total}) / \text{Grand total}$$

Note: calculate the expected values for each cell. Then apply in chi square formula

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

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Chi-squared Test

Solution:

Expected values for Table

	Too Thin	OK	Too Thick	Total
Machine 1	15.84	96.48	7.68	120.00
Machine 2	26.40	160.80	12.80	200.00
Machine 3	13.20	80.40	6.40	100.00
Machine 4	10.56	64.32	5.12	80.00
Total	66.00	402.00	32.00	500.00

Observed values

	Too thin	OK	Too thick
Machine 1	10	102	8
Machine 2	34	161	5
Machine 3	12	79	9
Machine 4	10	60	10

$$\begin{aligned}\chi^2 &= \frac{(10 - 15.84)^2}{15.84} + \dots + \frac{(10 - 5.12)^2}{5.12} \\ &= \frac{34.1056}{15.84} + \dots + \frac{23.8144}{5.12} \\ \chi^2 &= 15.5844\end{aligned}$$

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Chi-squared Test

Solution:

- Since there are four rows and three columns, the number of degrees of freedom is $(4 - 1)(3 - 1) = 6$.
- To obtain the P -value, we consult the chi-square table. Looking under six degrees of freedom, we find that the upper 2.5% point is 14.449, and the upper 1% point is 16.812.
- Therefore $0.01 < P < 0.025$. We reject H_0 (since no matter what the p -val is less than 0.05) (default alpha is 5%).
- therefore It is reasonable to conclude that the machines differ in the proportions of pins that are too thin, OK, or too thick.

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Chi-squared Test

The Chi-Square Test for Independence:

Example:

The cylindrical steel pins in Example 6.21 are subject to a length specification as well as a diameter specification. With respect to the length, a pin may meet the specification, or it may be too short or too long. A total of 1021 pins are sampled and categorized with respect to both length and diameter specification. The results are presented in the following table. Test the null hypothesis that the proportions of pins that are too thin, OK, or too thick with respect to the diameter specification do not depend on the classification with respect to the length specification.

Observed Values for 1021 Steel Pins

Length	Diameter			Total
	Too Thin	OK	Too Thick	
Too Short	13	117	4	134
OK	62	664	80	806
Too Long	5	68	8	81
Total	80	849	92	1021

Example:

Formulate the Hypotheses:

- **Null Hypothesis (H_0):** The proportions of pins that are too thin, OK, or too thick with respect to the diameter specification do not depend on the length specification.
- **Alternative Hypothesis (H_1):** The proportions of pins that are too thin, OK, or too thick with respect to the diameter specification depend on the length specification.

Calculate the expected values

The expected frequency E_{ij} for each cell under the assumption of independence is calculated using:

$$E_{ij} = \frac{(\text{Row Total}_i \times \text{Col Total}_j)}{\text{Grand Total}}$$

For each combination of row (length category) and column (diameter category), calculate the expected frequencies.

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Chi-squared Test

Expected Values for 1021 Steel Pins

Length	Diameter			Total
	Too Thin	OK	Too Thick	
Too Short	10.50	111.43	12.07	134.0
OK	63.15	670.22	72.63	806.0
Too Long	6.35	67.36	7.30	81.0
Total	80.0	849.0	92.0	1021.0

Observed Values for 1021 Steel Pins

Length	Diameter			Total
	Too Thin	OK	Too Thick	
Too Short	13	117	4	134
OK	62	664	80	806
Too Long	5	68	8	81
Total	80	849	92	1021

$$\begin{aligned}\chi^2 &= \frac{(13 - 10.50)^2}{10.50} + \dots + \frac{(8 - 7.30)^2}{7.30} \\ &= \frac{6.25}{10.50} + \dots + \frac{0.49}{7.30} = 7.46\end{aligned}$$

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Chi-squared Test

- Since there are three rows and three columns, the number of degrees of freedom is $(3 - 1)(3 - 1) = 4$.
- To obtain the P -value, we consult the chi-square table Looking under four degrees of freedom, we find that the upper 10% point is 7.779. We conclude that $P > 0.10$.
- We reject null hypothesis
- There is no evidence that the length and thickness are related.

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Chi-squared Test-Practice Problems (independence test)

- At an assembly plant for light trucks, routine monitoring of the quality of welds yields the following data:

Can you conclude that the quality varies among shifts?

- State the appropriate null hypothesis.
- Compute the expected values under the null hypothesis.
- Compute the value of the chi-square statistic.
- Find the P-value. What do you conclude?

	Number of Welds		
	High Quality	Moderate Quality	Low Quality
Day Shift	467	191	42
Evening Shift	445	171	34
Night Shift	254	129	17

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Chi-squared Test-Practice Problems (homogeneity test)

- A survey of adults with diabetes. Each respondent was categorized by gender and income level.

Can you conclude that the proportions in the various income categories differ between men and women?

	Poor	Near Poor	Low Income	Middle Income	High Income
Men	156	77	253	513	604
Women	348	152	433	592	511

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Mcqs

The Chi-Square Test of Independence is used to:

- a) Compare means of two groups
- b) Test whether two categorical variables are associated
- c) Compare medians of two groups
- d) Test whether variances are equal

Answer b)

Q2: The Chi-Square Test of Homogeneity is used to:

- a) Compare means of independent samples
- b) Test whether the distribution of a categorical variable is the same across different populations
- c) Test relation ship between two quantitative variables
- d) Compare paired observations

Answer b)



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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

HYPOTHESIS and INFERENCE

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Fixed Level Testing
Type I and Type II Errors

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Fixed Level Testing



- A hypothesis test measures the plausibility of the null hypothesis by producing a P -value.
- The smaller the P –value, the less plausible the null. We have pointed out that there is no scientifically valid dividing line between plausibility and implausibility, Sometimes, however, a decision has to be made.
- Fixed-level testing is just like the hypothesis testing we have been discussing so far, except that a firm rule is set ahead of time for rejecting the null hypothesis.

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Fixed Level Testing



To conduct a fixed-level test:

- Choose a number α , where $0 < \alpha < 1$. This is called the significance level, or the level, of the test.
- Compute the P -value in the usual way.
- If $P \leq \alpha$, reject H_0 . If $P > \alpha$, do not reject H_0 .

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Fixed Level Testing



Why we need fixed-level hypothesis testing

- **Controls the probability of a Type I error**

A **Type I error** occurs when we *wrongly reject* a true null hypothesis.

By fixing a level (say, $\alpha = 0.05$), we cap the risk of making this false rejection

This makes hypothesis testing reliable and fair — everyone knows what risk of false alarm we are willing to tolerate.

- **Ensures consistent decision-making**

Without a fixed α , two people analyzing the same data could draw different conclusions.

Setting α (e.g., 0.01, 0.05, or 0.10) provides a standard threshold for decision-making:

- If $p \leq \alpha \rightarrow \text{reject } H_0$
- If $p > \alpha \rightarrow \text{fail to reject } H_0$

This uniform rule removes subjectivity.

Balances evidence and risk

- A smaller α (like 0.01) demands stronger evidence to reject H_0 , reducing false positives but increasing false negatives (Type II errors).
- A larger α (like 0.10) allows more flexibility but increases the risk of false rejection. Thus, α defines how cautious or liberal we are in making decisions.

Supports comparison across studies

When multiple studies use the same significance level (usually $\alpha = 0.05$), their results can be compared meaningfully — which is essential in research, meta-analysis, and policy-making.

Balances evidence and risk

- A smaller α (like 0.01) demands stronger evidence to reject H_0 , reducing false positives but increasing false negatives (Type II errors).
- A larger α (like 0.10) allows more flexibility but increases the risk of false rejection. Thus, α defines how cautious or liberal we are in making decisions.

Supports comparison across studies

When multiple studies use the same significance level (usually $\alpha = 0.05$), their results can be compared meaningfully — which is essential in research, meta-analysis, and policy-making.

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Fixed Level Testing



Type I error (False Positive): Healthy person diagnosed as sick (false alarm).

The test says the person **has COVID-19**, but in reality, they **don't**.

Type II error (False Negative): Sick person diagnosed as healthy (missed detection).

The test says the person **does NOT have COVID-19**, but they **actually do**.

Context	More Dangerous Error
Medical diagnosis	Type II
Legal decision	Type I
Scientific research	Type I
Quality control / Safety testing	Type II (faulty items may go unnoticed)

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Fixed Level Testing



Example:

The mean wear in a sample of 45 steel balls was $\bar{X} = 673.2\mu m$, and the standard deviation was $s = 14.9\mu m$. Let μ denote the population mean wear. A test of $H_1: \mu > 675$ versus $H_0: \mu \leq 675$ yielded a P -value of 0.209. Can we reject H_0 at the 25% level? Can we reject H_0 at the 5% level?

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Fixed Level Testing



Solution:

- The P -value of 0.209 is less than 0.25
- So if we had chosen a significance level of $\alpha = 0.25$, we would reject H_0 .
- Thus we reject H_0 at the 25% level.
- Since $0.209 > 0.05$, we do not reject H_0 at the 5% level.

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Fixed Level Testing



Example:

A process for a certain type of ore is designed to reduce the concentration of impurities to less than 2%. It is known that the standard deviation of impurities for processed ore is 0.6%. Let μ represent the mean impurity level, in percent, for ore specimens treated by this process.

- The impurity of 80 ore specimens is measured, and a test of the hypothesis $H_0: \mu \geq 2$ versus $H_1: \mu < 2$ will be performed.
 - a. If the test is made at the 5% level, what is the rejection region?
 - b. If the sample mean impurity level is 1.85, will H_0 be rejected at the 10% level?

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Fixed Level Testing



Solution 2 (a)

$H_0: \mu \geq 2$ versus $H_1: \mu < 2$

- Null distribution of $\bar{X}$:

$$\bar{X} \sim N(2, 0.6^2/80)$$

- $\alpha = 0.05$
- $Z = -1.645$ (Critical value)

$$\Rightarrow \bar{X} = \frac{Z * S}{\sqrt{n}} + 2$$

$$\Rightarrow \bar{X} = \frac{-1.645 * 0.6}{\sqrt{80}} + 2$$

$$\Rightarrow \bar{X} = 1.89$$

Hence, H_0 will be rejected if $\bar{X} \leq 1.890$. The rejection region consists of all values of $\bar{X}$ less than or equal to 1.890.

Solution Using Rejection Region Approach:

$H_0: \mu \geq 2$ versus $H_1: \mu < 2$

- Null distribution of $\bar{X}$:

$$\bar{X} \sim N(2, 0.6^2/80)$$

- $\alpha = 0.10$
- $Z = -1.28$ (Critical value)

$$\Rightarrow \bar{X} = \frac{Z * S}{\sqrt{n}} + 2$$

$$\Rightarrow \bar{X} = \frac{-1.28 * 0.6}{\sqrt{80}} + 2$$

$$\Rightarrow \bar{X} = 1.9141$$

- Since $1.85 < 1.9141$

$\Rightarrow H_0$ will be rejected at the 10% level

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Fixed Level Testing



Solution Using P Value Approach:

$H_0: \mu \geq 2$ versus $H_1: \mu < 2$

- Null distribution of $\bar{X}$:

$$\bar{X} \sim N(2, 0.6^2/80)$$

- $\alpha = 0.10$
 - Finding z-score for 1.85
 - $Z = (1.85 - 2) / (0.6/\sqrt{80}) = -2.24$
 - $\Rightarrow P = P(Z < -2.24) = 0.0125$
- $\Rightarrow P < \alpha \Rightarrow H_0$ will be rejected at the 10% level

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Fixed Level Testing

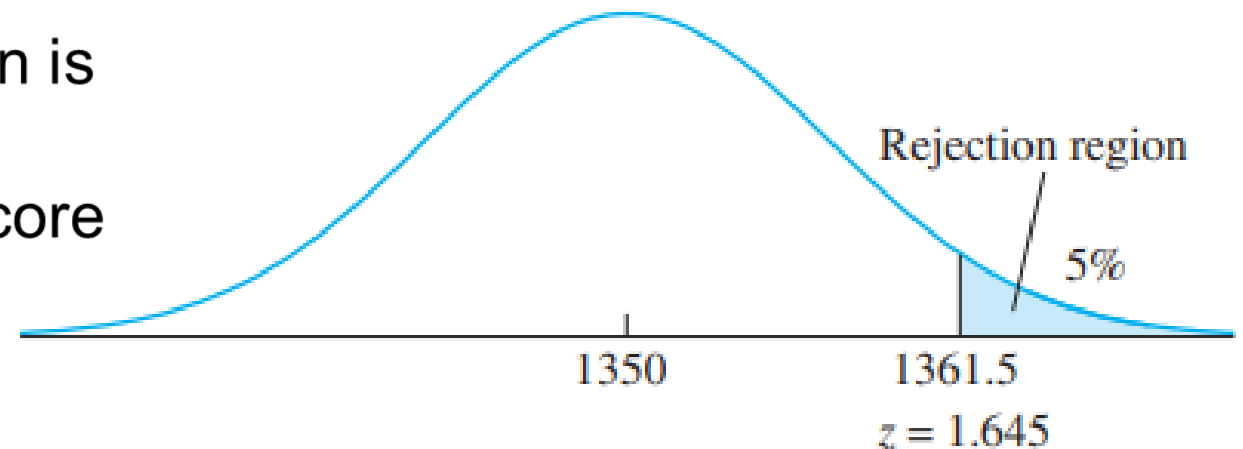


Example:

- A new concrete mix is being evaluated. 100 concrete blocks made with the new mix are sampled, the sample mean compressive strength is $\bar{X}$, it is tested on the following hypothesis
- $H_0 : \mu \leq 1350 \text{ Mpa}$
- $H_1 : \mu > 1350 \text{ Mpa}$
- If population standard deviation is 70 Mpa. Find critical point and rejection region if significance level of the test is 5%.

Solution

- $H_0 : \mu \leq 1350$ versus $H_1 : \mu > 1350$
 $\alpha = 0.05$
- we reject H_0 if p-val is less than 0.05
where p-val is area to the right of $\bar{X}$ (critical x-val).
- The critical point of area for rejection is upper 5% i.e 0.05
- We calculate the corresponding z-score from z table
- $Z_{0.05} = 1.645$ (critical value)



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Fixed Level Testing



Solution

- Null distribution :

$$\begin{aligned} & N(1350, 70^2/100) \\ \Rightarrow \bar{X} &= \frac{Z * s}{\sqrt{n}} + \mu \\ \Rightarrow \bar{X} &= \frac{1.645 * 70}{\sqrt{100}} + 1350 \\ \Rightarrow \bar{X} &= 1361.5 \end{aligned}$$

Hence, H_0 will be rejected if $\bar{X} \geq 1361.5$. The rejection region consists of all values of $\bar{X}$ greater than or equal to 1361.5.

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Fixed Level Testing



Example:

In a hypothesis test to determine whether a scale is in calibration, the null hypothesis is

$$H_0 : \mu = 1000$$

and the null distribution of $\bar{X}$ is $N(1000, 0.26^2)$.

Find the rejection region if the test will be conducted at a significance level of 5%.

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Fixed Level Testing



Solution

- $H_0 : \mu = 1000$ $\alpha = 0.05$ (since = sign $\rightarrow$ two tailed test)
- $N(1000, 0.26^2)$ $0.26^2 = \text{sample variance} = s^2/n$
 $\frac{s}{\sqrt{n}} = 0.26$
- H_0 will be rejected if $\bar{X}$ is in either the upper or the lower 2.5% of the null distribution (sum of two regions = 5%)
- Corresponding z-score for 0.025 in z table (we only calculate for one since normal distribution is symmetrical, i.e. other side z-score is same but reverse in sign)
- $Z_{0.025} = \pm 1.96$

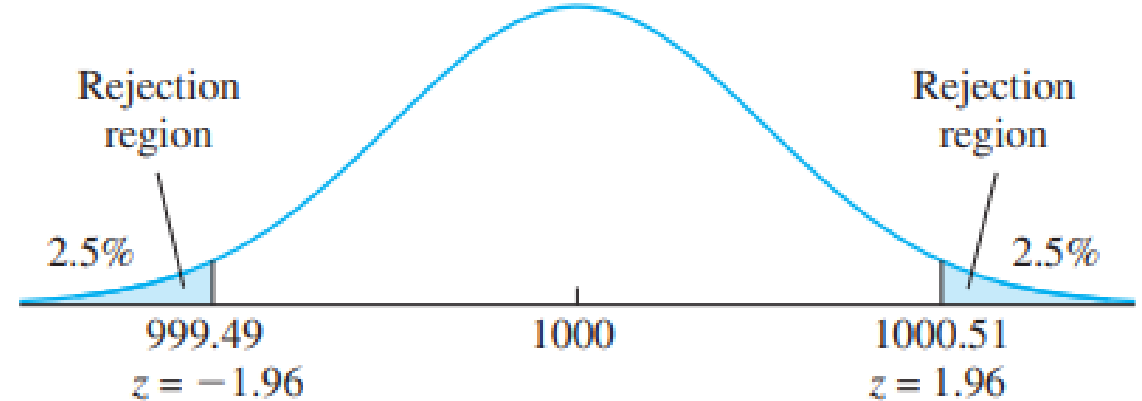
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Fixed Level Testing

Solution

$$\bar{x} = \frac{Z * S}{\sqrt{n}} + \mu$$
$$\Rightarrow X1 = 0.26 * 1.96 + 1000$$
$$\Rightarrow X1 = 1000.51$$

$$\Rightarrow X2 = 0.26 * -1.96 + 1000$$
$$\Rightarrow X2 = 999.49$$



Therefore the critical point is 999.49 and 1000.51

The rejection region consists of all values of $\bar{X}$ greater than or equal to 1000.51 or lesser than equal to 999.49

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Type I and Type II Errors



- Suppose we test $H_0: \mu = 15$
 $H_1: \mu > 15$
- And we reject H_0 at $\alpha = 0.05$

One of two things will occur:

1. The null hypothesis is false and we made the correct decision
2. The null hypothesis is true and we made a type I error

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Type I and Type II Errors



Suppose we test $H_0: \mu = 15$

$$H_1: \mu > 15$$

And we do not reject H_0 at $\alpha = 0.05$

One of two things will occur:

1. The null hypothesis is true and we made the correct decision
2. The null hypothesis is false and we made a type II error

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Type I and Type II Errors



Statistical Errors

		The truth	
		H0 True	H0 False
Your Research	Accept H0	Correct Decision $1-\alpha$	Type II Error β
	Reject H0	Type I error α	Correct Decision $1-\beta$

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Type I and Type II Errors



- A fixed-level test results in a firm decision, there is a chance that the decision could be the wrong one.
- There are exactly two ways in which the decision can be wrong.
- One can reject H_0 when it is in fact true. This is known as a type I error.
- Or, one can fail to reject H_0 when it is false. This is known as a type II error.

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Type I and Type II Errors



Consider a criminal trial:

- We test the hypothesis H_0 : The defendant did not commit the crime
 H_1 : The defendant committed the crime
- Type I error: Convicting a person who in reality did not commit the crime
- Type II error : Acquitting a person who in reality , committed a crime

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Type I and Type II Errors



- If α is the significance level that has been chosen for the test, then the probability of a type I error is never greater than α .

When conducting a fixed-level test at significance level α , there are two types of errors that can be made. These are

- Type I error: Reject H_0 when it is true/plausible.
- Type II error: Fail to reject H_0 when it is false.

The probability of a type I error is never greater than α .

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Type I and Type II Errors



Example:

- A hypothesis test is to be performed, and it is decided to reject the null hypothesis
- If $P \leq 0.10$.
- If H_0 is in fact true, what is the maximum probability that it will be rejected?

Solution:

- The maximum probability of rejecting H_0 when true is the level $\alpha = 0.10$.

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Type I and Type II Errors



Example:

A vendor claims that no more than 10% of the parts she supplies are defective. Let p denote the actual proportion of parts that are defective. A test is made of the hypotheses $H_0: p \leq 0.10$ versus $H_1: p > 0.10$. For each of the following situations, determine whether the decision was correct, a type I error occurred, or a type II error occurred.

Example:

- a. The claim is true, and H_0 is rejected.
- b. The claim is false, and H_0 is rejected.
- c. The claim is true, and H_0 is not rejected.
- d. The claim is false, and H_0 is not rejected.

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Type I and Type II Errors



Example:

- a. The claim is true, and H_0 is rejected.
- b. The claim is false, and H_0 is rejected.
- c. The claim is true, and H_0 is not rejected.
- d. The claim is false, and H_0 is not rejected.

Ans: (a) Type I error (b) Correct decision (c) Correct decision (d) Type II error

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Recall- type I error and type II error



Hypothesis testing : H_0 vs H_1

Statistical Errors

		The truth	
		H0 True	H0 False
Your Research	Accept H_0	Correct Decision $1-\alpha$	Type II Error β
	Reject H_0	Type I error α	Correct Decision $1-\beta$

Power of test

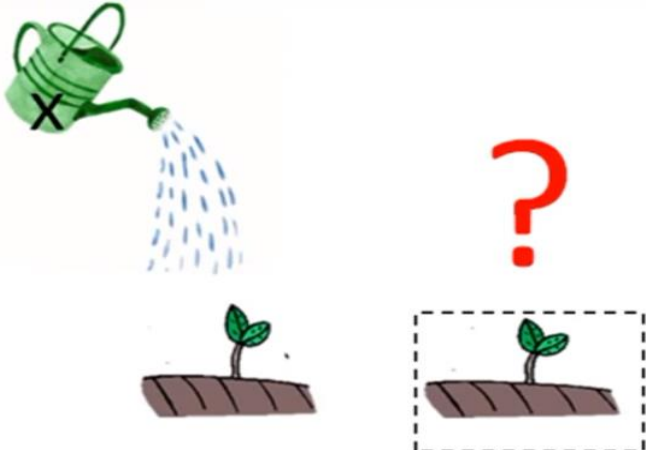
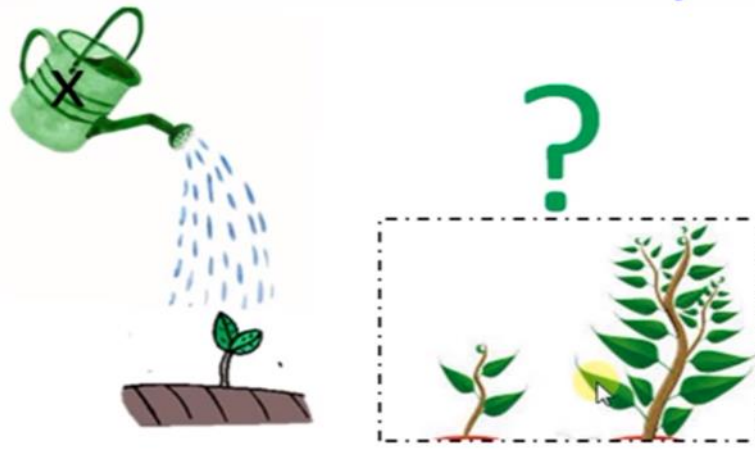
- **Power of a Hypothesis Test :**

The power of a test is the probability of rejecting H_0 when it is false.

$$\begin{aligned}\text{Power} &= 1 - P(\text{type II error}) \\ &= 1 - \beta.\end{aligned}$$

80% power means you have 80% chance of getting a significant results when the effect is real.

Effect of bio-fertilizer 'x' on plant growth

 H_0: Application of bio-fertilizer 'x' <u>do not</u> increase plant growth. Null hypothesis	 H_1: Application of bio-fertilizer 'x' increase plant growth. Alternative hypothesis
---	--

Power is 0.80(or 80%) → there is an 80% chance of rejecting the null hypothesis(false) when conducting the study.

Power of test

Why is Power Important?

- The power of a test is important because it measures the probability of correctly rejecting a false null hypothesis, meaning it's the test's ability to detect a real effect when one exists.
- A high-powered test reduces the risk of a Type II error (a false negative), preventing researchers from mistakenly concluding there is no effect when one actually exists.
- This ensures that research is more likely to find significant results that are actually there, avoiding wasted resources and the failure to identify important findings.

Power of test

How large the power must be for a test ?

- As with P-values, there is no scientifically valid dividing line between sufficient and insufficient power.
- In general, tests with power greater than 0.80 or perhaps 0.90 are considered acceptable, but there are no well-established rules of thumb.

Power of test

Analysis of power is performed:

1) Before gathering data

To determine the **minimal sample size** needed to have desired power in statistical testing (to detect a particular effect size).

2) After gathering data

To determine the **magnitude of power** that your statistical test will have given the sample parameters (**n** and **s**) and the magnitude of the effect that you want to detect.

Note: Statistical power has relevance only when the null is false.

Power of test

-

Power calculations are generally done **before data are collected**.

The purpose of a power calculation is to determine whether or not a hypothesis test, when performed, is likely to reject H_0 in the event that H_0 is false.



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POWER OF TEST

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Power of test(Computation) Factor Affecting the power of the test)

SIVASANKARI V

Department of Science & Humanities

Computing the power involves two steps:

1. Compute the rejection region.
2. Compute the probability that the test statistic falls in the rejection region if the alternate hypothesis is true.

This is the power.

Example of a power calculation

Assume that a new chemical process has been developed that may increase the yield over that of the current process. The current process is known to have a **mean yield of 80** and a **standard deviation of 5**, where the units are the percentage of a theoretical maximum. If the mean yield of the new process is shown to be greater than 80, the new process will be put into production.

Let μ denote the mean yield of the new process. It is proposed to run the new process 50 times and then to test the hypothesis

$H_0: \mu \leq 80$ versus $H_1: \mu > 80$ at a significance level of 5%.

Problem 1 :

Find the power of the 5% level test of

$H_0 : \mu \leq 80$ versus $H_1 : \mu > 80$

for the mean yield of the new process under the alternative $\mu = 81$, assuming $n = 50$ and $\sigma = 5$.

Solution:

Null distribution of $\bar{X}$:

$$\bar{X} \sim N\left(\mu, \sigma_{\bar{X}}^2\right) \text{ where } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Power of test

Null distribution of $\bar{X}$:

$$\bar{X} \sim N(80, 0.707^2)$$

The **critical point** has a z-score of 1.645, so its value is $\bar{X} = 80 + (1.645)(0.707) = 81.16$.

The rejection region consists of all values of $\bar{X} \geq 81.16$



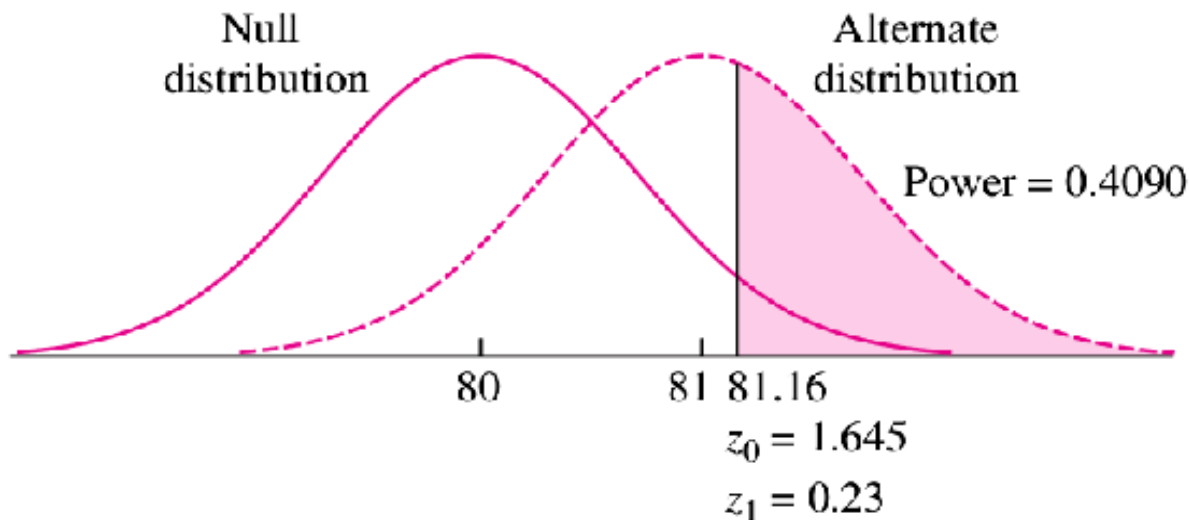
Power of test

Alternate distribution of $\bar{X}$:

$$\bar{X} \sim N(81, 0.707^2)$$

(The alternate distribution is obtained by shifting the null distribution to chosen value of μ .)

Power is the probability that $\bar{X}$ will fall into the rejection region if the alternate hypothesis $\mu = 81$ is true.



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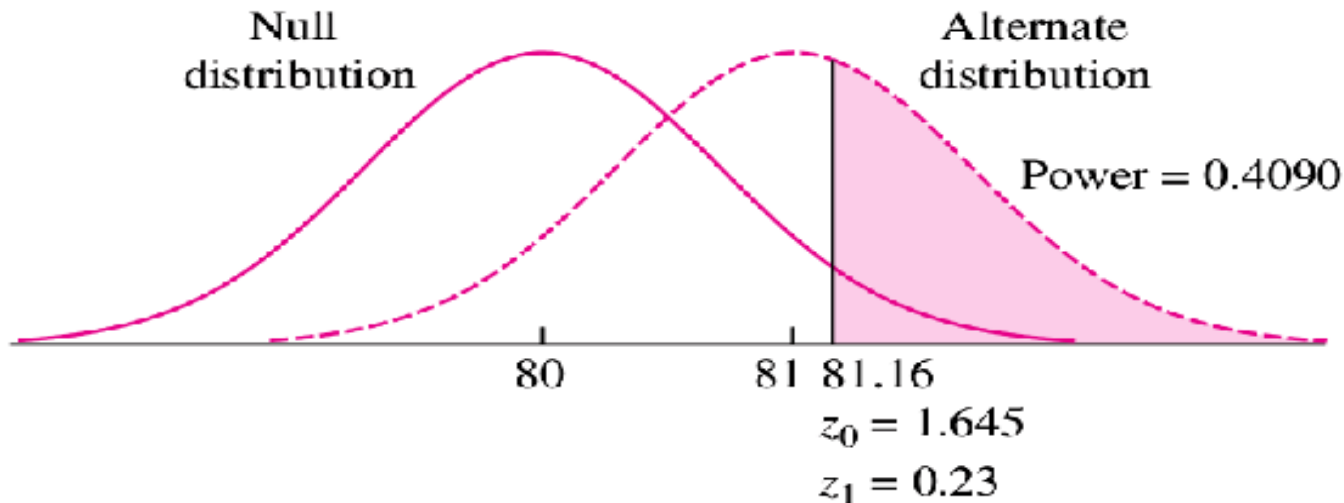
Computing the power

z -Score under H_1 for the critical point 81.16 is

$$z = \frac{\bar{X} - \mu}{\sigma} = \frac{81.16 - 81}{0.707} = 0.23$$

The area to the right of $z = 0.23$ is **0.4090**.

This is the **power** of the test.



Power of test

Conclusion:

A power of **0.4090** is very low.

- It means that if the mean yield of new process is actually equal to 81, there is only a 41% chance that the proposed experiment will detect the improvement over the old process and allow the new process to be put into production.
- It would be unwise to invest time and money to run this experiment, since it has a large chance to fail.

Calculation of Power

Problem 2 :

Find the power of the 5% level test of

$H_0 : \mu \leq 80$ versus $H_1 : \mu > 80$

for the mean yield of the new process under the alternative $\mu = 82$, assuming $n = 50$ and $\sigma = 5$.

Solution:

Null distribution of $\bar{X}$:

$$\bar{X} \sim N\left(\mu, \sigma_{\bar{X}}^2\right) \text{ where } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

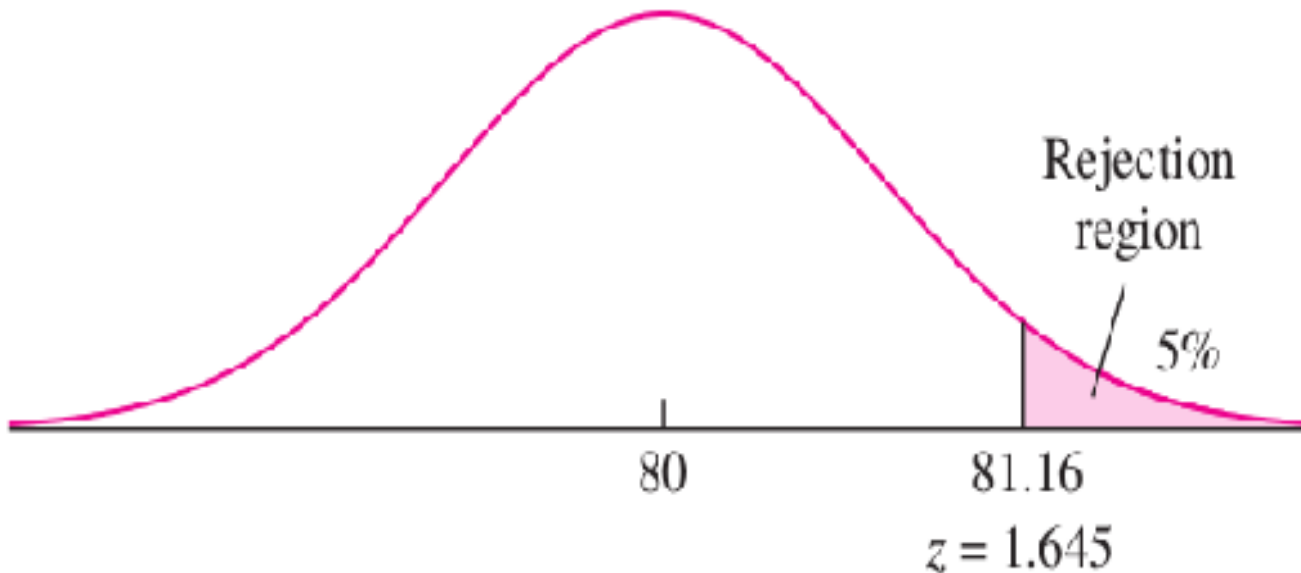
Power of test

Null distribution of $\bar{X}$:

$$\bar{X} \sim N(80, 0.707^2)$$

The **critical point** has a z-score of 1.645, so its value is $\bar{X} = 80 + (1.645)(0.707) = 81.16$.

The rejection region consists of all values of $\bar{X} \geq 81.16$



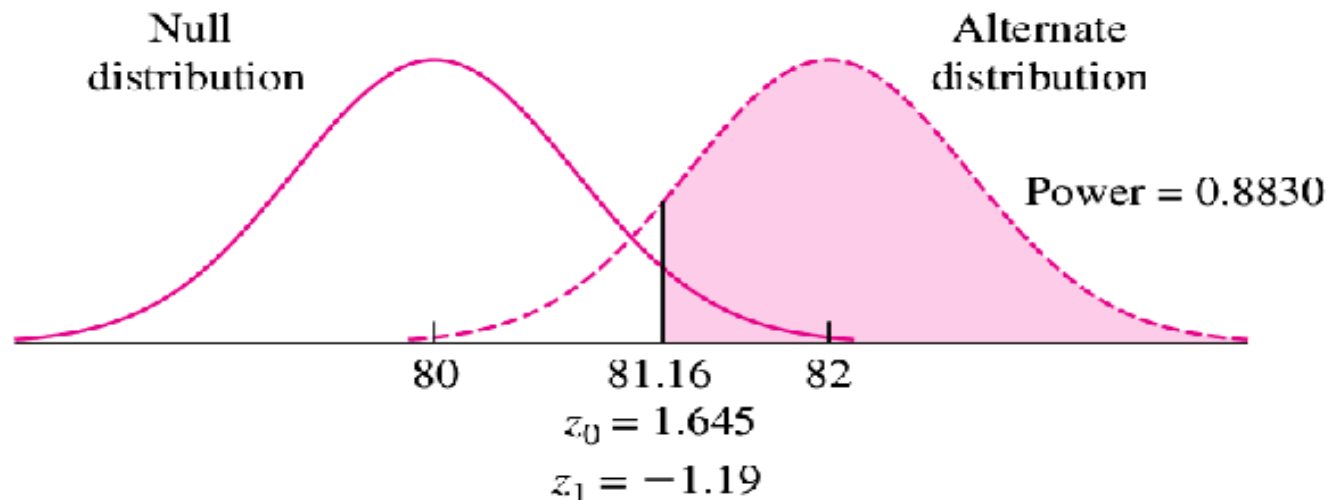
Power of test

Alternate distribution of $\bar{X}$:

$$\bar{X} \sim N(82, 0.707^2)$$

(The alternate distribution is obtained by shifting the null distribution to chosen value of μ .)

Power is the probability that $\bar{X}$ will fall into the rejection region if the alternate hypothesis $\mu = 82$ is true.



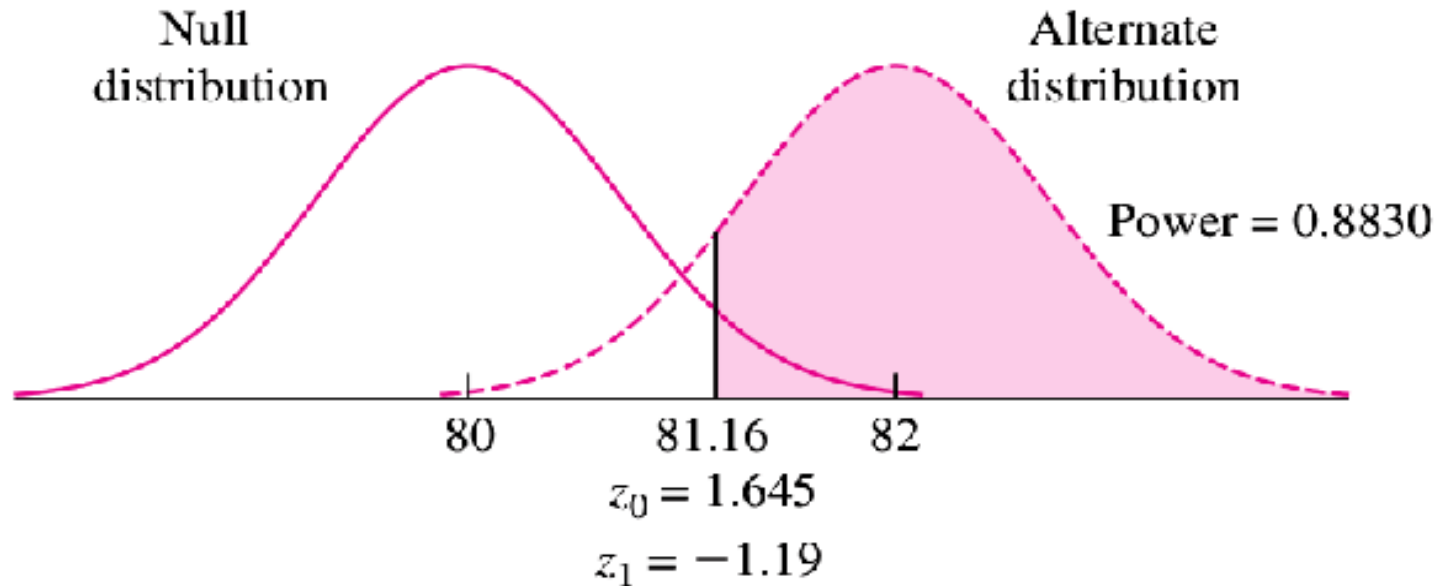
Computing the power

z -Score under H_1 for the critical point 81.16 is

$$z = \frac{\bar{X} - \mu}{\sigma} = \frac{81.16 - 82}{0.707} = -1.19$$

The area to the right of $z = -1.19$ is **0.8830**.

This is the **power** of the test.



Power of test

Conclusion

A power of **0.8830** is high.

- It means that if the mean yield of new process is actually equal to 82, there is a 88.30% chance that the proposed experiment will detect the improvement over the old process and allow the new process to be put into production.
- It would be a wise decision to invest time and money to run this experiment, since it has a large chance to succeed.

In order to compute the power, it is necessary to specify a particular value of μ , because:

power is different for different values of μ

□ if μ is close to H_0 : the power will be small.

□ if μ is far from H_0 : the power will be large.

- When power is not large enough, it can be increased by increasing the sample size.
- When planning an experiment, one can determine the sample size necessary to achieve a desired power.
- Knowing the significance level and the required power allows a researcher to determine a minimum sample size needed for the study.

Power of test

Problem 3 :

In testing the hypothesis $H_0 : \mu \leq 80$ versus $H_1 : \mu > 80$

regarding the mean yield of the new process, how many times must the new process be run so that a test conducted at a significance level of 5% will have power 0.90 against the alternative $\mu = 81$, if it is assumed that $\sigma = 5$?

Solution:

Let n represent the necessary sample size.

Null distribution of $\bar{X}$:

$$\bar{X} \sim N\left(\mu, \sigma_{\bar{X}}^2\right) \text{ where } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Critical point : $80 + 1.645 \left(\frac{5}{\sqrt{n}}\right)$

Consider the **alternate distribution of $\bar{X}$** .

Given Power is 0.90. The power of the test is the area of the rejection region under the alternate curve. This area must be 0.90.

Therefore, Z-score is -1.28.

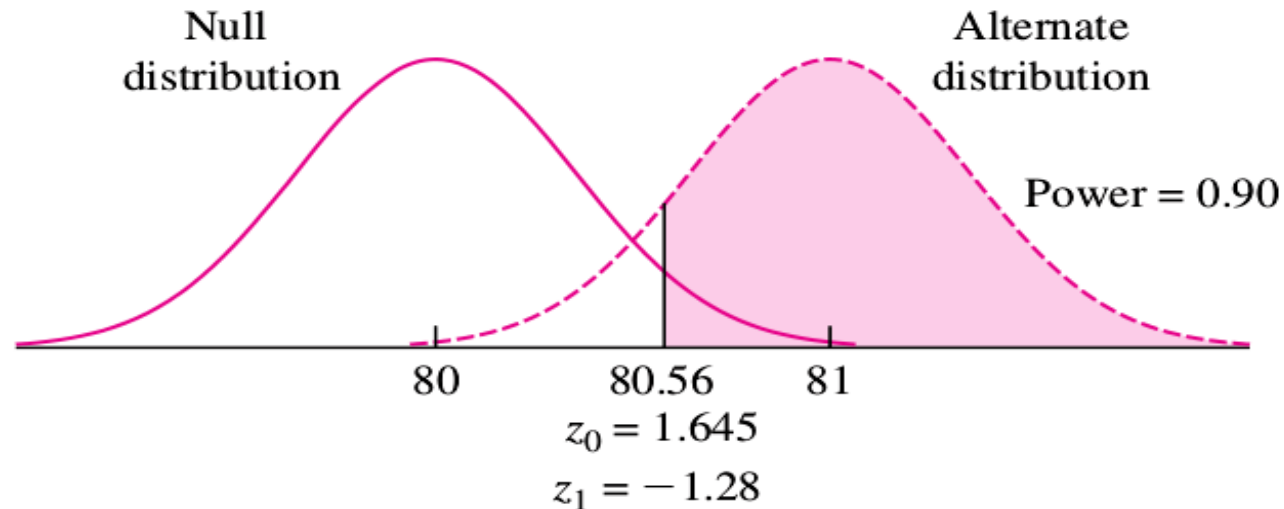
Critical point : $81 - 1.28 \left(\frac{5}{\sqrt{n}}\right)$

We now have two different expression for the critical point. Since there is only one critical point, these two expressions are equal.

Power of test

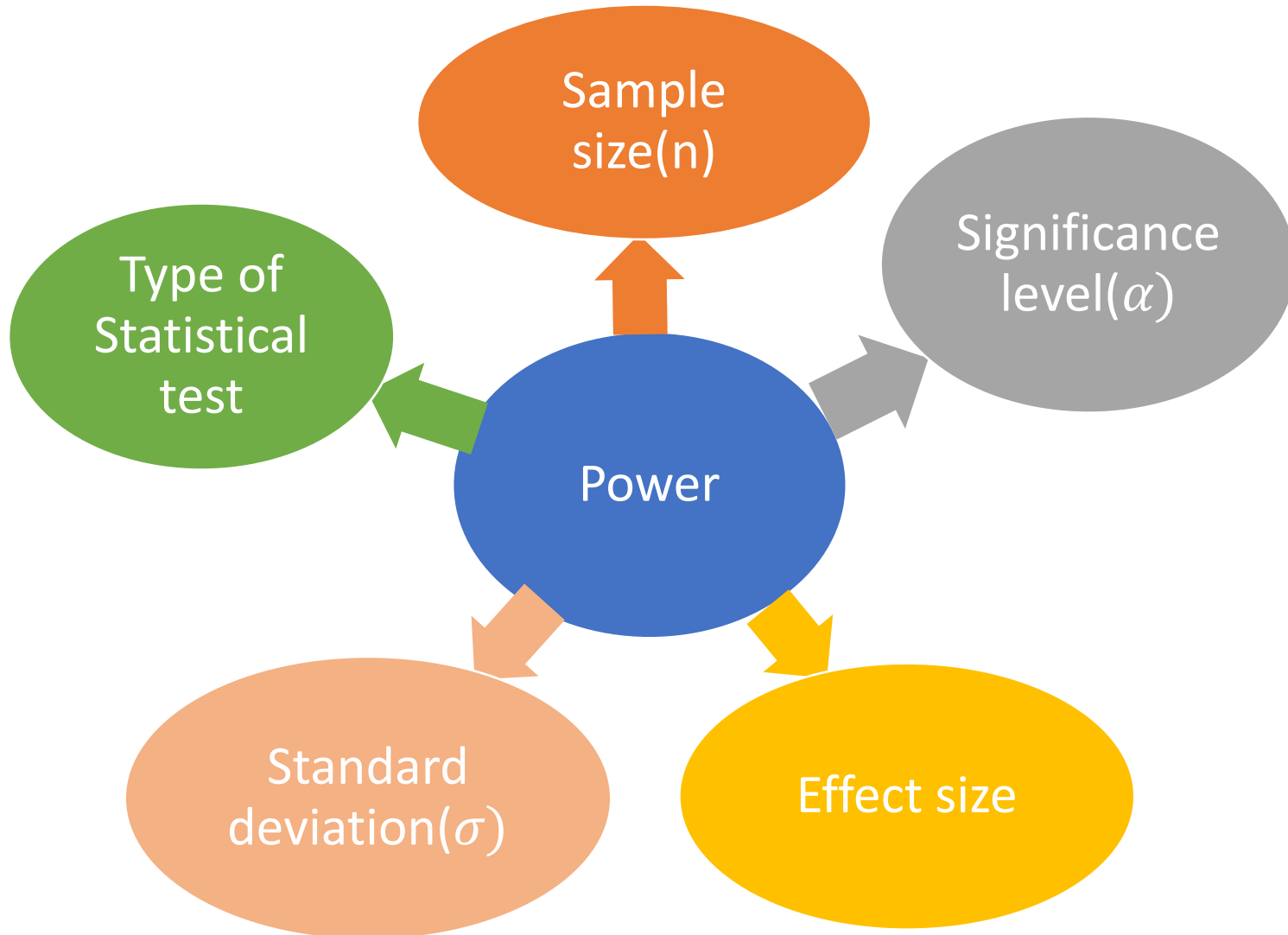
Set them equal and solve for n

$$80 + 1.645 \left(\frac{5}{\sqrt{n}} \right) = 81 - 1.28 \left(\frac{5}{\sqrt{n}} \right)$$
$$\rightarrow n \approx 214.$$



The critical point is 80.56 (The critical point can be computed by substituting this value for n into either side of the equation).

Factors affecting Statistical Power of test



Factors affecting Statistical Power of test – Sample size

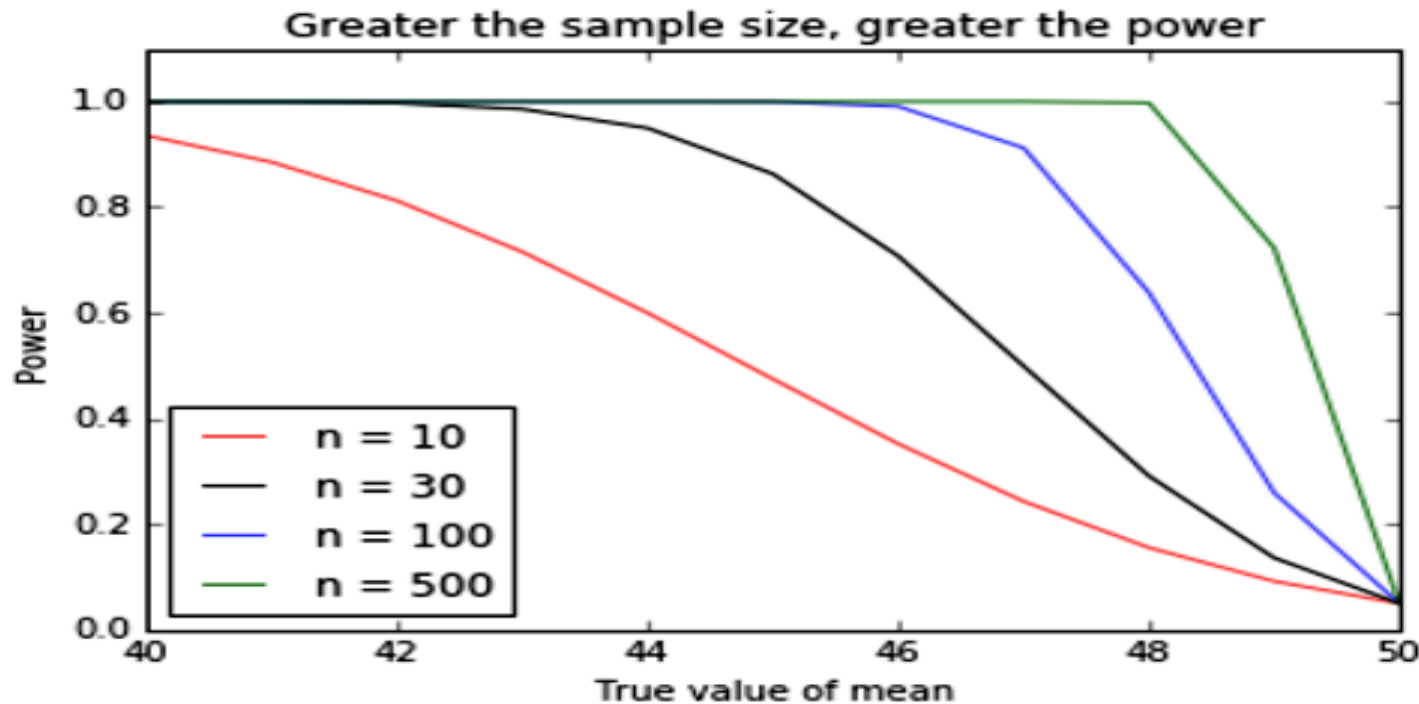
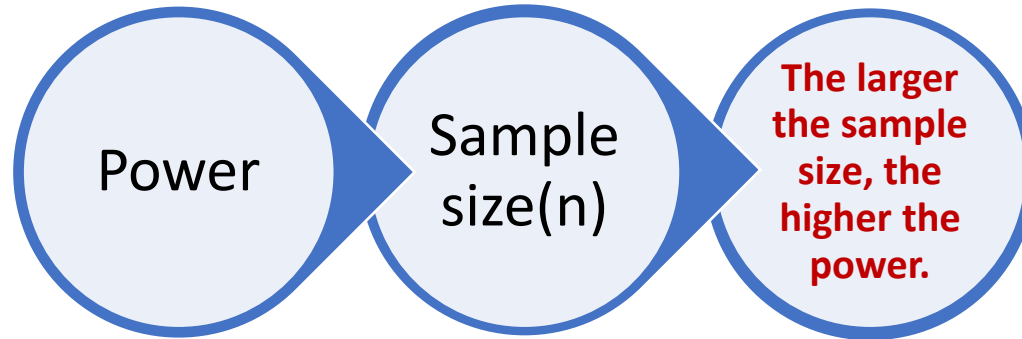
Example:

A random sample of n people's weight whose mean and standard deviation are 168 lbs and 7.2 lbs. Can we conclude that the mean of the population is 165lb?

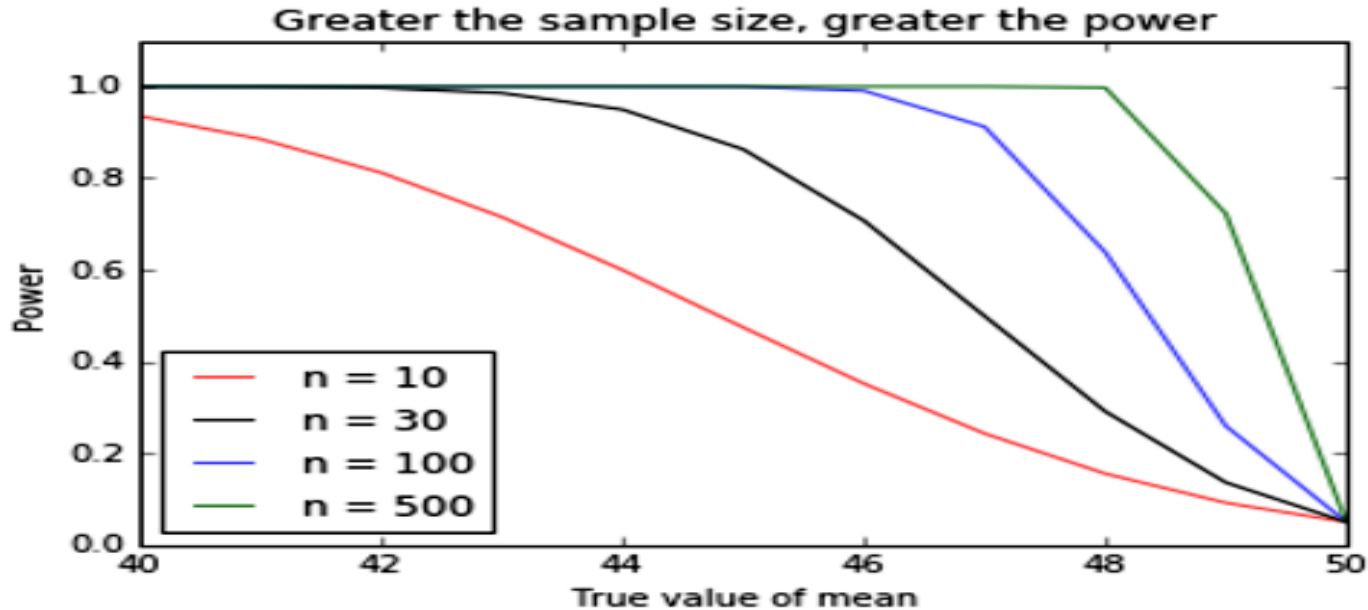
$$H_0: \mu = 165$$

$$H_1: \mu \neq 165$$

$$z = \frac{168 - 165}{7.2/\sqrt{n}} = \frac{(168 - 165)\sqrt{n}}{7.2}$$



Factors affecting Statistical Power of test – Sample size



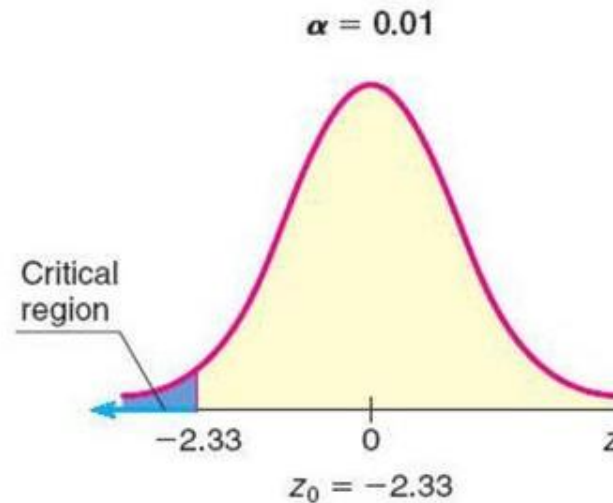
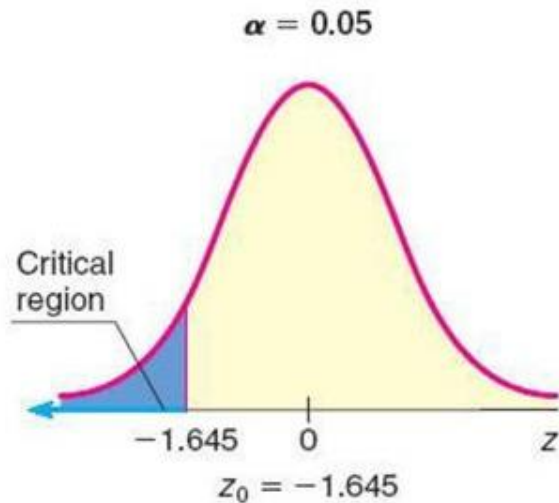
The above figure shows that the larger the sample size, the higher the power. Since sample size is typically under an experimenter's control, increasing sample size is one way to increase power. However, it is sometimes difficult and/or expensive to use a large sample size.

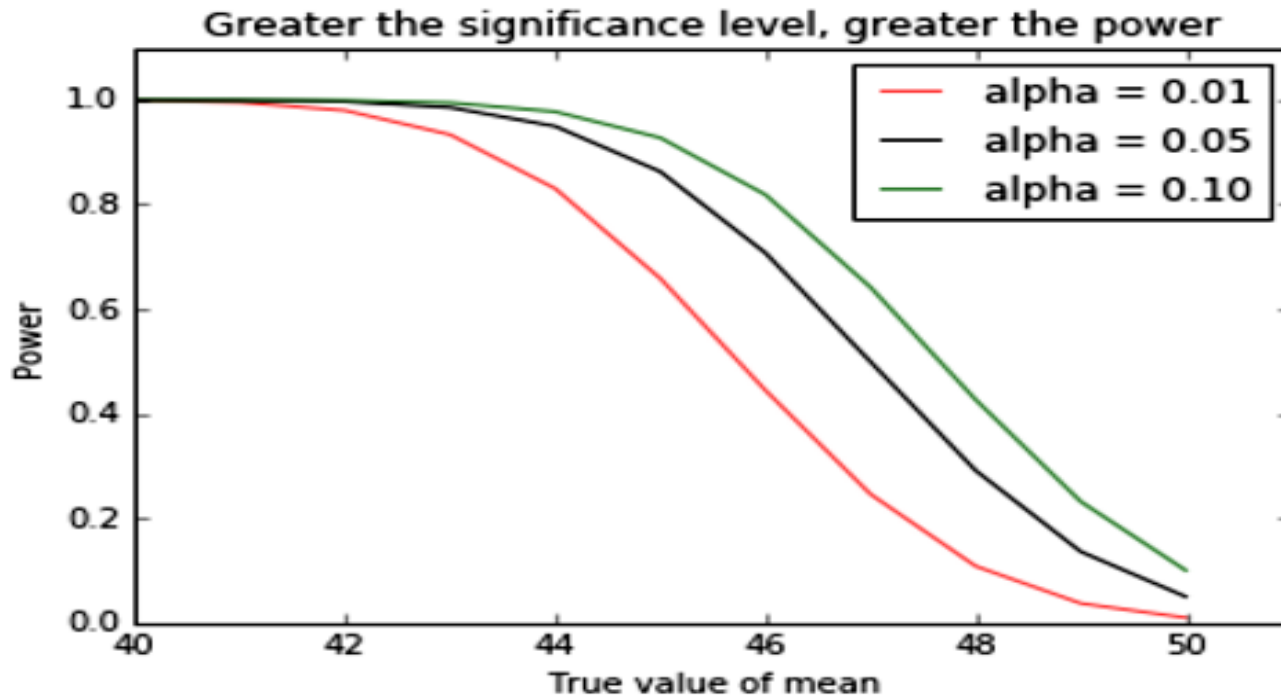
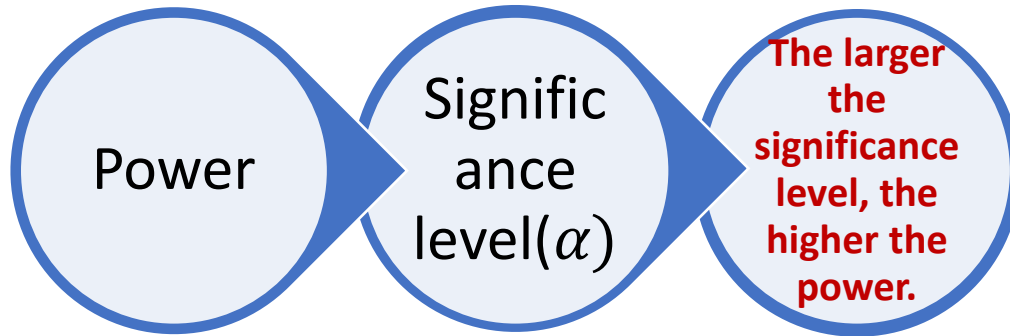
Factors affecting Statistical Power of test – Significance level

Critical Values z_0 for $\alpha = 0.05$ and $\alpha = 0.01$: Left-tailed Test

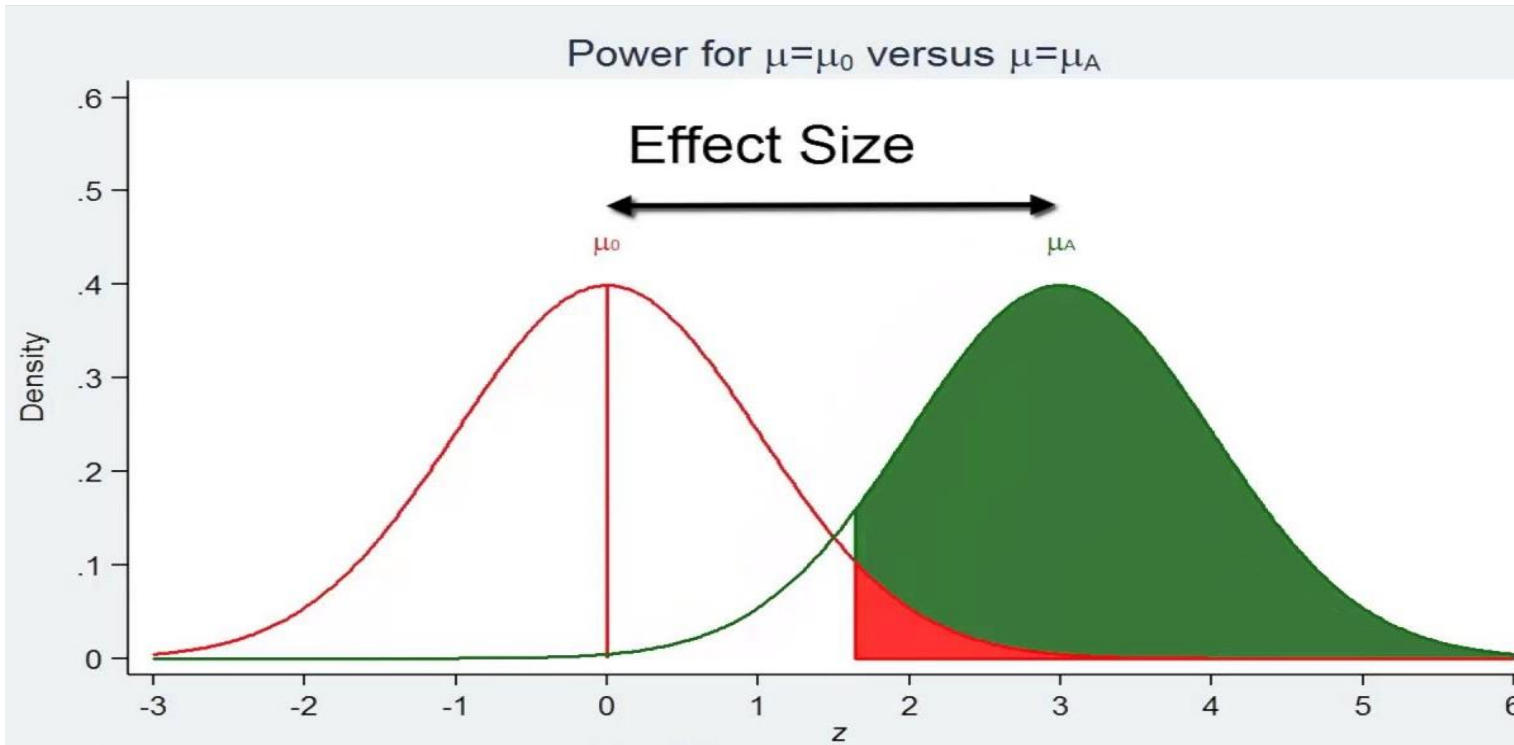
Level of significance

For a left-tailed test
 $H_1: \mu < k$
Critical value z_0
Critical region:
all $z < z_0$





Effect size = True Mean - Hypothesized Mean
 $= \mu_A - \mu_0$



Example of a power calculation

Assume that a new chemical process has been developed that may increase the yield over that of the current process. The current process is known to have a **mean yield of 80** and a **standard deviation of 5**, where the units are the percentage of a theoretical maximum. If the mean yield of the new process is shown to be greater than 80, the new process will be put into production.

Let μ denote the mean yield of the new process. It is proposed to run the new process 50 times and then to test the hypothesis

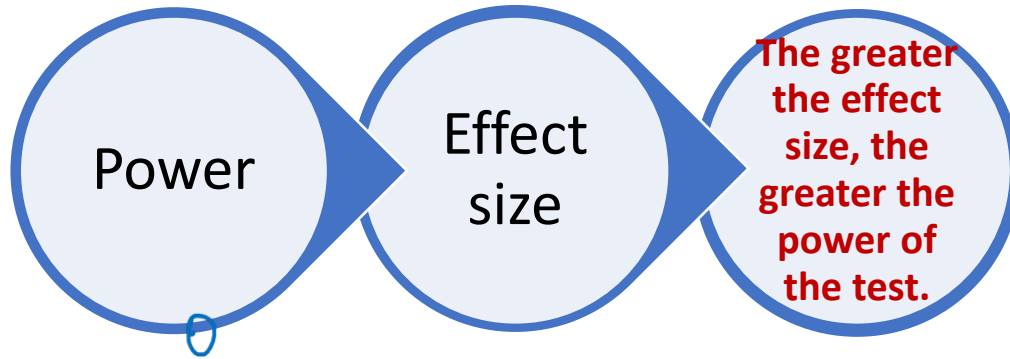
$H_0: \mu \leq 80$ versus $H_1: \mu > 80$ at a significance level of 5%.

if μ is close to μ_0 : the power will be small

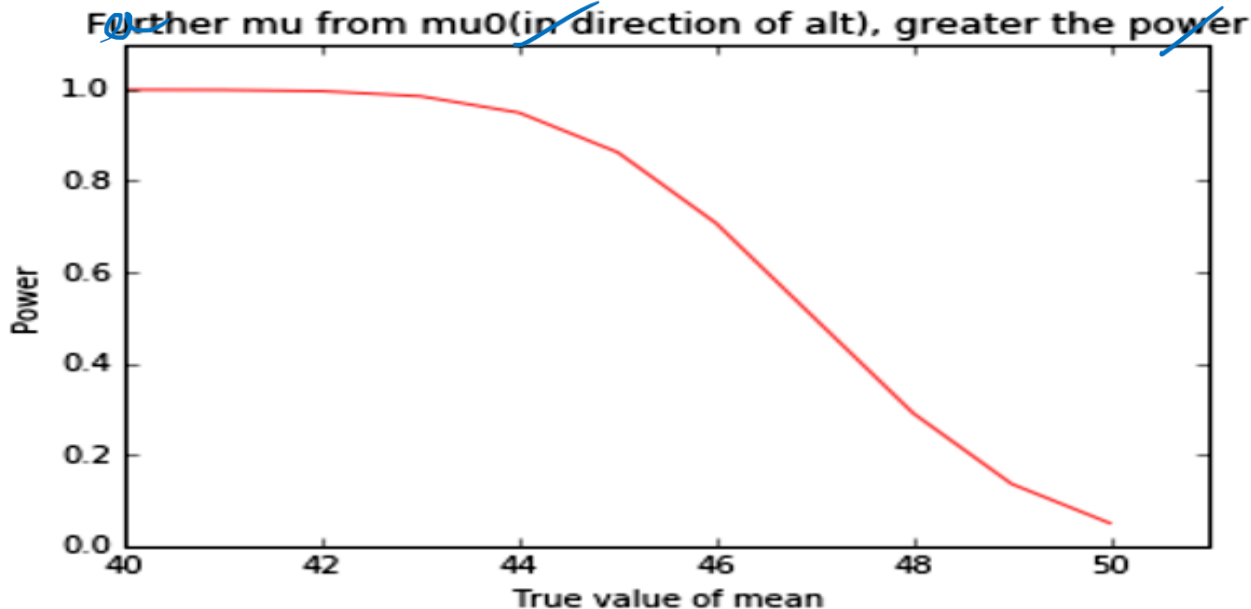
(when $\mu = 81$, Power=0.4090)

if μ is far from μ_0 : the power will be large

(when $\mu = 82$, Power=0.8830)



Effect size = True value - Hypothesized value



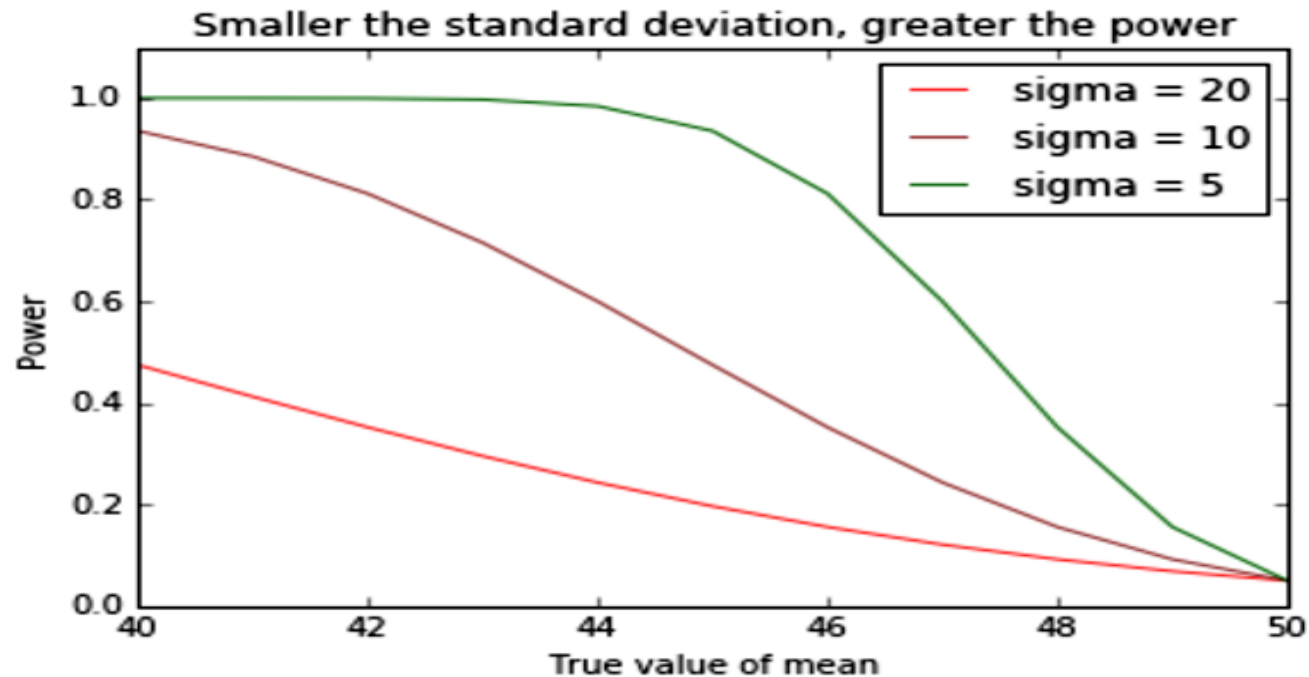
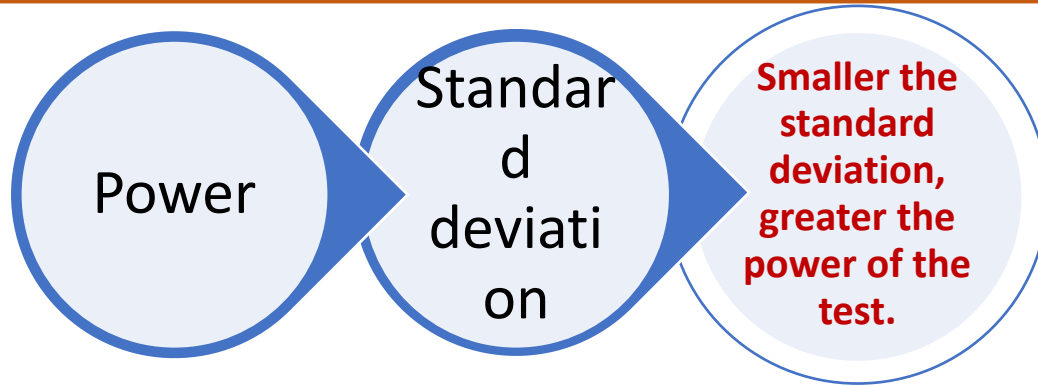
Factors affecting Statistical Power of test –Standard deviation

Example:

A random sample of 200 people's weight whose mean is 168 lbs.
Can we conclude that the mean of the population is 165lb?

$$H_0: \mu = 165$$
$$H_1: \mu \neq 165$$

$$z = \frac{168 - 165}{\sigma / \sqrt{200}} = \frac{(168 - 165)\sqrt{200}}{\sigma}$$



STANDARD DEVIATION

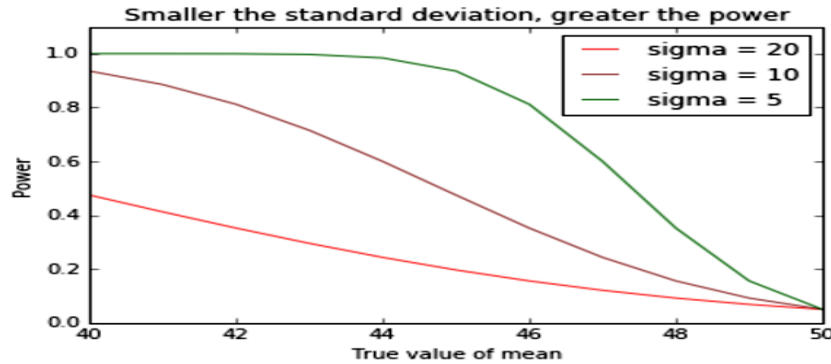
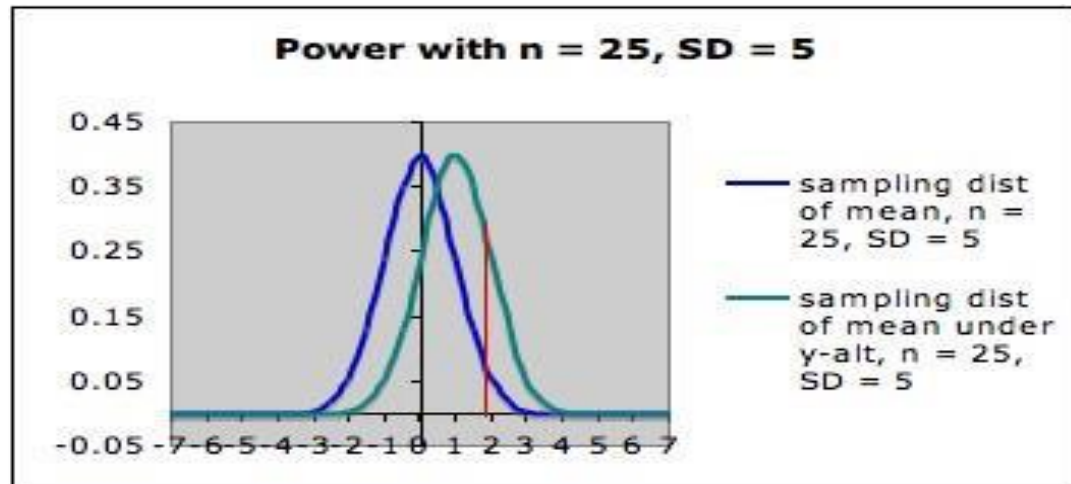
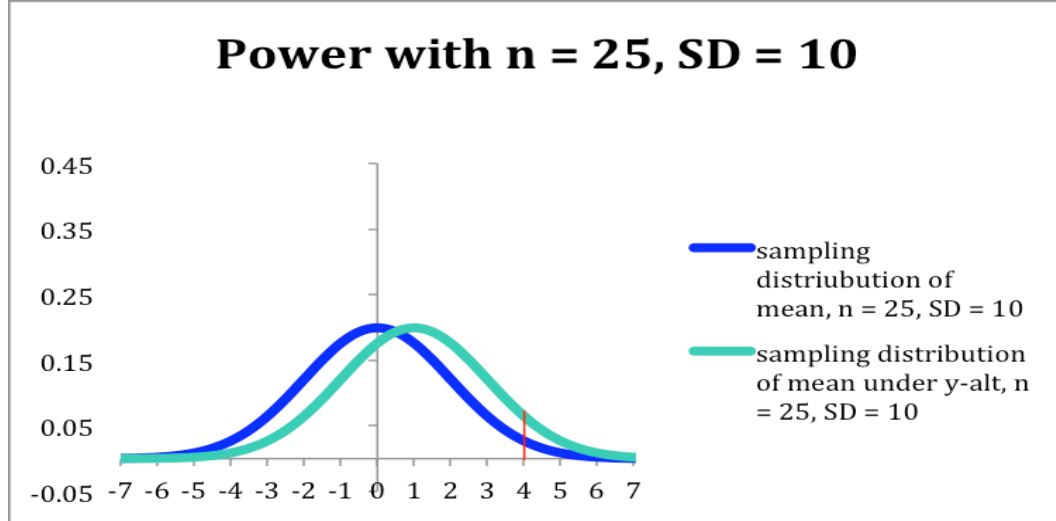
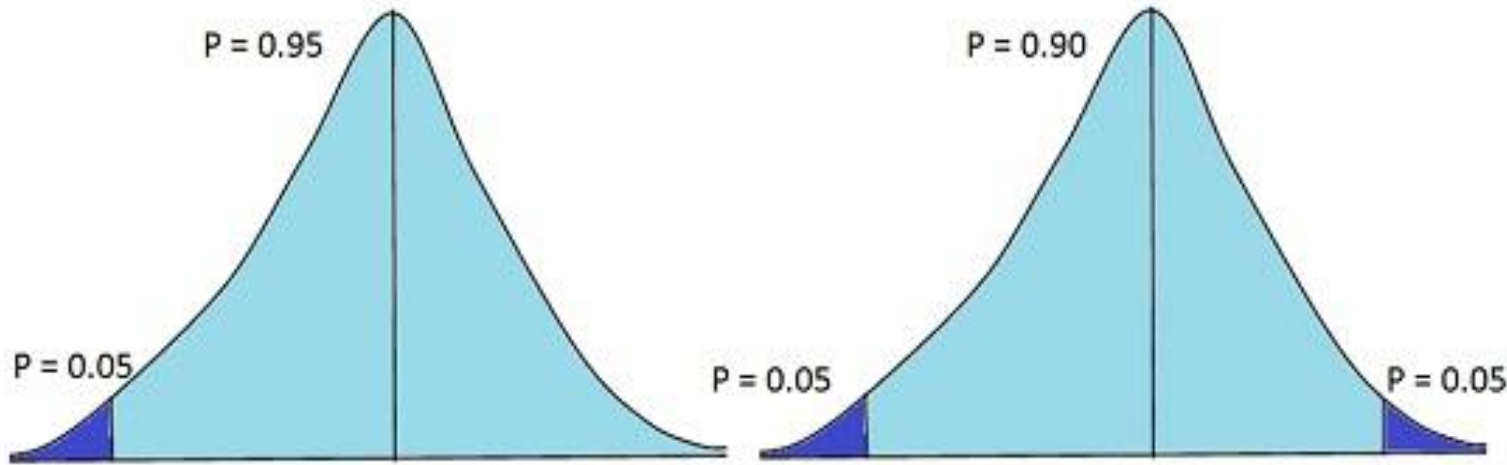


Figure also shows that power is higher when the standard deviation is small than when it is large. For all values of N , power is higher for the standard deviation of 10 than for the standard deviation of 15 (except, of course, when $N = 0$). Experimenters can sometimes control the standard deviation by sampling from a homogeneous population of subjects, by reducing random measurement error, and/or by making sure the experimental procedures are applied very consistently.

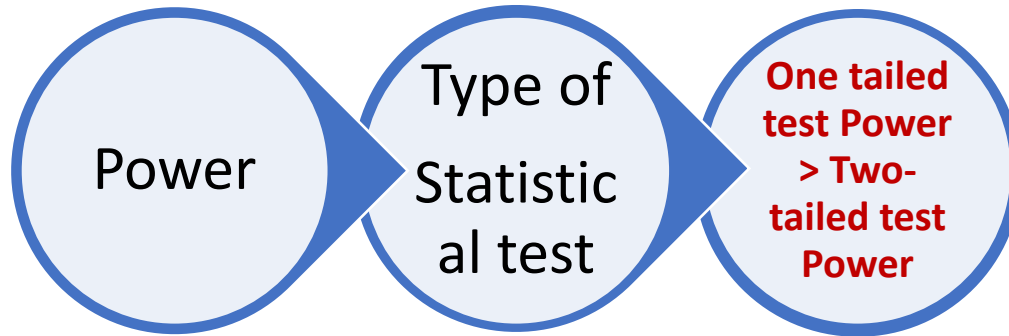
Factors affecting Statistical Power of test – Standard deviation



In each picture, the area under the *green* curve to the right of the red line is the power of the test against the alternate depicted. Note that this area is *larger* in the second picture (the one with smaller standard deviation) than in the first picture.



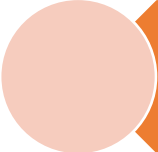
One-tailed Test Vs Two-tailed Test



ONE- VERSUS TWO-TAILED TESTS

Power is higher with a *one-tailed* test than with a *two-tailed* test as long as the hypothesized direction is correct. A one-tailed test at the 0.05 level has the same power as a two-tailed test at the 0.10 level. A one-tailed test, in effect, raises the significance level.

Factors affecting Statistical Power of test



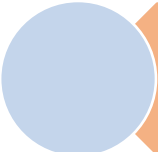
Greater the sample size, the higher the power.



The larger the significance level, the higher the power.



The greater the effect size, the greater the power of the test..



Smaller the standard deviation, greater the power of the test.



One tailed test Power > Two-tailed test Power

True or False/ Mcqs

1) A test has power 0.90 when $\mu = 15$.

True or false:

- a. The probability of rejecting H_0 when $\mu = 15$ is 0.90.
- b. The probability of making a correct decision when $\mu = 15$ is 0.90.
- c. The probability of making a correct decision when $\mu = 15$ is 0.10.
- d. The probability that H_0 is true when $\mu = 15$ is 0.10.

2) If the sample size remains the same, and the level α increases, then the power will

-
- a) Increase b) decrease c) remains constant d) both increase and decreases

True or False/ Mcqs

1) A test has power 0.90 when $\mu = 15$.

True or false:

- a. The probability of rejecting H_0 when $\mu = 15$ is 0.90.
- b. The probability of making a correct decision when $\mu = 15$ is 0.90.
- c. The probability of making a correct decision when $\mu = 15$ is 0.10.
- d. The probability that H_0 is true when $\mu = 15$ is 0.10.

- (a) True. This is the definition of power.
- (b) True. When H_0 is false, making a correct decision means rejecting H_0 .
- (c) False. The power is 0.90, not 0.10.
- (d) False. The power is not the probability that H_0 is true.

2) If the sample size remains the same, and the level α increases, then the power will

- a) Increase b) decrease c) remains constant d) both increase and decreases

Answer a) increase. If the level increases, the probability of rejecting H_0 increases, so in particular, the probability of rejecting H_0 when it is false increases.



THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Simple Linear Regression

Dr. Karthiyayini

Department of Science and Humanities

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

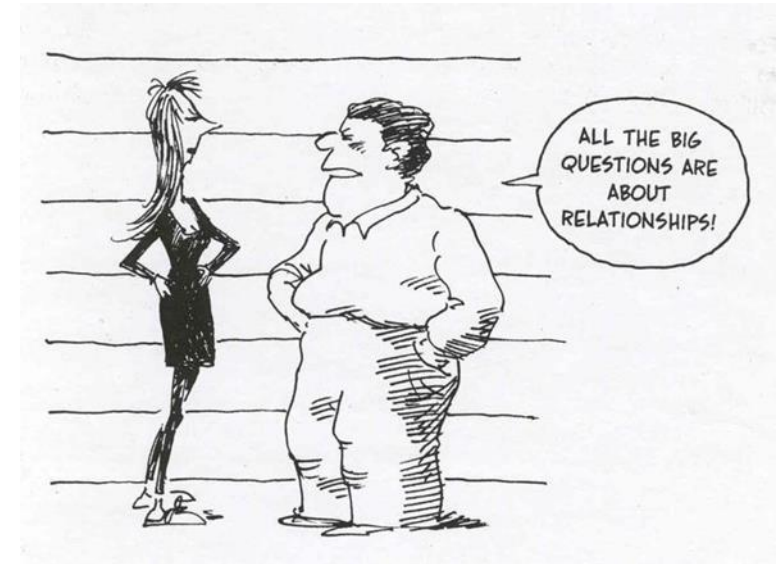
Correlation, Simple Linear Regression:

Dr. Karthiyayini

Department of Science & Humanities

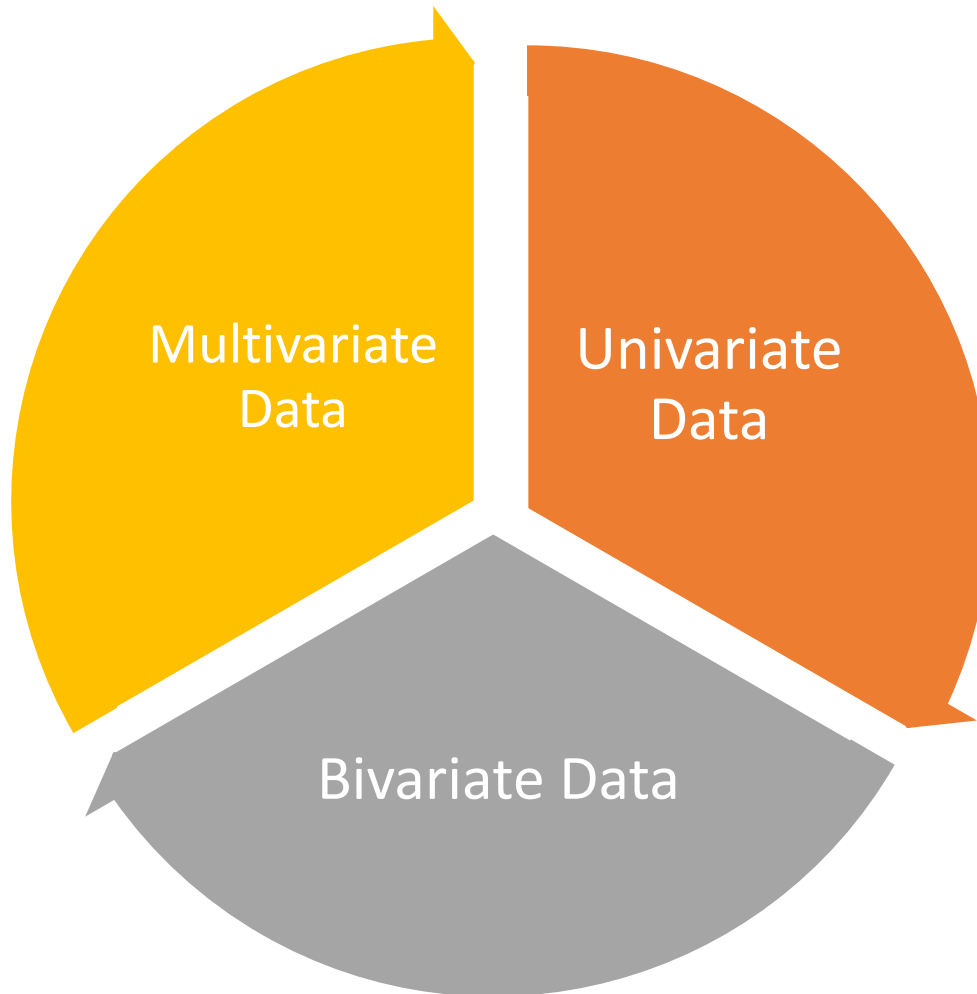
So far we have done statistics on one variable at a time. We are now interested in **relationships** between two variables and how to use one variable to **predict** another variable.

- Does weight depend on height?
- Does blood pressure level predict life expectancy?
- Does screen time correlate with eye strain?
- Is temperature related to electricity consumption (AC usage)?



- ❖ Classification of Data
- ❖ What is Correlation ?
- ❖ Pearson's Correlation Coefficient

Types of Data



Univariate & Bi - Variate Data

The analysis of Univariate data can be done using :

Analytical Techniques :

- Central tendency measures (mean, median and mode)
- Dispersion or Spread of data (range, minimum, maximum quartiles, variance and standard deviation)
- Frequency distribution tables

Visualization techniques :

- Histograms
- Pie Charts
- Frequency Polygon
- Bar Charts

The analysis of Bivariate data can be done using :

Analytical Technique :

- Correlation Co-efficient
- Regression Analysis

Visualization Technique :

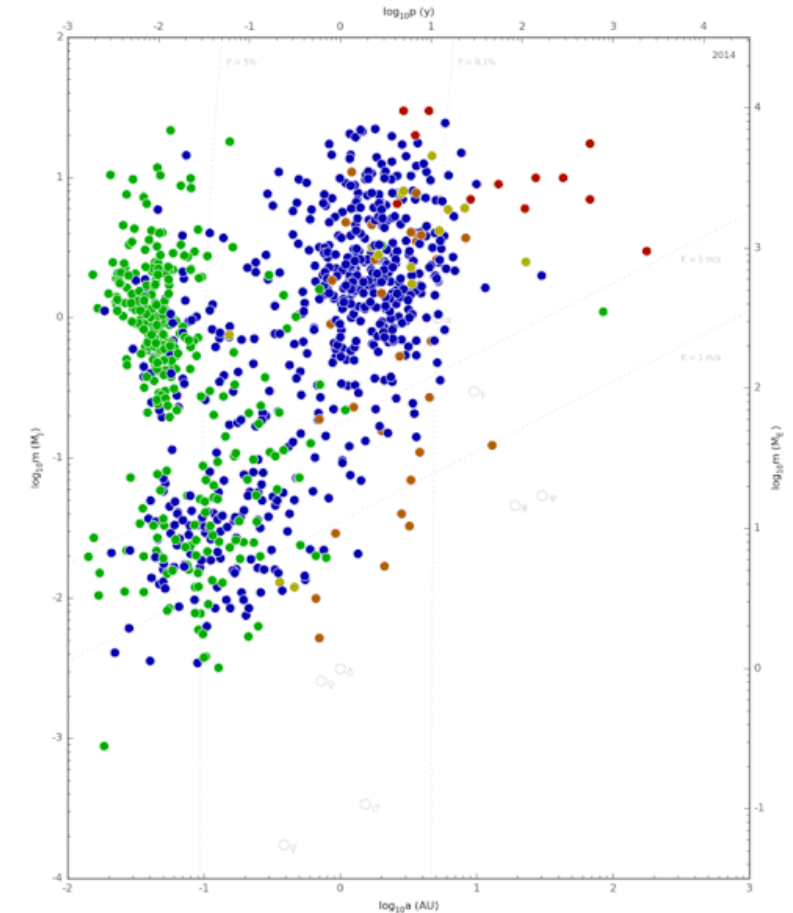
- Scatter Plot

Bi - Variate Analysis

- ❖ Bivariate analysis means the analysis of bivariate data; used to find out if there is a relationship between two sets of values.
- ❖ It usually involves the variables X and Y and is represented as an ordered pair (X, Y) .
- ❖ X represents the independent variable and Y represents the dependent variable.

Scatter Plots

- ❖ Scatter Plot is a mathematical diagram that plots pairs of data on an X-Y graph in order to reveal the relationship between the data sets.
- ❖ [Scatter plots](#) give you a visual idea of the pattern that your variables follow.
- ❖ Scatter plots can show you visually the strength of the relationship between the variables, the direction of the relationship between the variables and whether any outliers exist.



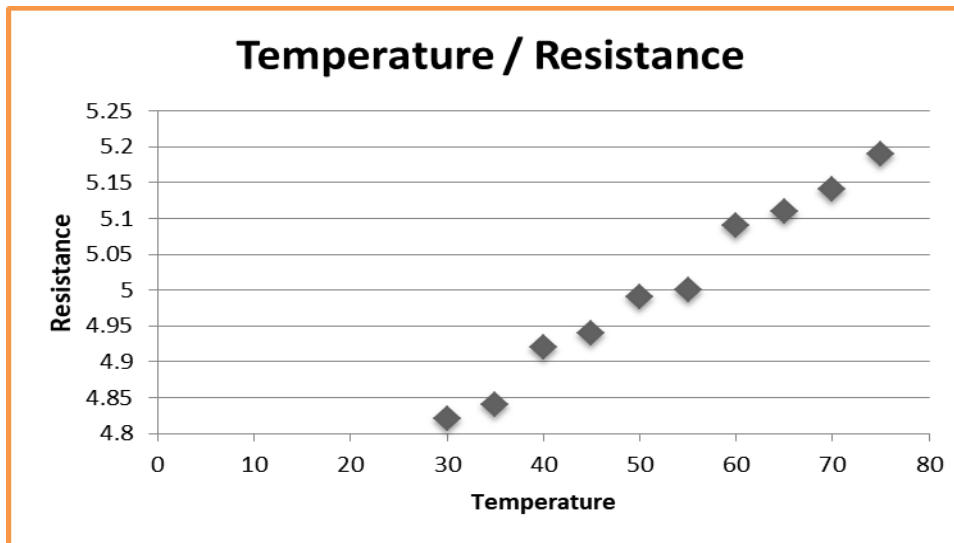
MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Example for Physics Lab :

❖ Variation of resistance with change in temperature of a Semiconductor/ Conductor.



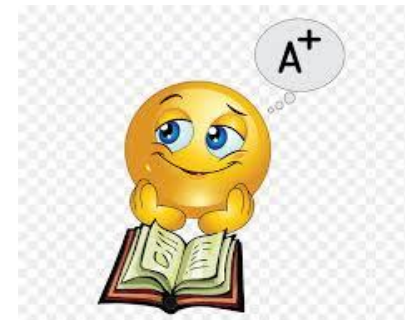
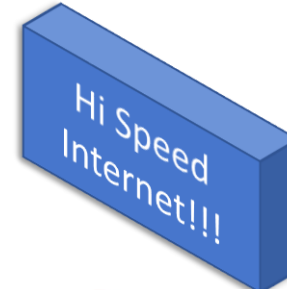
Temperature	Resistance
55	5
45	4.94
35	4.84
65	5.11
75	5.19
70	5.14
60	5.09
50	4.99
40	4.92
30	4.82



The resistance decreases with increase in temperature in a Semiconductor whereas in a Conductor, the resistance increases with an increase in the temperature.

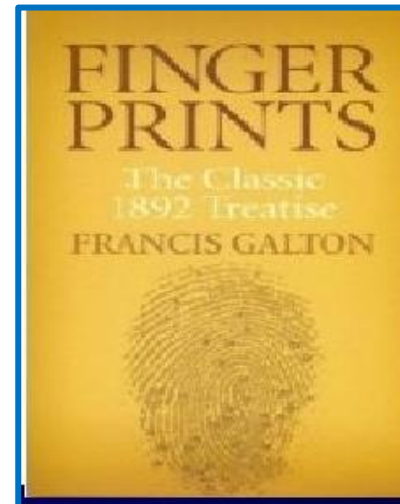
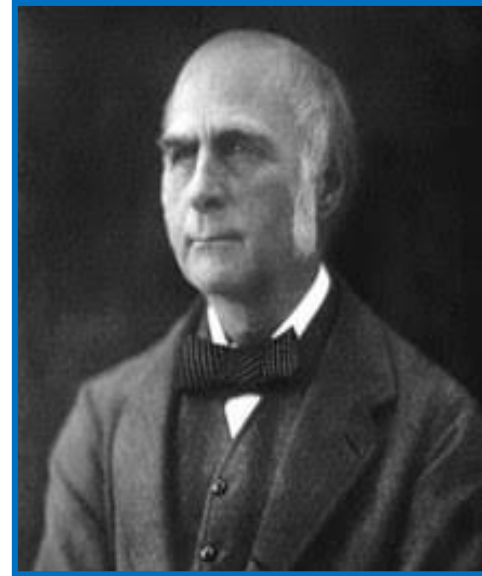
Correlation

- ❖ Does height have an impact on the performance of a player in a Basket ball match?
- ❖ Is there a relationship between internet bandwidth and time taken for data transfer?
- ❖ Are Height and Weight of an individual related?
- ❖ Does no. of hours effort have an impact on CGPA scored?



Brief history of Correlation

- ❖ Sir Francis Galton, (16 February 1822 – 17 January 1911).
- ❖ He was an English [Victorian era statistician](#) and a Fellow of the Royal Society.
- ❖ Galton produced over 340 papers and books.
- ❖ In 1892, he published the book “Finger Prints” and proposed the use of fingerprints as a means of personal identification.



MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Galton's case study!!



Is there any relation between
the *height of an individual* and
the *length of his forearm*???



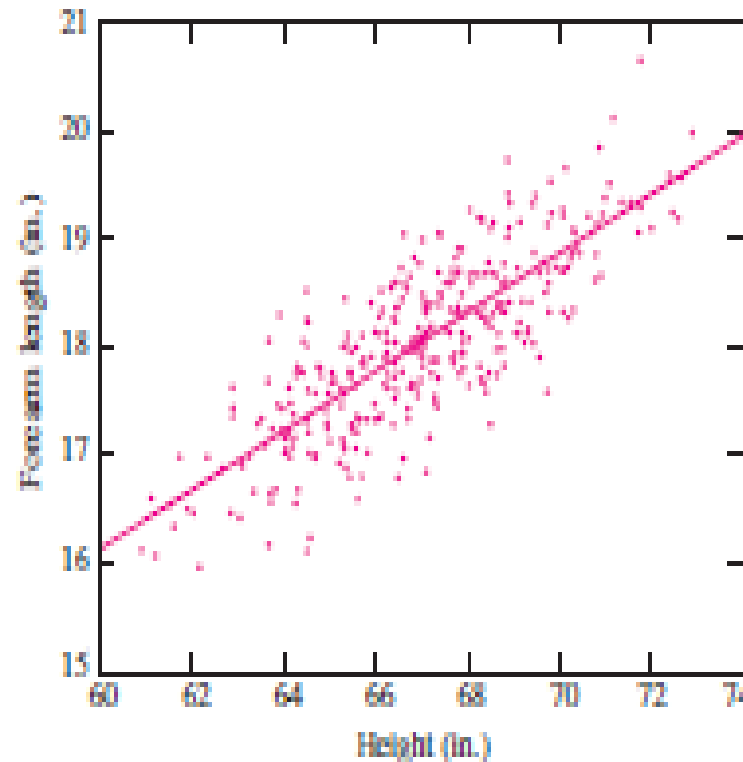
Sir Francis Galton introduced the concept of 'Correlation' in 1888 with a paper discussing how to measure the relationship between two variables.

Case Study : Galtons

- ❖ The data set that he considered consisted of the *heights and forearm lengths of 348 adult men*.

(He measured the distance from the elbow to the tip of the middle finger which is called as a cubit)

- ❖ Let the *height of the i th man* be $= x_i$
- ❖ Let the *length of the forearm of the i th man* be $= y_i$
- ❖ Then Galton's data consists of *348 ordered pairs (x_i, y_i)*



Correlation Coefficient

- ❖ Let (x_i, y_i) = ordered pairs that represent points on a scatter plot.
- ❖ $\bar{x}$ = mean of the 'x' values
- ❖ $\bar{y}$ = mean of the 'y' values
- ❖ S_x = standard deviation of 'x' values
- ❖ S_y = standard deviation of 'y' values
- ❖ Correlation Co-efficient is given by

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{S_x} \right) \left(\frac{y_i - \bar{y}}{S_y} \right)$$

$$\Rightarrow r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

The Correlation Coefficient is the average of the product of z-scores

Pearson's Correlation Coefficient

Example Problem :

The Psychological tests of intelligence and of engineering ability were applied to 10 students. Here is a record of ungrouped data showing intelligence ratio (I.R) and engineering ratio (E.R). Calculate the Correlation Coefficient?

Student	A	B	C	D	E	F	G	H	I	J
I.R	105	104	102	101	100	99	98	96	93	92
E.R	101	103	100	98	95	96	104	92	97	94

Solution :

Students	IR (x)	ER (y)	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
1	105	101	6	3	36	9	18
2	104	103	5	5	25	25	25
3	102	100	3	2	9	4	6
4	101	98	2	0	4	0	0
5	100	95	1	-3	1	9	-3
6	99	96	0	-2	0	4	0
7	98	104	-1	6	1	36	-6
8	96	92	-3	-6	9	36	18
9	93	97	-6	-1	36	1	6
10	92	94	-7	-4	49	16	28

$$r = \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}}$$

Solution :

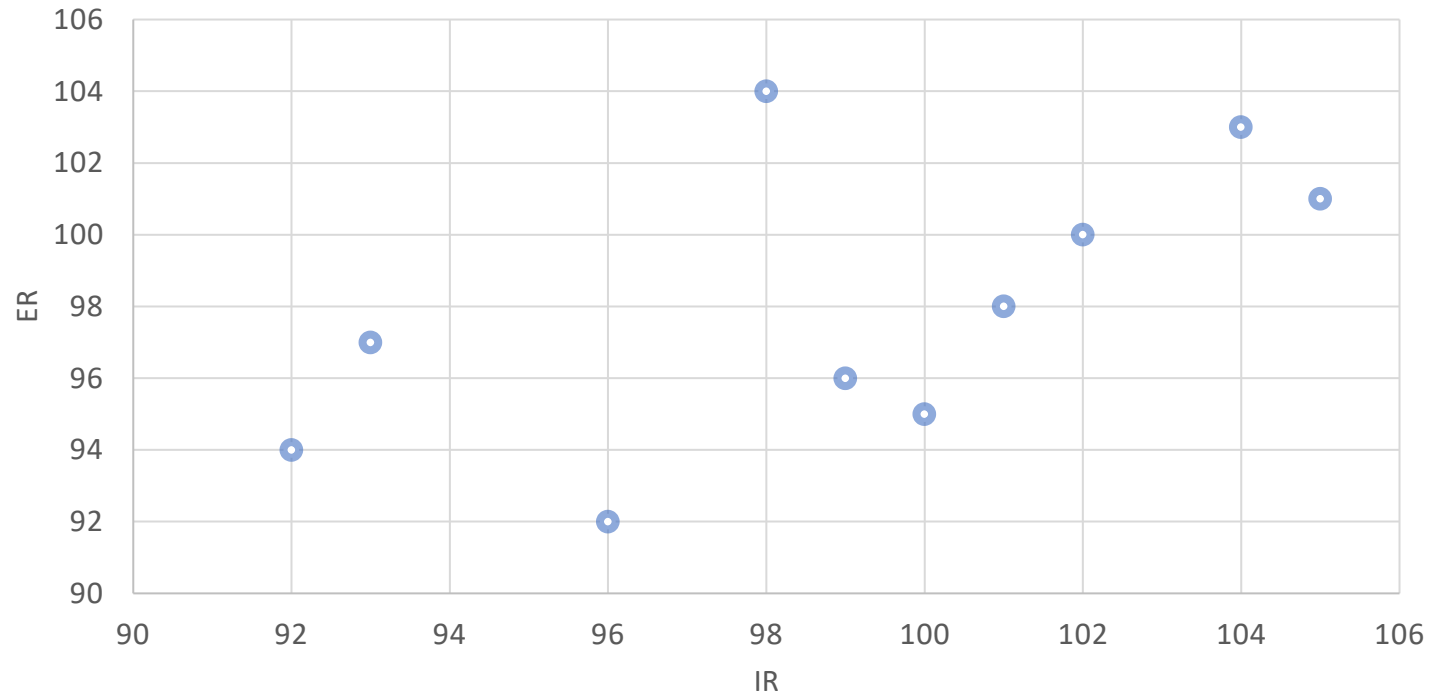
Students	IR (x)	ER (y)	$X = x - \bar{x}$	$Y = y - \bar{y}$	X^2	Y^2	XY
1	105	101	6	3	36	9	18
2	104	103	5	5	25	25	25
3	102	100	3	2	9	4	6
4	101	98	2	0	4	0	0
5	100	95	1	-3	1	9	-3
6	99	96	0	-2	0	4	0
7	98	104	-1	6	1	36	6
8	96	92	-3	-6	9	36	18
9	93	97	-6	-1	36	1	6
10	92	94	-7	-4	49	16	28
	990	980	0	0	170	140	92

$$\bar{x} = 99 \quad \bar{y} = 98$$

$$\begin{aligned}
 r &= \frac{\sum XY}{\sqrt{\sum X^2} \sqrt{\sum Y^2}} \\
 &= \frac{92}{\sqrt{170} \sqrt{140}} \\
 &= 0.59
 \end{aligned}$$

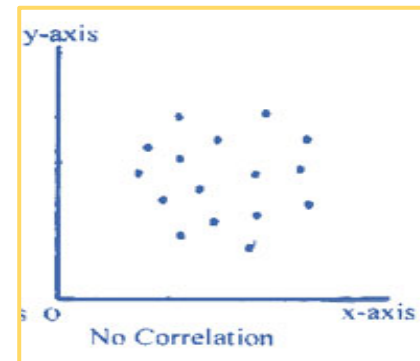
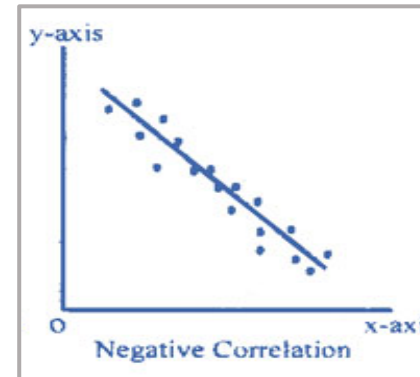
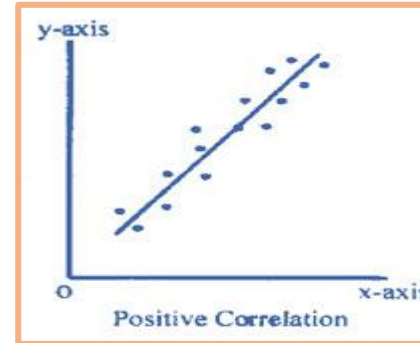
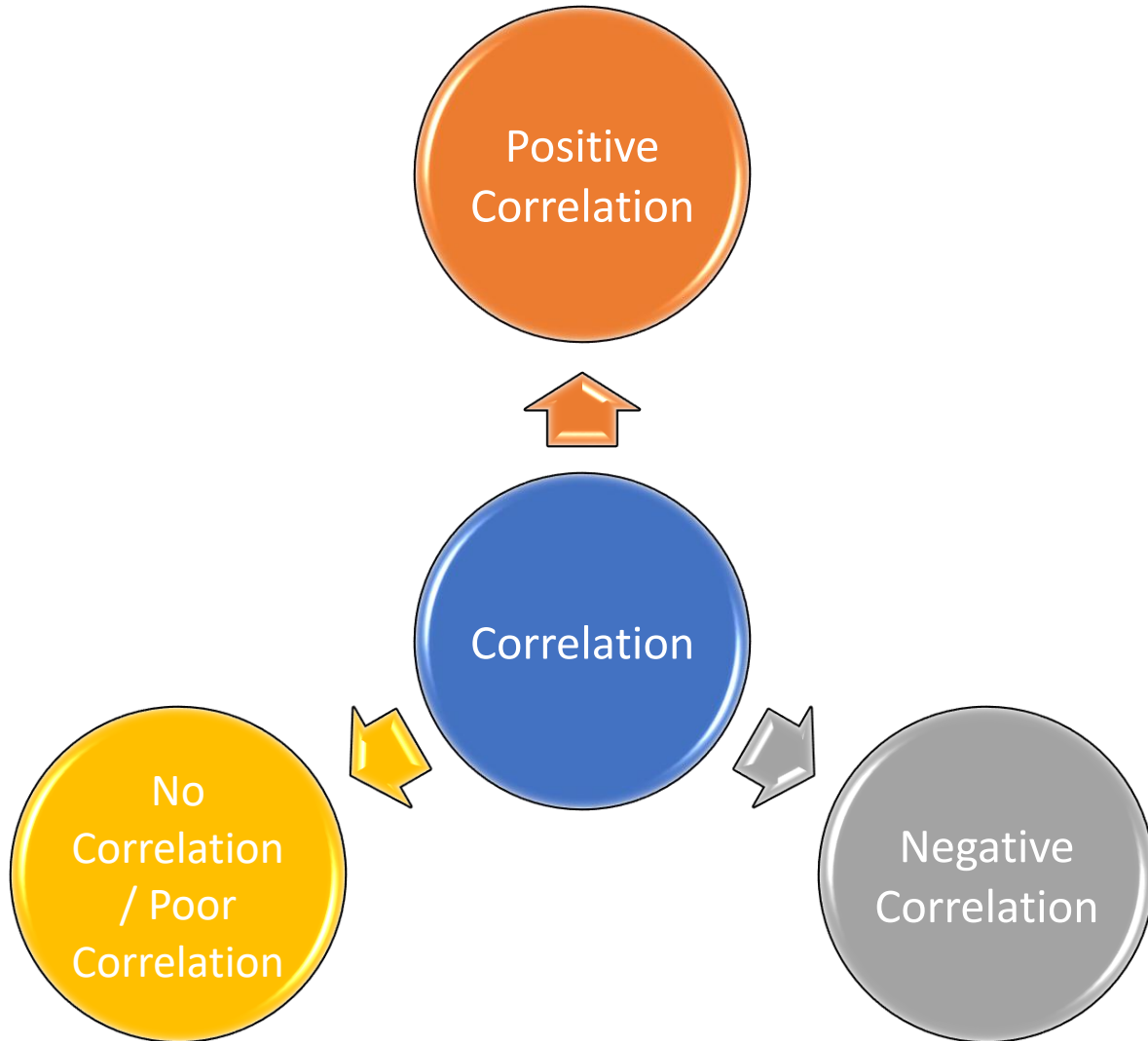
Scatter Plot

Intelligence Quotient / Engineering Ability



Correlation coefficient $r=0.59$

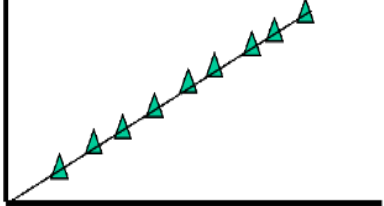
Classification of Correlation



Interpretation of Correlation Coefficient

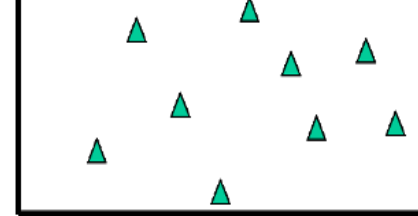
$r = +1$:

Perfect + correlation



$r = \pm 1 \Rightarrow$
Perfect Positive
/
Perfect Negative
Correlation

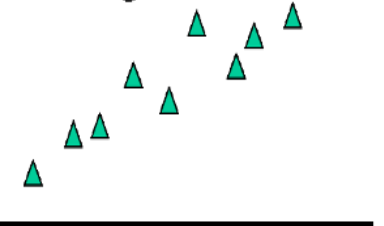
r close to 0: Weak or
no association



$r = 0 \Rightarrow$
No Correlation

r close to +1:

strong + association



$0 < r < 1 \Rightarrow$
Positive
Correlation

$-1 < r < 0 \Rightarrow$
Negative
Correlation

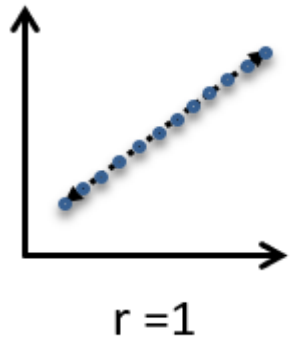
r close to -1:

strong - association

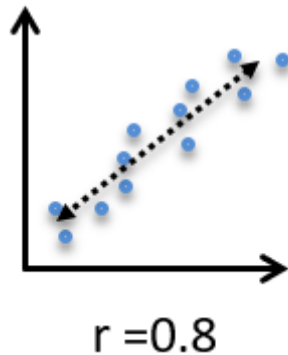


Examples of various levels of correlation

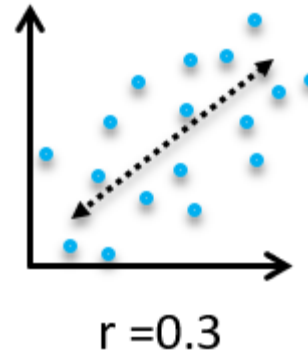
Perfect
Positive
Correlation



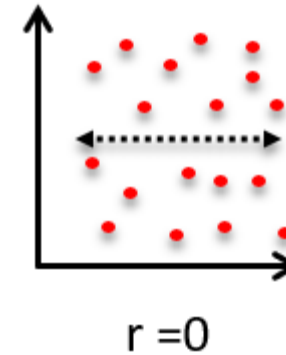
Highly
Positive
Correlation



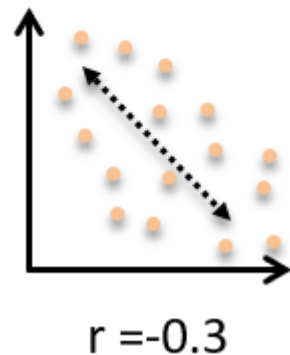
Low
Positive
Correlation



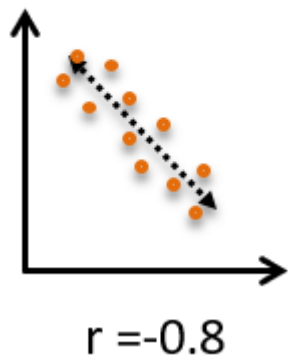
No
Correlation



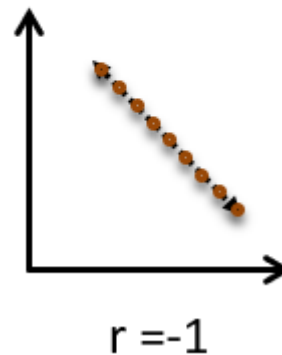
Low
Negative
Correlation



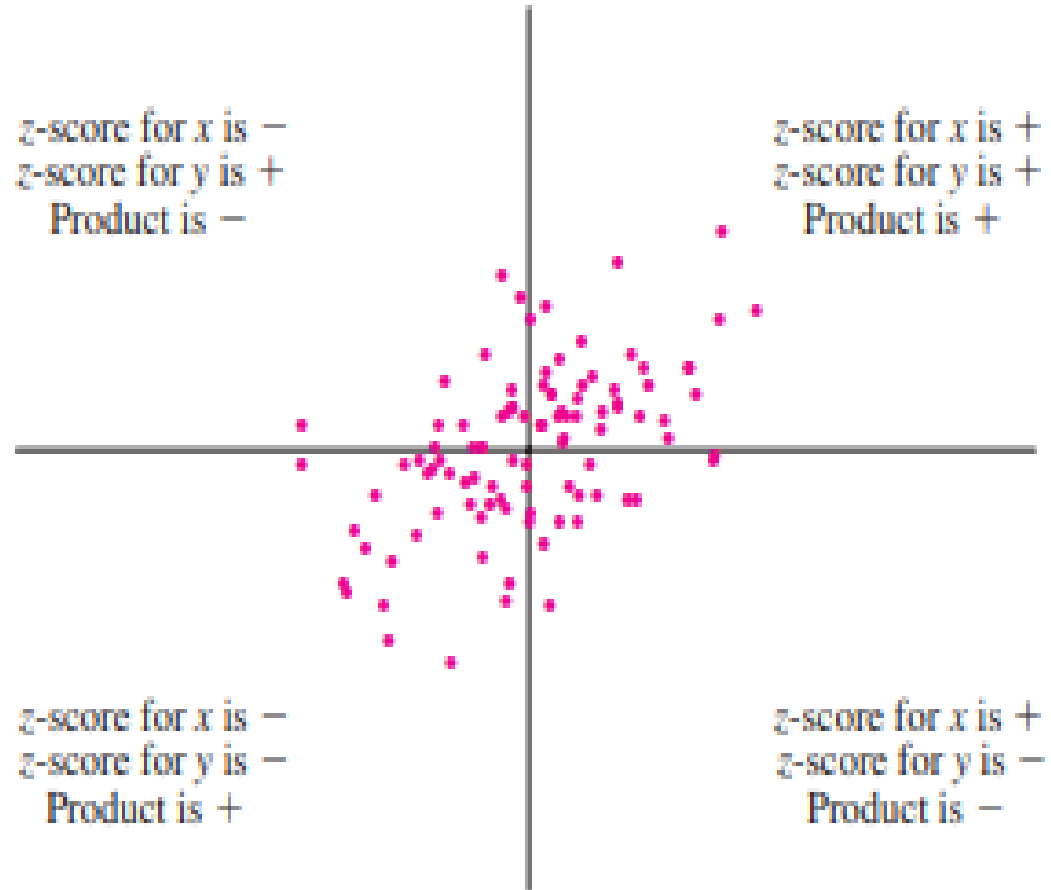
Highly
Negative
Correlation



Perfect
Negative
Correlation



How the Correlation coefficient works!!



More about Correlation Coefficient

Correlation Coefficient



Sample Correlation ' r '



Population Correlation ' ρ '

Note that Sample Correlation is not only used to measure the strength of a relationship but is also used [to construct Confidence intervals and perform Hypothesis testing](#) on the population correlation.

The Correlation coefficient remains unaltered in the following cases :

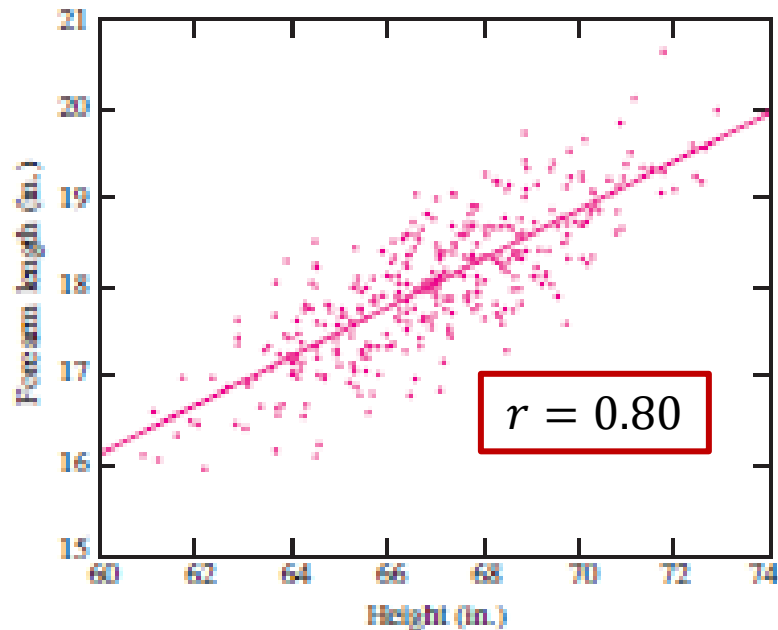
- ❖ Interchanging the values of x and y .
- ❖ Adding a constant to each value of a variable
- ❖ Multiplying each value of a variable by a positive constant.

The Correlation Coefficient is unitless!!

❖ Consider the Correlation Co-efficient given by,

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{S_x} \right) \left(\frac{y_i - \bar{y}}{S_y} \right)$$

have the same units (pointing to $x_i - \bar{x}$ and S_x)
have the same units. (pointing to $y_i - \bar{y}$ and S_y)



The Correlation Coefficient Is Unit less

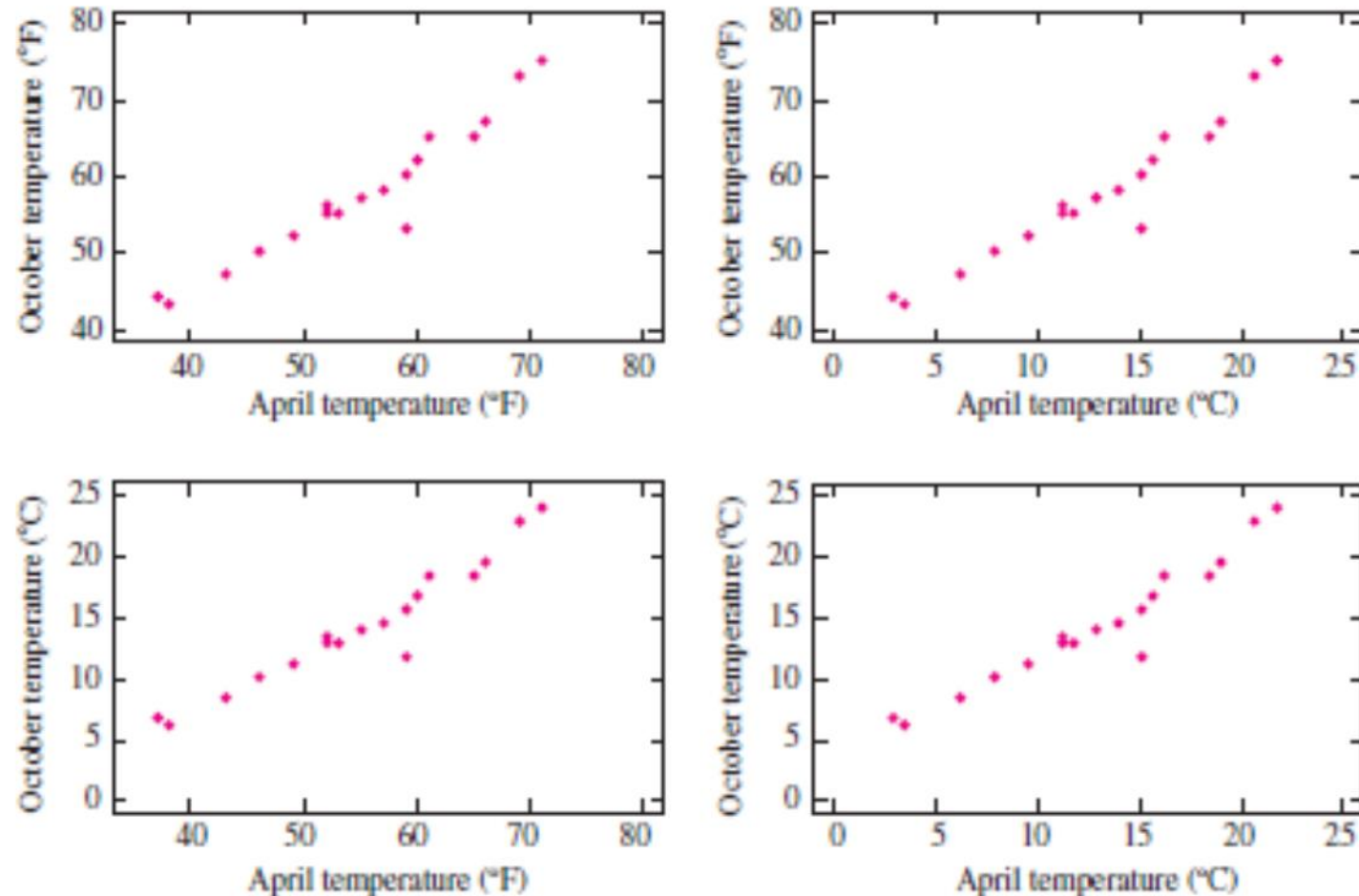
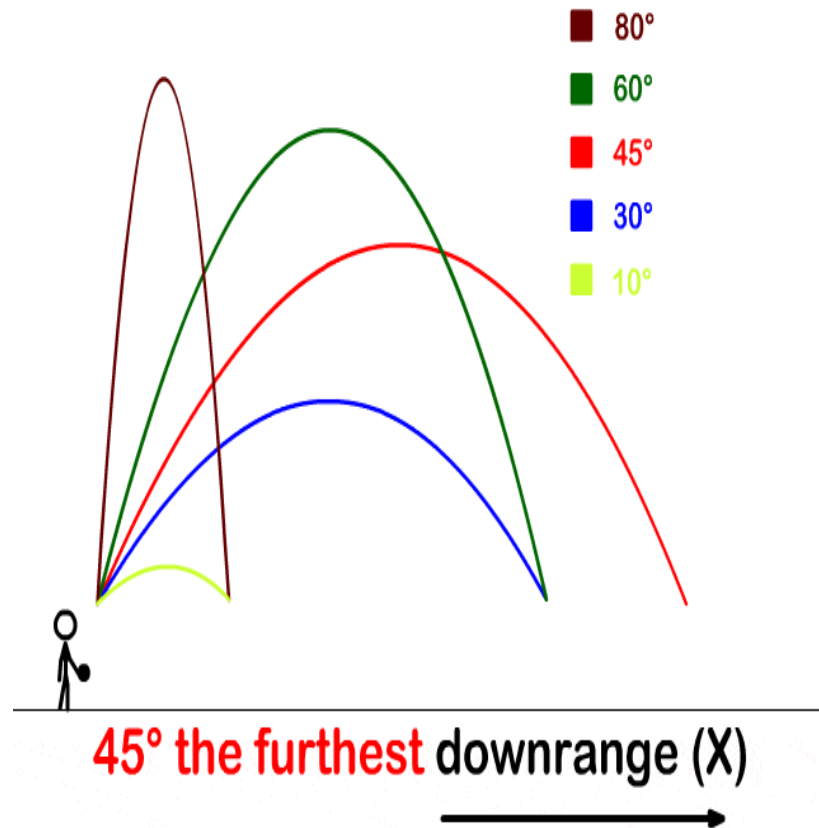
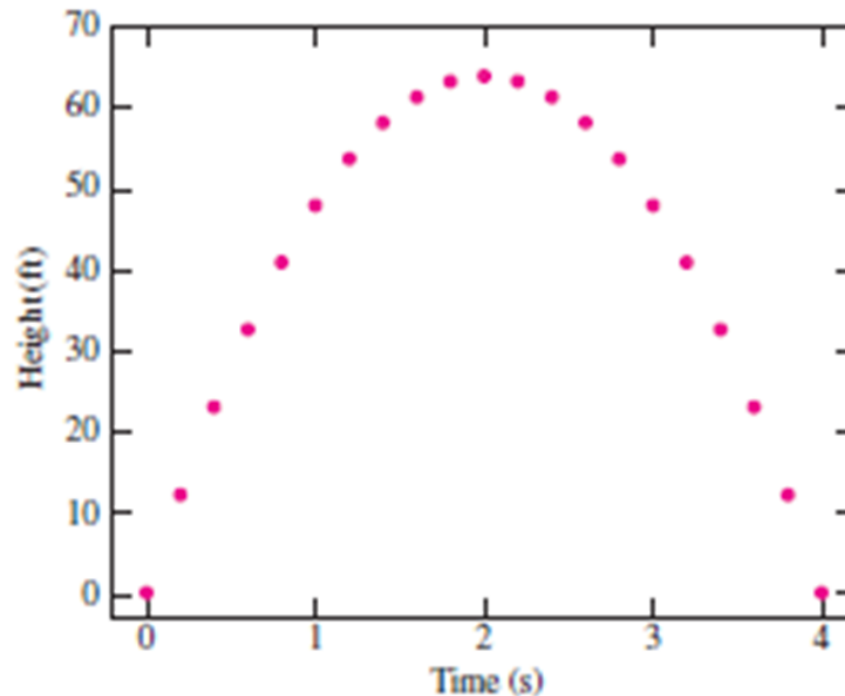


FIGURE 7.6 Mean April and October temperatures for several U.S. cities. The correlation coefficient is 0.96 for each plot; the choice of units does not matter.

Correlation Coefficient measures only Linear Association



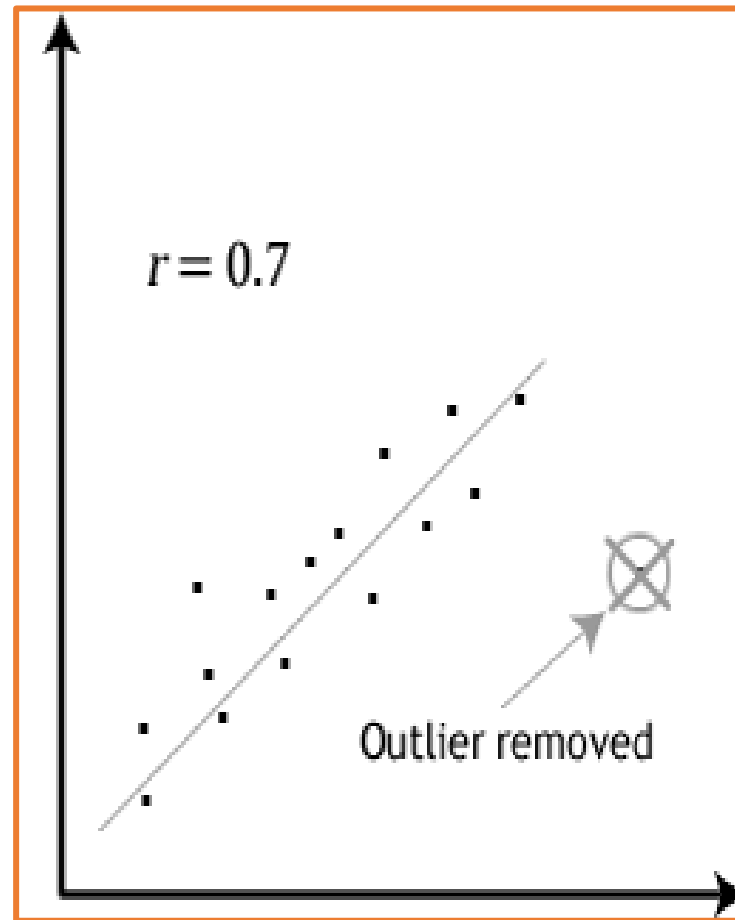
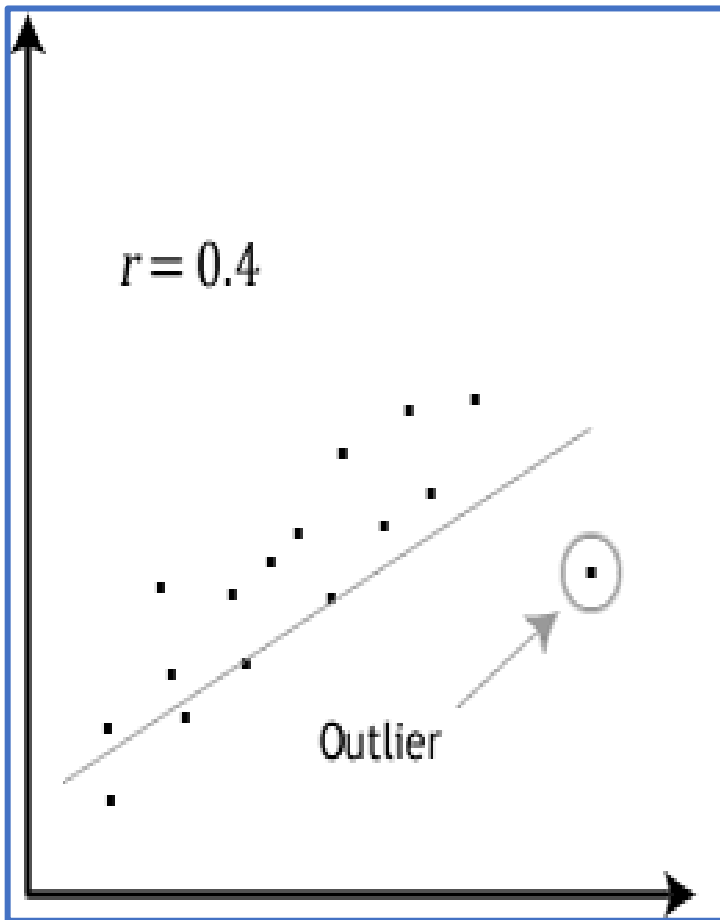
- ❖ Initial velocity = 64ft/s
- ❖ Equation : $y = 64x - 16x^2$
- ❖ Correlation coefficient $r = 0$



Some more properties of the Correlation Coefficient

- the correlation between x and y is equal to 0.
- Is something wrong?
- No. The value of 0 for the correlation indicates that there is no *linear* relationship between x and y , which is true.
- The relationship is purely *quadratic*.
- The correlation coefficient should only be used when the relationship between the x and y is linear.
- Otherwise the results can be misleading.

Correlation Coefficient - Misleading when outliers are present



MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Anscombe's Quartet

#1

X	Y
10.0	8.04
8.0	6.95
13.0	7.58
9.0	8.81
11.0	8.33
14.0	9.96
6.0	7.24
4.0	4.26
12.0	10.84
7.0	4.82
5.0	5.68

#2

X	Y
10.0	9.14
8.0	8.14
13.0	8.74
9.0	8.77
11.0	9.26
14.0	8.10
6.0	6.13
4.0	3.10
12.0	9.13
7.0	7.26
5.0	4.74

#3

X	Y
10.0	7.46
8.0	6.77
13.0	12.74
9.0	7.11
11.0	7.81
14.0	8.84
6.0	6.08
4.0	5.39
12.0	8.15
7.0	6.42
5.0	5.73

#4

X	Y
8.0	6.58
8.0	5.76
8.0	7.71
8.0	8.84
8.0	8.47
8.0	7.04
8.0	5.25
19.0	12.50
8.0	5.56
8.0	7.91
8.0	6.89

Anscombe's Quartet Summary Statistics

Each of the four datasets in Anscombe's Quartet consists of 11 (x, y) points. Despite having nearly identical statistical properties, they display very different relationships when plotted.

Statistical Property	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Mean of x	9	9	9	9
Mean of y	7.5	7.5	7.5	7.5
Variance of x	11	11	11	11
Variance of y	4.12	4.12	4.12	4.12
Correlation between x and y	0.816	0.816	0.816	0.816
Linear Regression Line	$y = 3 + 0.5x$	$y = 3 + 0.5x$	$y = 3 + 0.5x$	$y = 3 + 0.5x$

- ❖ The summary statistics (ie., Mean, Variance, Pearson's correlation coefficient and the Linear Regression) for all the four datasets is the same.
- ❖ But the datasets are significantly different and visually distinct. This can be observed in their respective scatter plots.

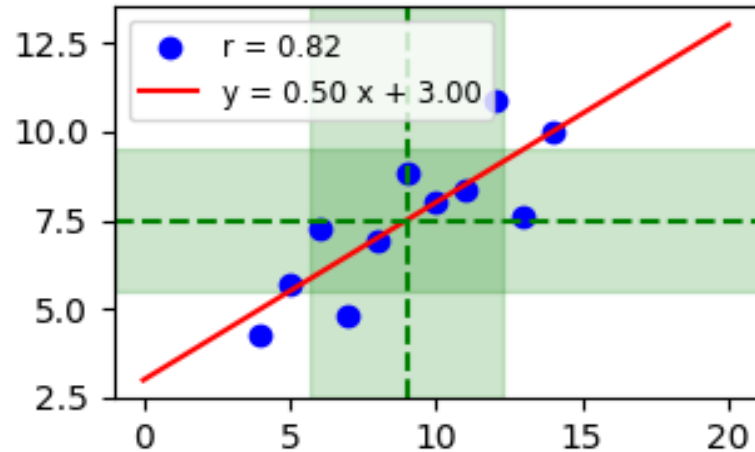
Anscombe's Quartet Summary Statistics

Each of the four datasets in Anscombe's Quartet consists of 11 (x, y) points. Despite having nearly identical statistical properties, they display very different relationships when plotted.

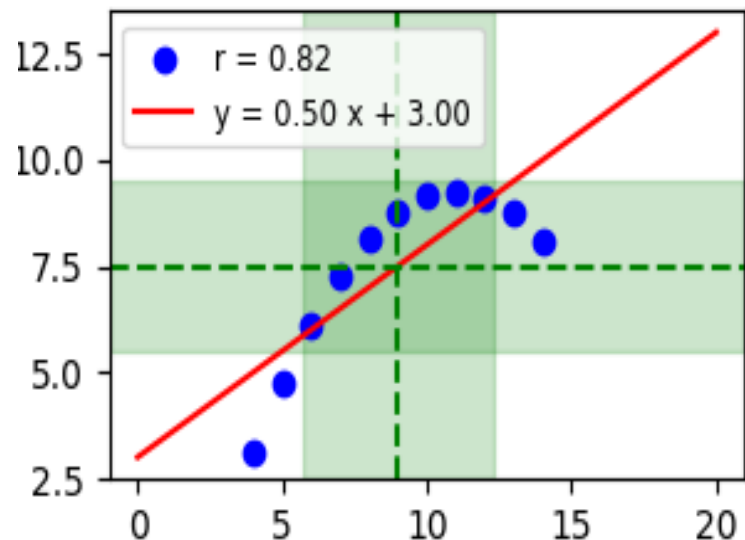
Statistical Property	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Mean of x	9	9	9	9
Mean of y	7.5	7.5	7.5	7.5
Variance of x	11	11	11	11
Variance of y	4.12	4.12	4.12	4.12
Correlation between x and y	0.816	0.816	0.816	0.816
Linear Regression Line	$y = 3 + 0.5x$	$y = 3 + 0.5x$	$y = 3 + 0.5x$	$y = 3 + 0.5x$

Identical statistics

Visual representation of the Anscombe's Quartet

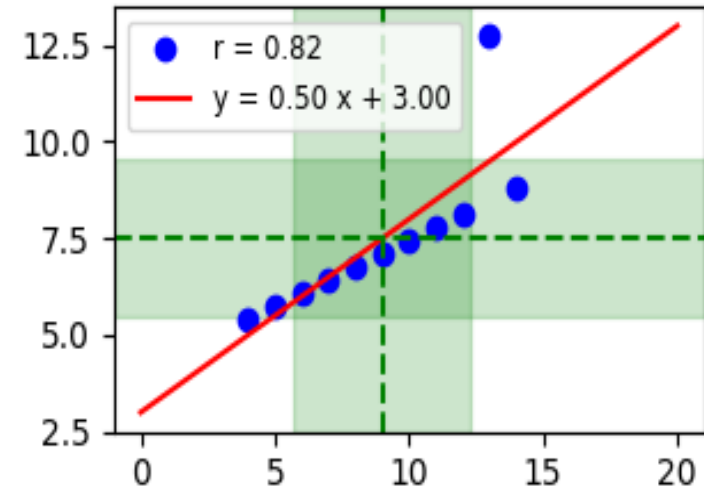


- ❖ The first [scatter plot](#) appears to be a simple [linear relationship](#), corresponding to two [variables](#)

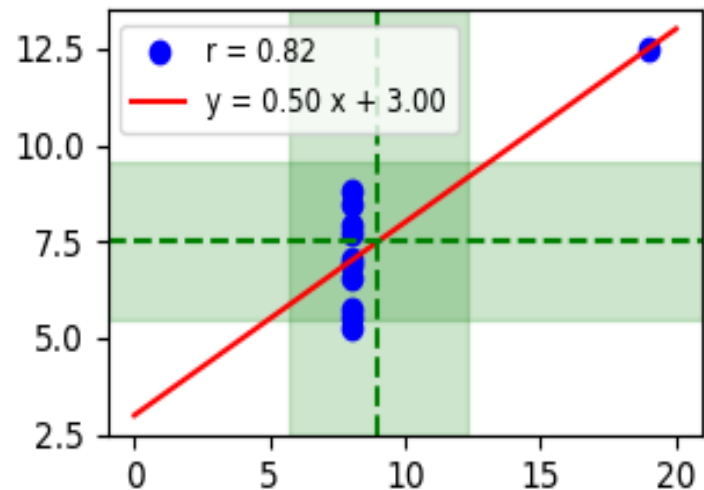


- ❖ The second graph is not distributed normally.
- ❖ Though a relationship between the two variables is obvious, it is not linear.
- ❖ The [Pearson correlation coefficient](#) is not relevant.

Visual representation of the Anscombe's Quartet



- ❖ In the third graph the linear relationship is perfect.
- ❖ But one [outlier](#) which exerts enough influence to lower the correlation coefficient from 1 to 0.816.



- ❖ The fourth example shows another example when one outlier is enough to produce a high correlation coefficient, even though the relationship between the two variables is not linear.

Conclusions from the Anscombe's Quartet

Takeaways from Anscombe's Quartet:

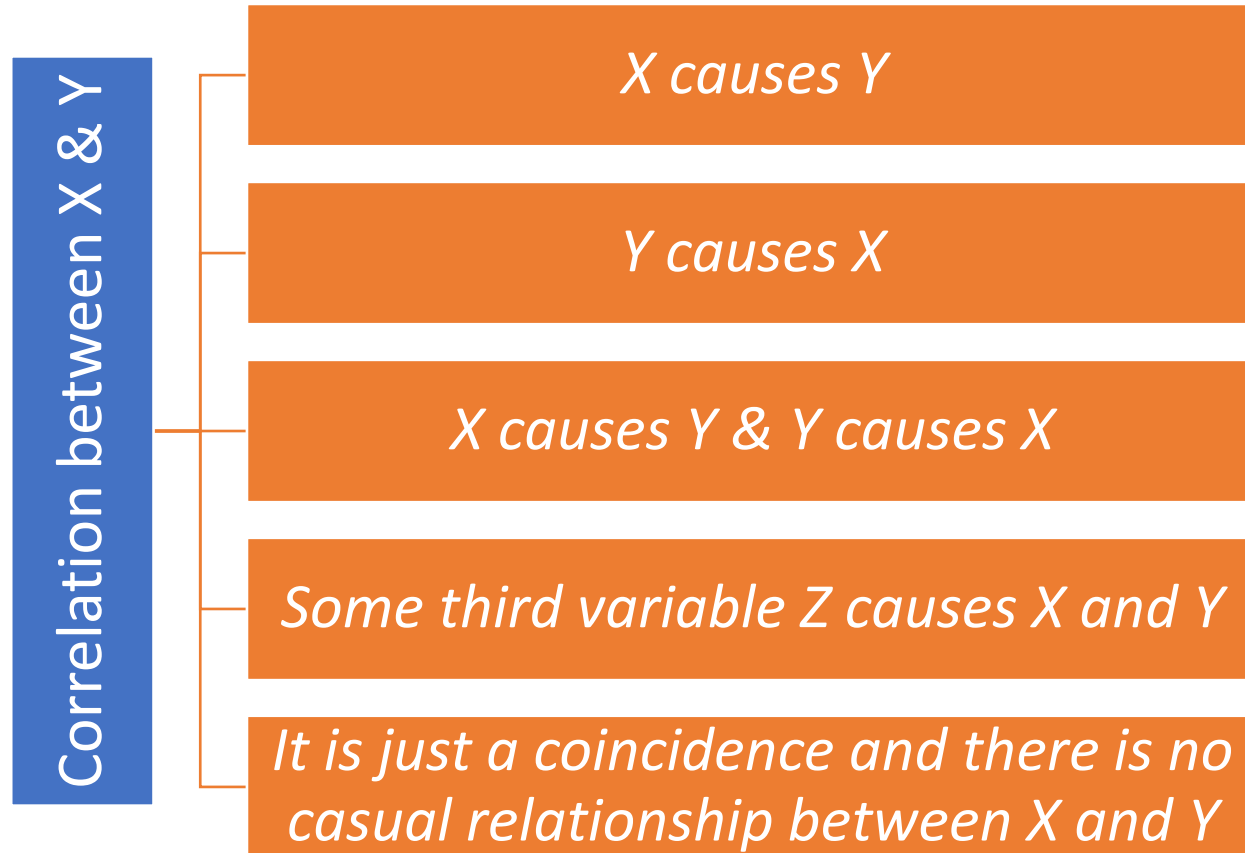
Visualize your data: Graphs reveal patterns that summary statistics can miss, such as nonlinear relationships or the influence of outliers.

Summary statistics aren't enough: The datasets show that datasets with identical statistics can have very different distributions and relationships.

Be cautious with outliers: A single outlier can strongly influence the results of statistical analyses, especially correlation and regression.

Anscombe's Quartet is a classic example used in statistics to highlight the importance of data visualization alongside numerical analysis.

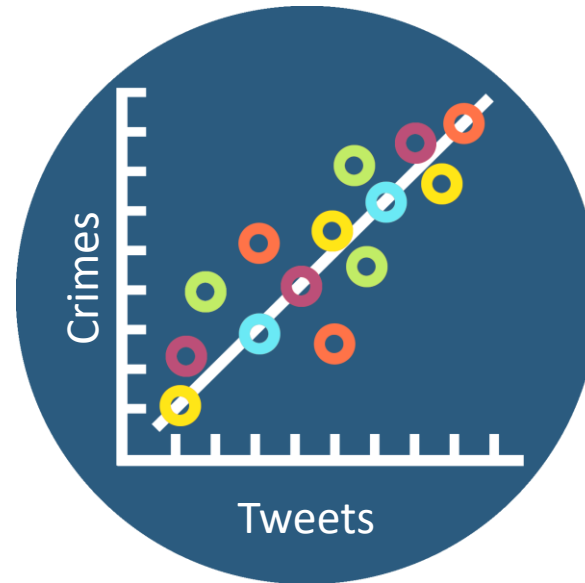
Remark :



Correlation is not causation!!

Example : Relationship between drug related tweets and crime rates (Strong Positive Correlation).

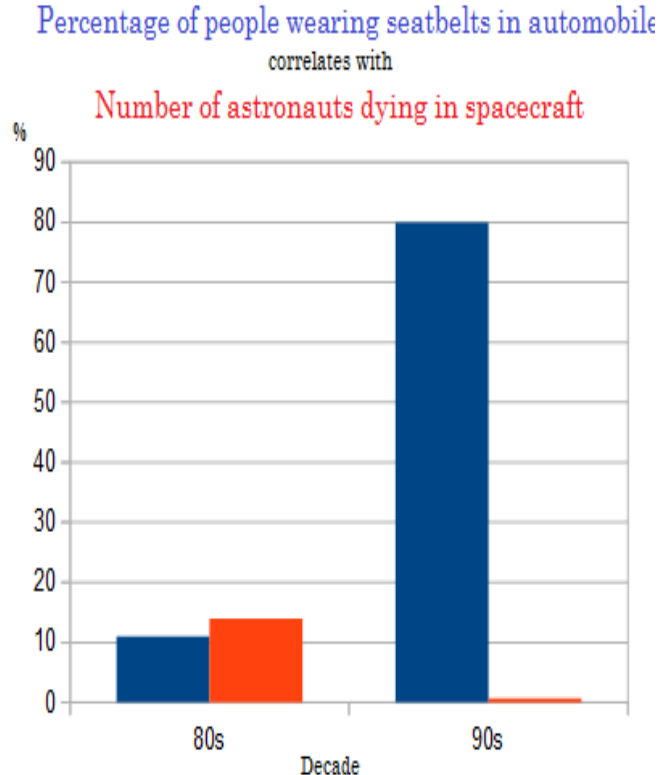
- ❖ A strong Positive relationship between tweets and crime has been found.
- ❖ But there is no evidence to suggest that *tweets are causing more crime* and *tweets about crime do not necessarily reflect the crime rate*.



Reference : The Relationship Between Social Media Data and Crime Rates in the United States Yan Wang¹ , Wenchao Yu¹ , Sam Liu² , and Sean D. Y Social Media + Society January-March 2019: 1–9

Correlation is not causation!!

Example 2. : Relationship between wearing seat belt and astronaut deaths (Strong Positive Correlation).



■ Seat belt use in automobiles
■ Astronaut deaths in spacecraft

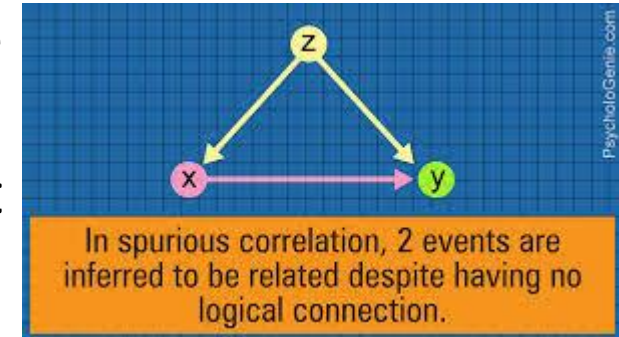


*Use your seatbelt and save
an astronaut life!*

- ❖ The graph shows that an increase in wearing car seat belt results in a lower number of astronaut deaths.
- ❖ Obviously there isn't a real correlation here: putting your seat belt on in a car has nothing to do with the odds of an accident in space.

Confounding Variable

- ❖ Confounding Variable is a variable that influences both the independent variable as well as the dependent variable causing a spurious correlation.
- ❖ This may interfere in your analysis and ruin your experiment by giving useless results.
- ❖ Confounding variables can cause two major problems:
 - Increase variance
 - Introduce bias.
- ❖ A confounding variables are like extra independent variables that are having a hidden effect on your dependent variables.
- ❖ A confounding variable can be what the actual cause of a correlation is, hence any studies must take these into account and find ways of dealing with them.



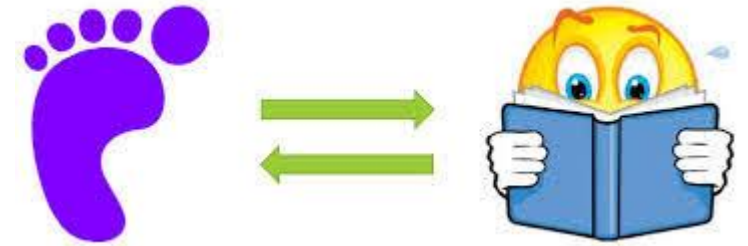
Confounding Variable!!

Example 1. : Relationship between reading ability and shoe size. (Strong Positive Correlation).

- ❖ You collect data on reading ability and shoe size
- ❖ You find that bigger the shoe size the better is the reading ability.
- ❖ Does that mean bigger shoe size leads to better reading abilities?
- ❖ Should children be hence fed growth hormones so that the reading abilities improve?
- ❖ Should children start focusing on their reading abilities to increase their shoe size?

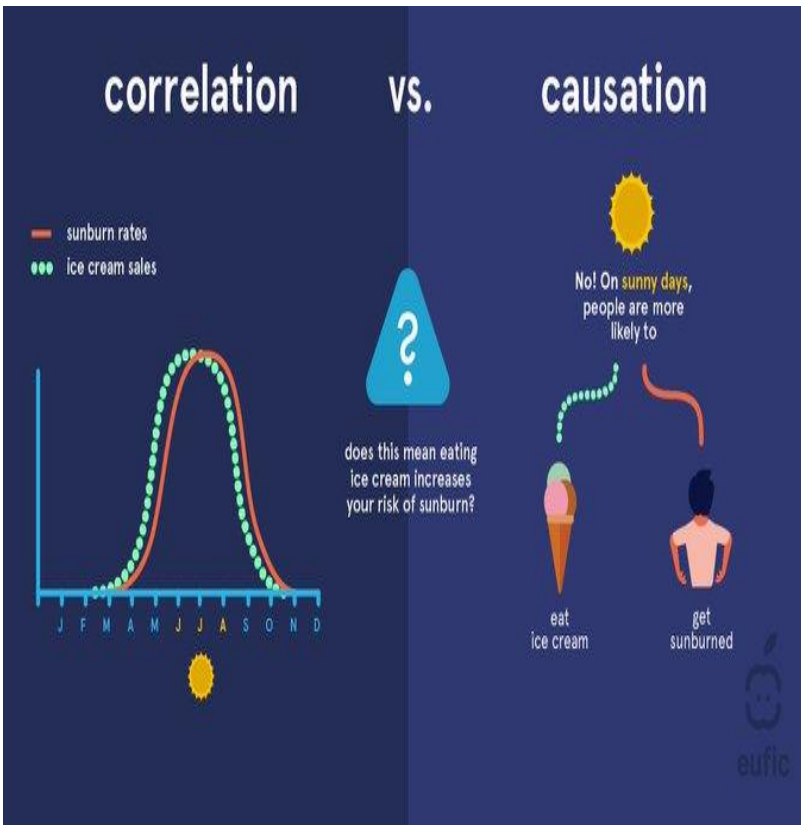
Confounding Variable : There is a third variable—a confounding variable—which causes the increase in both reading ability and shoe size.

Age : As the child's age increase, the foot size increases and also the reading ability increases since the child goes to higher classes.



Confounding Variable!!

Example 2. : Relationship between sun burns and ice – cream consumption. (Strong Positive Correlation).



- ❖ You collect data on sunburns and ice cream consumption.
- ❖ You find that higher ice cream consumption is associated with a higher probability of sunburn.
- ❖ Does that mean,

Possibility #1: Sun burns cause consumption of ice cream.

Possibility #2: Eating ice cream causes sun burns.

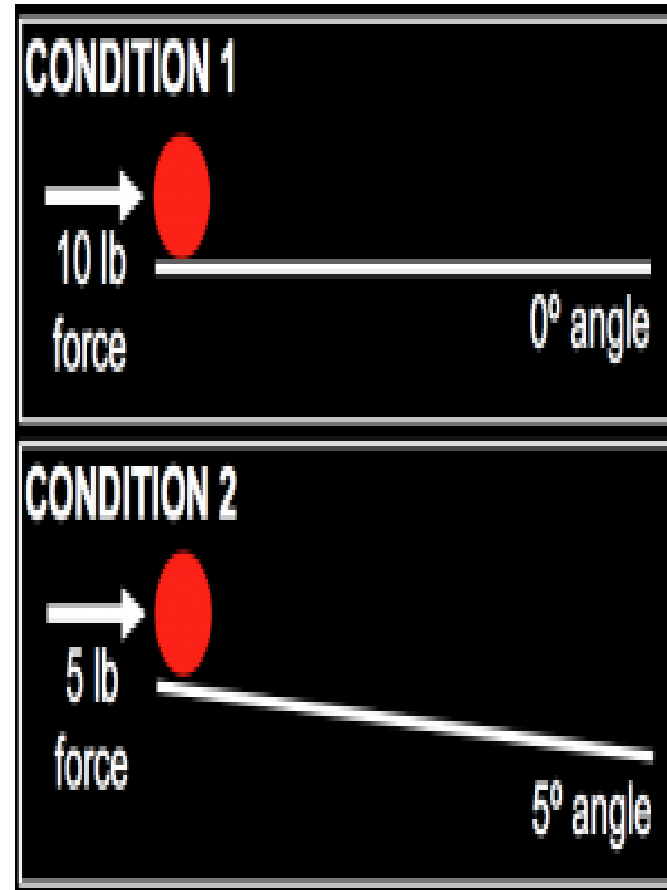
Possibility #3: There is a third variable—a confounding variable—which causes the increase in both ice cream sales and sun burn.

Confounding Variable : Hot temperatures

Confounding Variable!!

Example 3. : Relationship between the force you apply to a ball and the distance the ball travels.

- ❖ Naturally, you predict that the more force you apply, the further the ball will travel.
- ❖ After you run your experiment, you observe that the ball travels further in Condition 2 than it does in Condition 1.
- ❖ In other words, you find that the less force you apply, the further the ball travels.
- ❖ Confounding variable : the angle of the slope.



Example Problem 1.

- An environmental scientist is studying the rate of absorption of a certain chemical into skin.
- She places differing volumes of the chemical on different pieces of skin and allows the skin to remain in contact with the chemical for varying lengths of time.
- She then measures the volume of chemical absorbed into each piece of skin.
- She obtains the results shown in the following table.

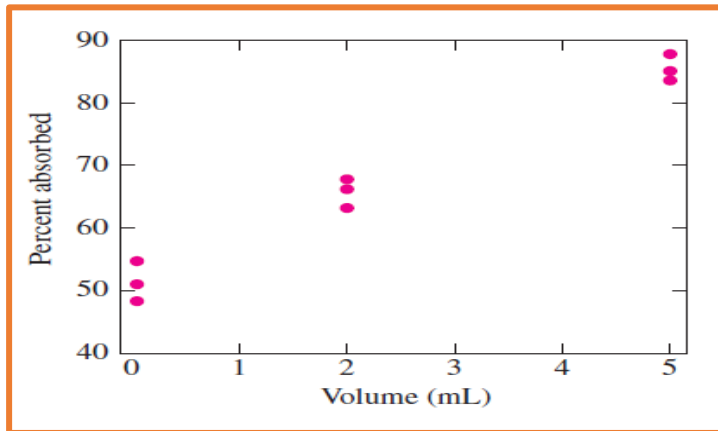
Volume (ml)	Time (h)	Percent Absorbed
0.05	2	48.3
0.05	2	51.0
0.05	2	54.7
2.00	10	63.2
2.00	10	67.8
2.00	10	66.2
5.00	24	83.6
5.00	24	85.1
5.00	24	87.8



Example Problem

Correlation between Volume & Percent Absorbed

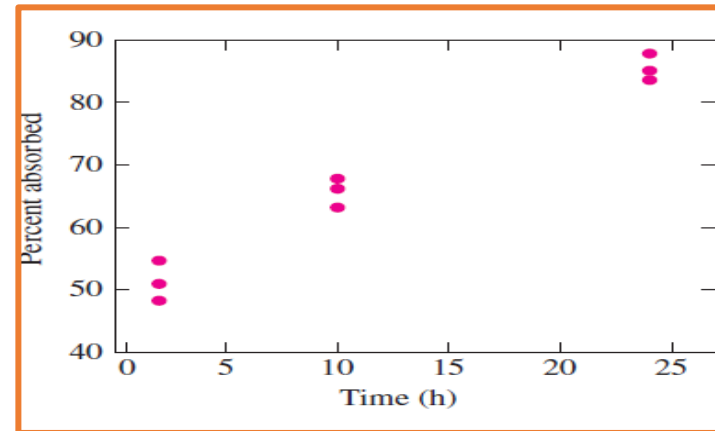
❖ Scatter Plot :



- ❖ Correlation , $r = 0.988$
- ❖ Positive Correlation
- ❖ Increasing the volume causes the percentage absorbed to increase.

Correlation between Time & Percent Absorbed

❖ Scatter Plot :



- ❖ Correlation , $r = 0.987$
- ❖ Positive Correlation
- ❖ Increasing the time that the skin is in contact with chemical causes the percentage absorbed to increase.

Are these conclusions Justified???

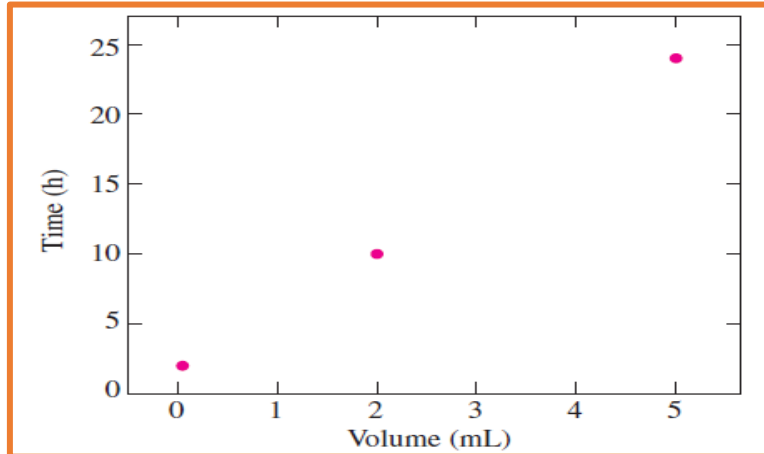


Example Problem

No! The conclusions are not justified!

Suggested Solution :

- ❖ The correlation between time & volume has to be explored.
- ❖ The Scatter plot :



- ❖ The correlation, $r = 0.999$
- ❖ Conclusion : These 2 variables are completely confounded.

Example Problem 2.

- The Scientist in Example Problem 1 has repeated the experiment, this time with a new design.
- The results are presented in the table.

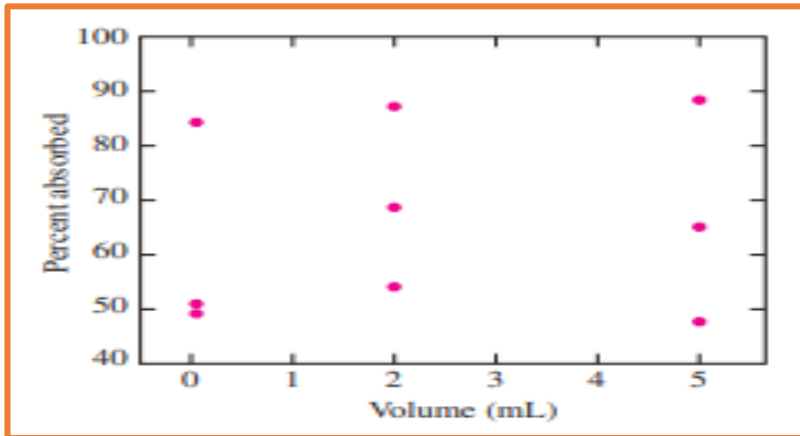
Volume (ml)	Time (h)	Percent Absorbed
0.05	2	49.2
0.05	10	51.0
0.05	24	84.3
2.00	2	54.1
2.00	10	68.7
2.00	24	87.2
5.00	2	47.7
5.00	10	65.1
5.00	24	88.4



Example Problem 2.

Correlation between Volume & Percent Absorbed

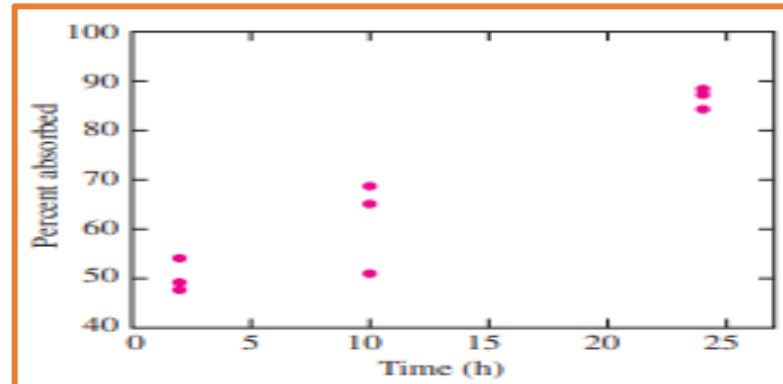
❖ Scatter Plot :



- ❖ Correlation, $r = 0.121$
- ❖ Weak Positive Correlation
- ❖ Hence increase of volume has little or no effect on the percentage absorbed.

Correlation between Time & Percent Absorbed

❖ Scatter Plot :



- ❖ Correlation, $r = 0.952$
- ❖ Strong Positive Correlation
- ❖ Increasing the time that the skin is in contact with the chemical will cause the percentage absorbed to increase.

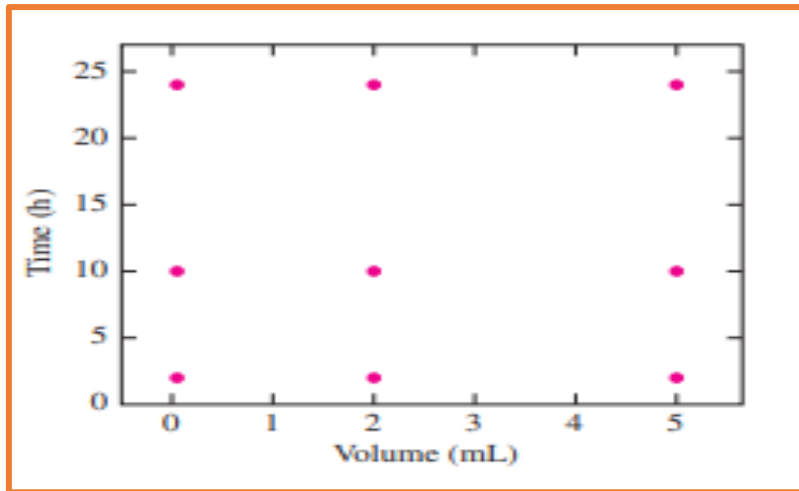
Are these conclusions Justified???



Example Problem 2.

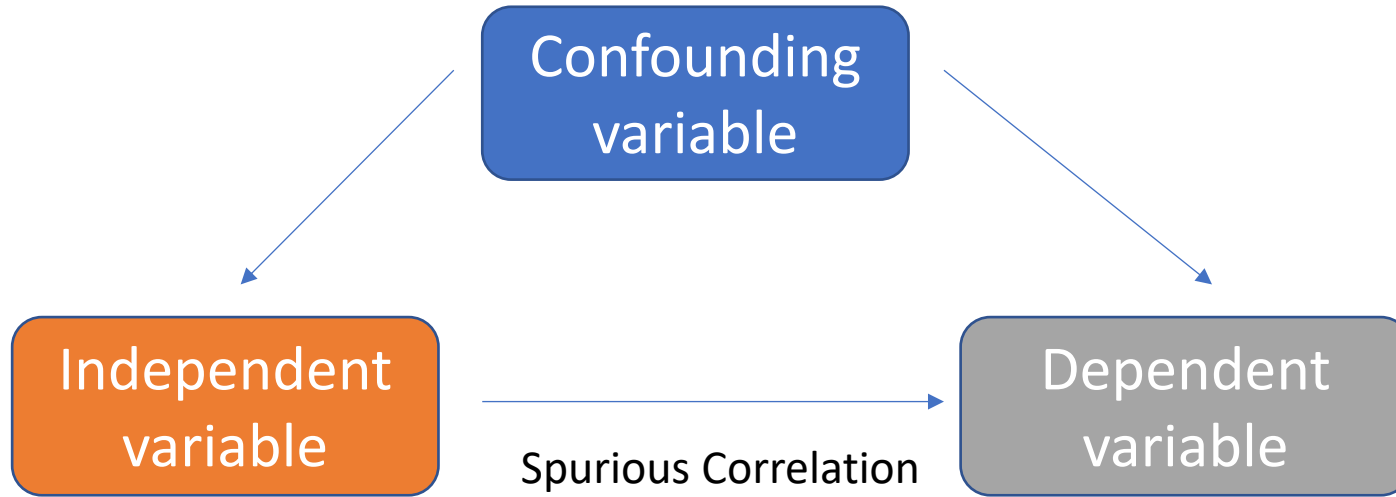
Correlation between Volume & Time

❖ Scatter Plot :



❖ *Time & Volume are not correlated in this case*

Controlled Experiments reduce the risk of Confounding



- ❖ One of the ways by which confounding can be avoided in controlled experiments by choosing values for certain factors in such a way that there exists no correlation between those factors.
- ❖ For instance in Example Problem 1. & 2. the environmental scientist reduced confounding by assigning values to volume and time such that they were uncorrelated.
- ❖ But this is not possible in all cases.

Controlled Experiments reduce the risk of Confounding

- ❖ The values of factors cannot be chosen by the observer in case of observational studies/ experiments.
- ❖ For instance, in studies involving public health issues like impact of environmental pollutants on human health, the observer cannot assign values to any of the factors.
- ❖ Hence it becomes difficult to avoid confounding.
- ❖ Example : People who live in areas with higher level pollutants may tend to have lower socio-economic status, which may affect their health.
- ❖ In observational studies to avoid or reduce confounding the study must be repeated a number of times under a variety of conditions before drawing reliable conclusions !!!

Then how can confounding be avoided in such cases ???

MCQ

An engineer computes the Pearson correlation coefficient between two variables and finds $r=0.95$. After removing one extreme outlier, the correlation drops to $r=0.60$. What does this suggest?

- A. The outlier was artificially inflating the strength of the linear relationship.
- B. The outlier was part of the natural trend and should not have been removed.
- C. The correlation is robust to outliers, so the change is likely due to random variation.
- D. The removal of the outlier improved the accuracy of the correlation coefficient.

Solution A



THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Simple Linear Regression

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Department of Computer Science and Engineering

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Department of Science and Humanities

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Simple Linear Regression: Correlation & Regression Analysis



❖ **Regression Analysis is basically the study of a set of data to make the best guess or some kind of prediction.**

- For Example : By studying a data which provides information of how much you eat and how much you weigh, you can conclude that there exists a relationship between the two.
- Regression analysis can help you to quantify that and can help you to predict how much you will weigh in 10 years time if you continue to put on weight at the same rate.

Impact of Global warming :

- ❖ Increase in rainfall resulting in Floods
- ❖ Increase in amount of dry land leading to droughts



Impact of Global Warming

Global Warming in Wet areas

Evaporation of water from land and sea

More rainfall

Increase in Floods

Global Warming in Dry areas

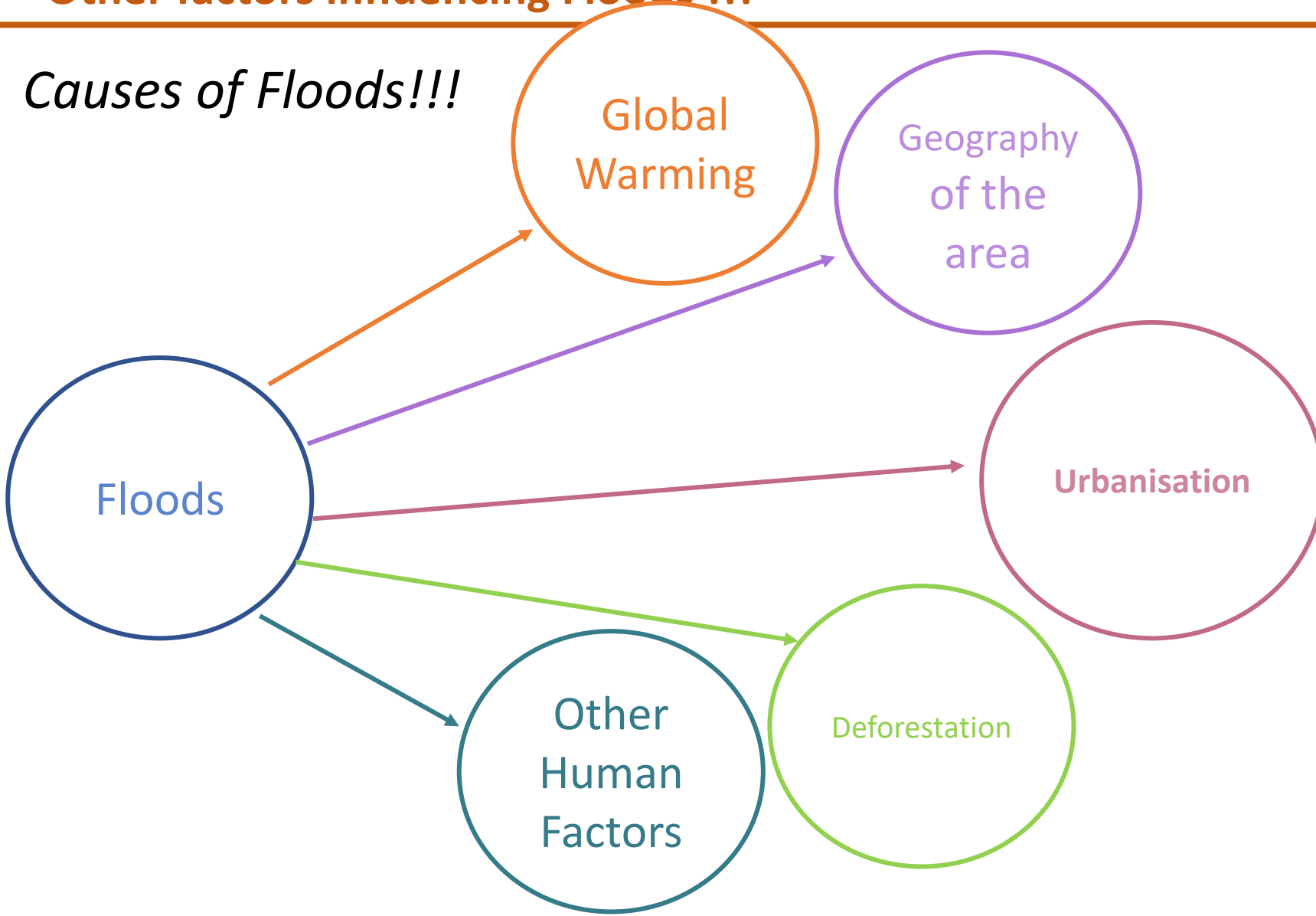
Increase in evaporation of water from land , water surfaces and plants

Dry areas become drier

Increase in droughts

Other factors influencing Floods !!!

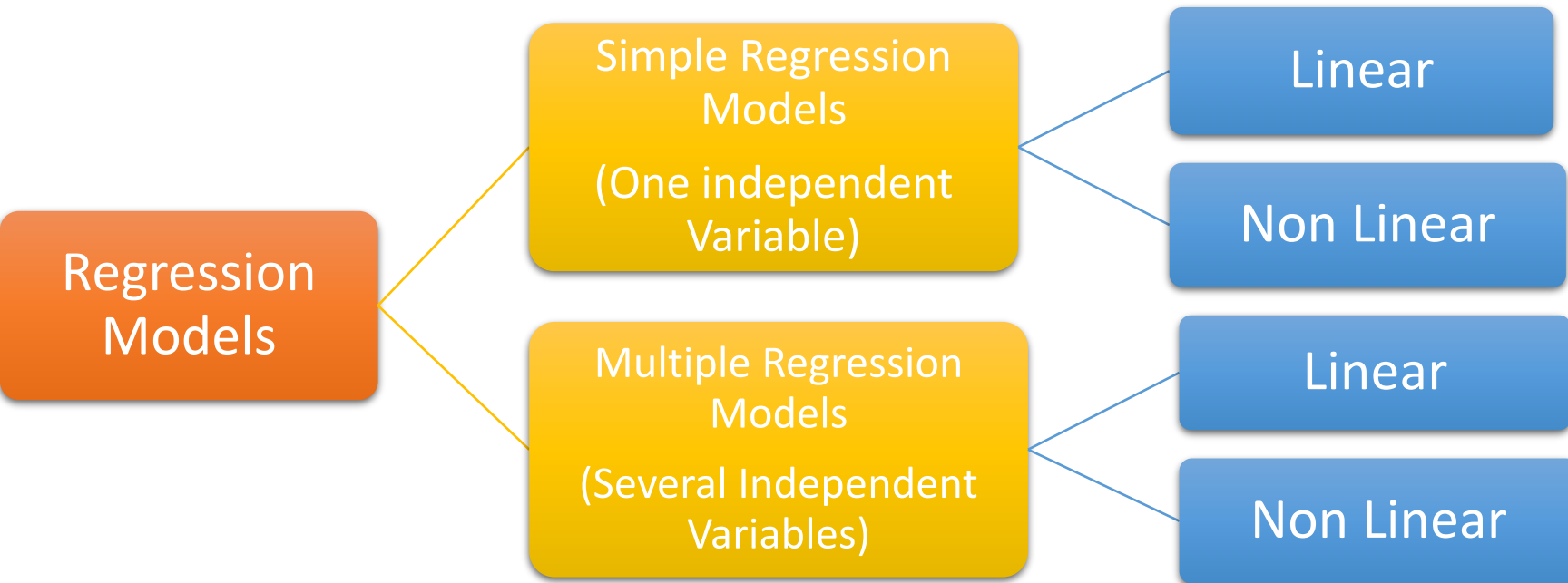
Causes of Floods!!!



Regression Analysis

- ❖ In statistical modeling, **regression analysis** is a set of statistical processes for estimating the relationships between a dependent variable and one or more independent variables.
- ❖ It is a way of mathematically sorting out which of those variables indeed have an impact
- ❖ Which factors matter most ?
- ❖ Which can we ignore ?
- ❖ How do the factors interact with each other?
- ❖ And most importantly, how certain are we about all these factors?

Regression Analysis – A Broad Classification

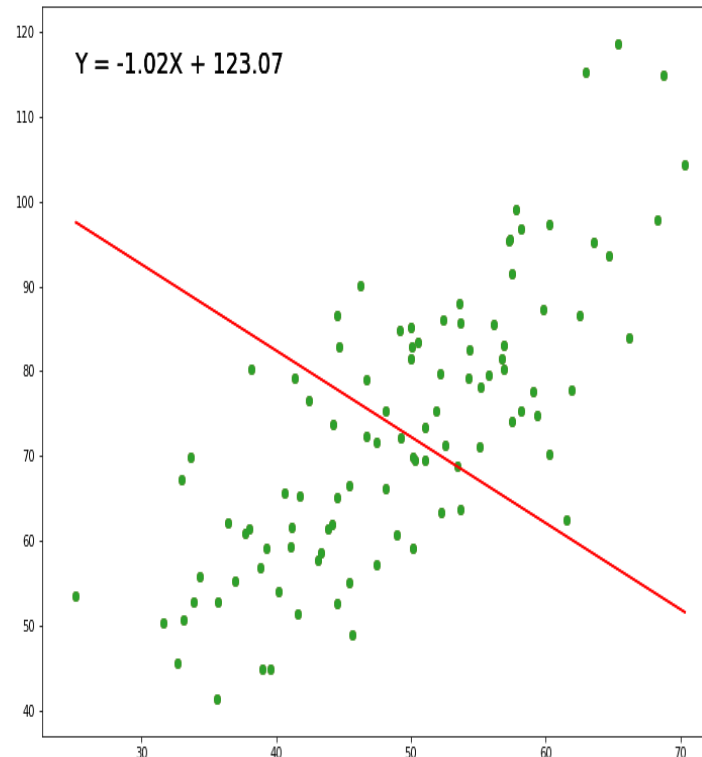


Some Inputs !!!

- ❖ Regression analysis is widely used for **prediction** and **forecasting**, where its use has substantial overlap with the field of **machine learning**.
- ❖ In some situations regression analysis can be used to infer **causal relationships** between the independent and dependent variables.
- ❖ The term "regression" was coined by **Francis Galton** in the nineteenth century to describe a biological phenomenon.
- ❖ The earliest form of regression analysis is **linear regression**, in which a researcher finds the line that most closely fits the data according to a specific mathematical criterion.
- ❖ This line is referred to as the line of **least squares**, which was published by **Legendre** in 1805, and by **Gauss** in 1809.

The Least – Squares Line

- ❖ When two variables have a linear relationship, the scatter plot tends to be clustered around a straight line.
- ❖ This line is referred to as the Least Squares Line.



The least squares line

- ❖ Consider the least square line given by,

$$y_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$$

where,

- $\widehat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- $\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$

Example :

❖ The details pertaining to the no. of hours spent by students in preparing for an entrance exam and the marks scored (on a scale of 0 – 100) is provided in the following table.

Using these values,

- Estimate the marks scored by a student who has spent 2.35 hours.
- Predict the marks that a student can score if he/she invests 20 hours.

SL No.	No. of hours spent	Marks Scored
1	6	82
2	10	88
3	2	56
4	4	64
5	6	77
6	7	92
7	0	23
8	1	41
9	8	80
10	5	59
11	3	47

Computing the least squares line

❖ We need to first obtain the least square line which is given by,

$$y = \hat{\beta}_0 + \hat{\beta}_1 x$$

- $\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
 ↓ ↘
 mean of y mean of x

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Example :

SL No.	No. of hours spent (x)	Marks Scored(y)	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
1	6	82	1.27	1.6129	17.55	22.33
2	10	88	5.27	27.7729	23.55	124.15
3	2	56	-2.73	7.4529	-8.45	23.06
4	4	64	-0.73	0.5329	-0.45	0.33
5	6	77	1.27	1.6129	12.55	15.97
6	7	92	2.27	5.1529	27.55	62.60
7	0	23	-4.73	22.3729	-41.45	195.97
8	1	41	-3.73	13.9129	-23.45	87.42
9	8	80	3.37	11.3569	15.55	50.88
10	5	59	0.27	0.0729	-5.45	-1.49
11	3	47	-1.73	2.9929	-17.45	30.15

MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Example :

SL No.	No. of hours spent (x)	Marks Scored(y)	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
1	6	82	1.27	1.6129	17.55	22.33
2	10	88	5.27	27.7729	23.55	124.15
3	2	56	-2.73	7.4529	-8.45	23.06
4	4	64	-0.73	0.5329	-0.45	0.33
5	6	77	1.27	1.6129	12.55	15.97
6	7	92	2.27	5.1529	27.55	62.60
7	0	23	-4.73	22.3729	-41.45	195.97
8	1	41	-3.73	13.9129	-23.45	87.42
9	8	80	3.37	11.3569	15.55	50.88
10	5	59	0.27	0.0729	-5.45	-1.49
11	3	47	-1.73	2.9929	-17.45	30.15
4.73 64.45				94.8459		611.37

Example :

From the table we have,

$$\bar{x} = 4.73 \quad ; \quad \bar{y} = 64.45$$

- $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = 611.37$
- $\sum_{i=1}^n (x_i - \bar{x})^2 = 94.8459$
- $\widehat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = 611.37 / 94.8459 = 6.49$
- $\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x} = 64.45 - [6.49 \times 4.73] = 33.7523$
- The equation of the least squares line is given by,
 $y_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i \Rightarrow 33.7523 + 6.79x$

Example :

- The equation of the least squares line is given by,
 $y = 33.7523 + 6.49x$
- i. To estimate the marks scored by a student who has spent 2.35 hours.

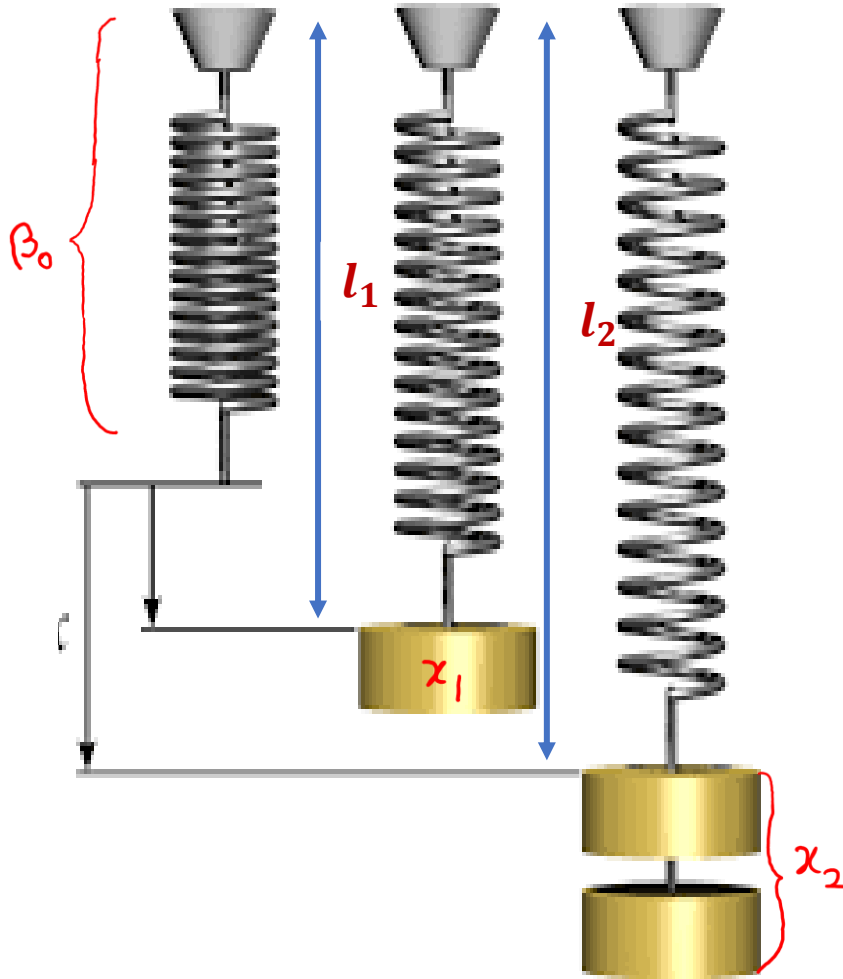
$$Y = 33.7523 + [6.35 \times 2.35] = 48.6748$$

- ii. To predict the marks that a student can score if he/she invests 20 hours.

$$Y = 33.7523 + [6.35 \times 20] = 163.5523$$

- ❖ How to compute the Least Squares Line
- ❖ Residuals and Errors
- ❖ Measuring Goodness of fit

How to compute the Least – Squares Line ???



Weights = x_i

Corresponding length = l_i

$$\begin{aligned}\therefore l_i &= \text{actual length of the spring} + \text{stretched length} \\ &= \beta_0 + \text{stretched length}\end{aligned}$$

Hooke's Law:

strain (deformation of spring) $\propto$ stress applied

$\Rightarrow$ stretched length $\propto$ weight applied

$= kx_i \rightarrow$ proportionality constant

$= \beta_1 x_i \rightarrow$ spring constant

$$\therefore \text{Length } l_i = \beta_0 + \beta_1 x_i \rightarrow (1)$$

How to compute the Least – Squares Line ???

Therefore, when weight x_i is applied, the corresponding length,

$$l_i = \beta_0 + \beta_1 x_i \longrightarrow (1)$$

- $\beta_0 \rightarrow$ initial length of the spring
- $\beta_1 \rightarrow$ spring constant
- $l_i \rightarrow$ true length

Observed / measured length $\neq$ true length due to errors.

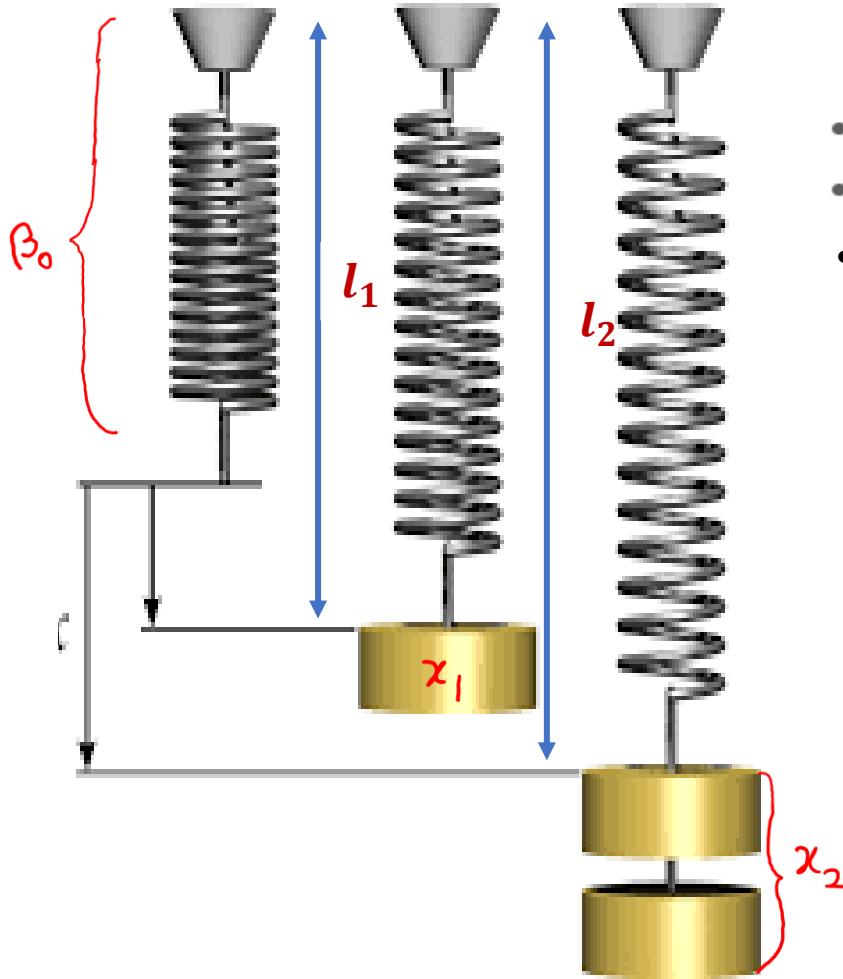
Let y_i = observed length corresponding to the weight x_i :

$$y_i = \text{true length} + \text{some error}$$

$$= l_i + \varepsilon_i$$

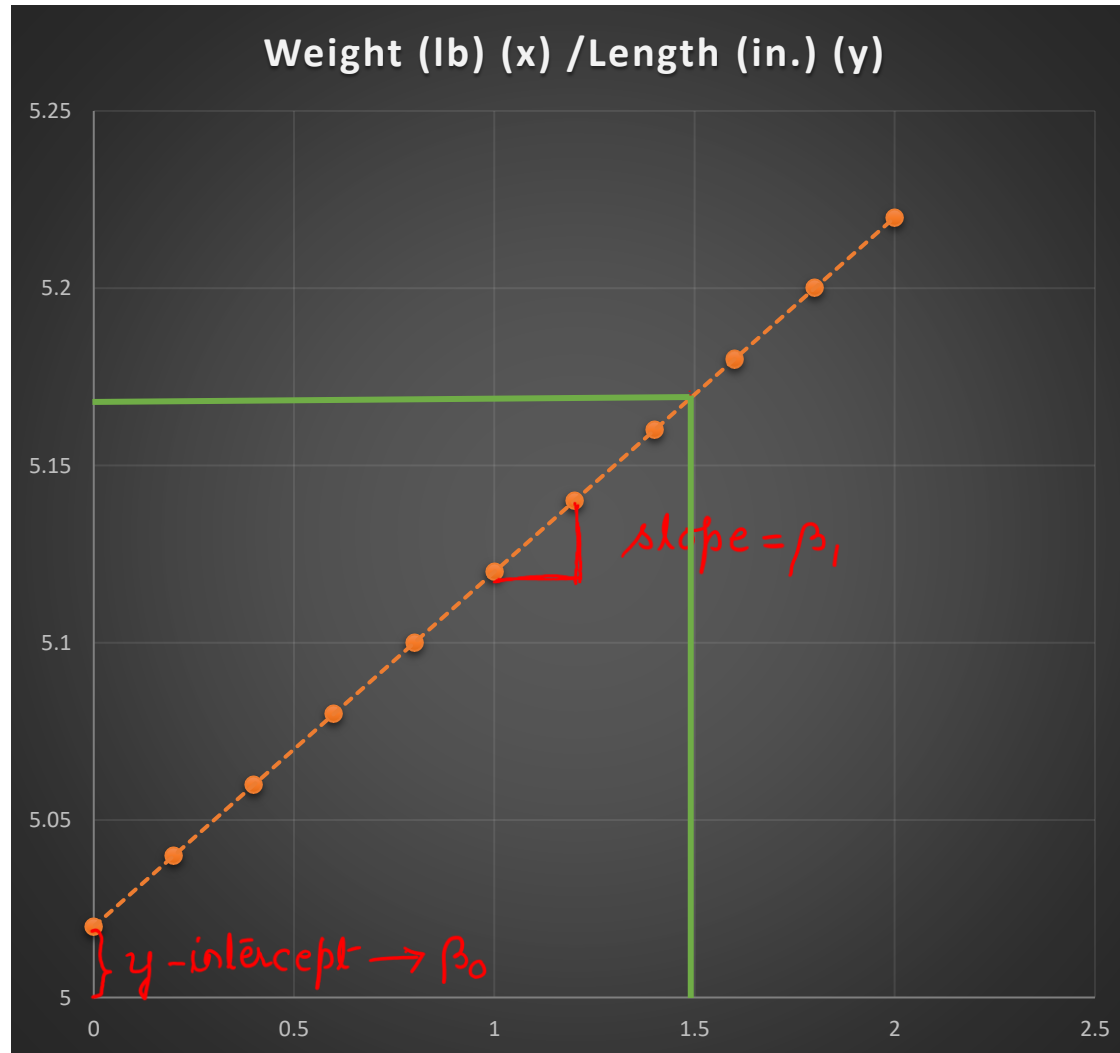
$$\therefore y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \longrightarrow (2) \quad \text{Linear Model}$$

- y_i : dependent variable
- x_i : independent variable
- ε_i : error corresponding to weight x_i
- β_0, β_1 : regression coefficients



Scenario # 1 : No Errors!!

Weight (<i>lb</i>) (<i>x</i>)	Length (<i>in.</i>) (<i>y</i>)
0.0	5.02
0.2	5.04
0.4	5.06
0.6	5.08
0.8	5.10
1.0	5.12
1.2	5.14
1.4	5.16
1.6	5.18
1.8	5.20
2.0	5.22



Observed / Measured length
= true length

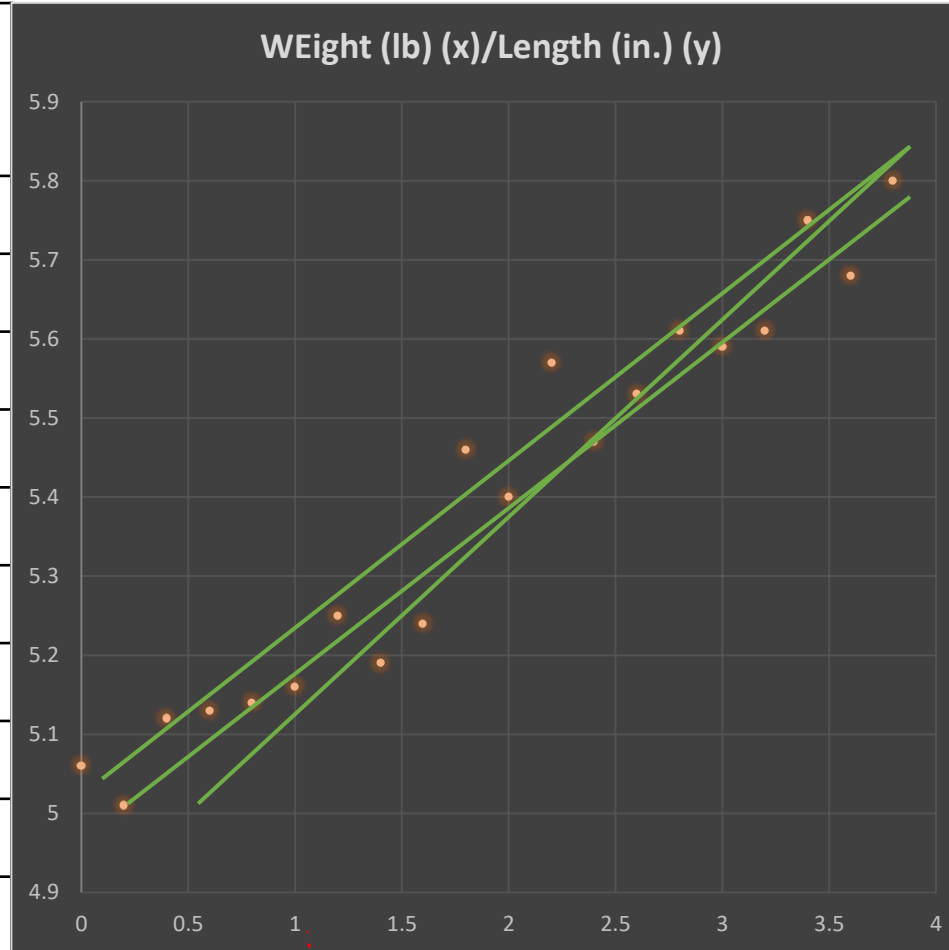
$$\therefore y_i = l_i$$

$$y_i = \beta_0 + \beta_1 x_i$$

- $\beta_0 \rightarrow y\text{-intercept}$
- $\beta_1 \rightarrow \text{slope}$

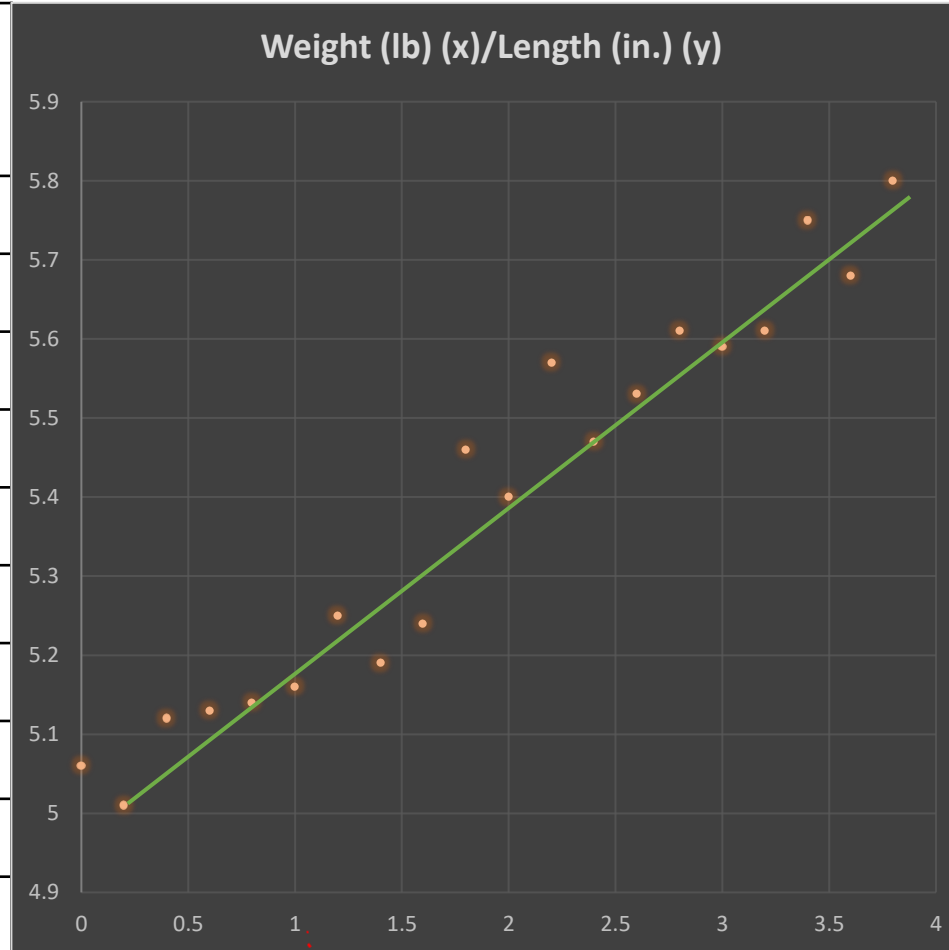
Scenario #2 : Measurement has Errors!!

Weight (<i>lb</i>) (<i>x</i>)	Length (<i>in.</i>) (<i>y</i>)	Weight (<i>lb</i>) (<i>x</i>)	Length (<i>in.</i>) (<i>y</i>)
0.0	5.06	2.0	5.40
0.2	5.01	2.2	5.57
0.4	5.12	2.4	5.47
0.6	5.13	2.6	5.53
0.8	5.14	2.8	5.61
1.0	5.16	3.0	5.59
1.2	5.25	3.2	5.61
1.4	5.19	3.4	5.75
1.6	5.24	3.6	5.68
1.8	5.46	3.8	5.80



Scenario #2 : Measurement has Errors!!

Weight (lb) (x)	Length (in.) (y)	Weight (lb) (x)	Length (in.) (y)
0.0	5.06	2.0	5.40
0.2	5.01	2.2	5.57
0.4	5.12	2.4	5.47
0.6	5.13	2.6	5.53
0.8	5.14	2.8	5.61
1.0	5.16	3.0	5.59
1.2	5.25	3.2	5.61
1.4	5.19	3.4	5.75
1.6	5.24	3.6	5.68
1.8	5.46	3.8	5.80

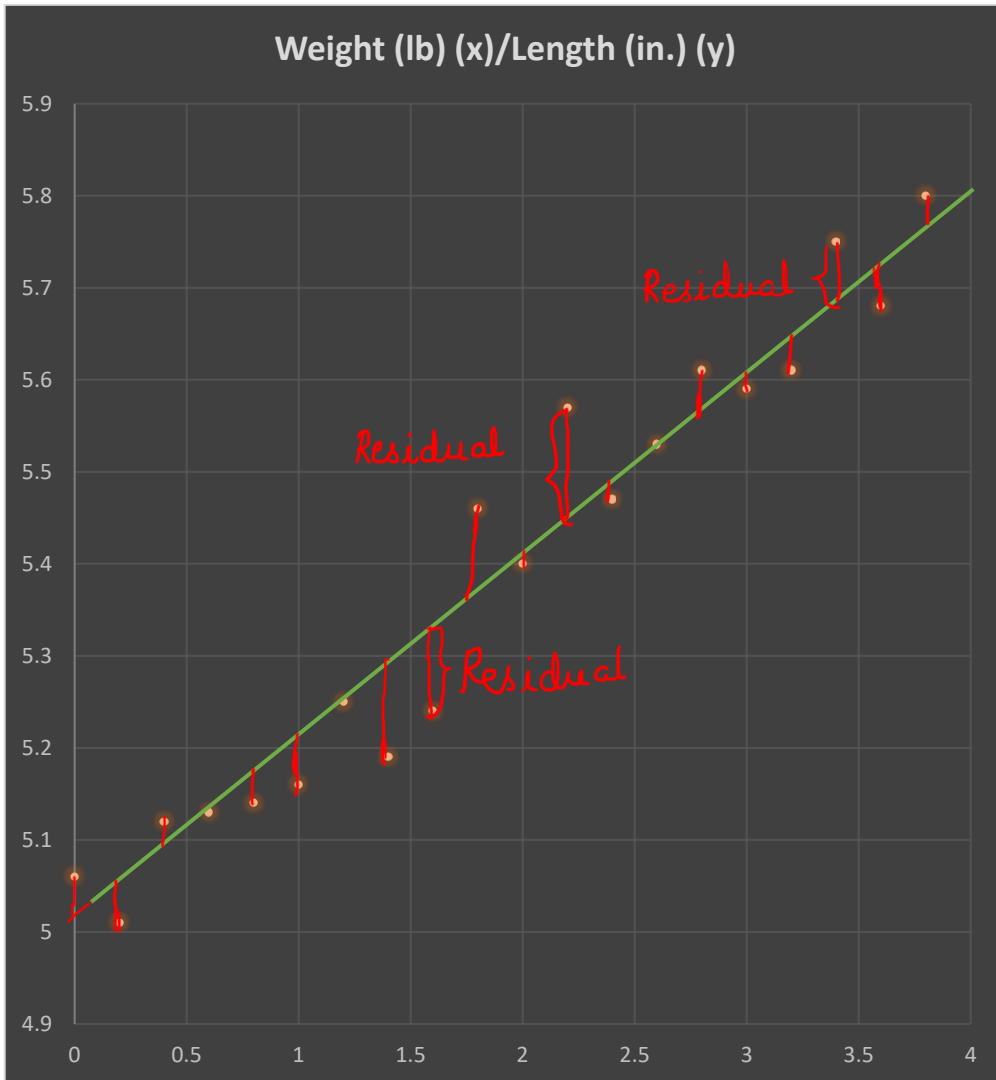


$$\sum_{i=1}^n e_i^2 \rightarrow \text{minimum}$$



Residual

Residual :



$$\begin{aligned} \diamond e_i &= y_{\text{observed}} - y_{\text{predicted}} \\ &= y_i - \hat{y}_i \end{aligned}$$

$$= y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

$e_i \rightarrow$ Residual associated with point (x_i, y_i)

Observed values of y
 $\rightarrow y$ values given in data

Predicted values of y
 $\rightarrow y$ values on the least-square line

Examples:

1. For $x = 1, y = 5.16$ and $\hat{y} = 5.21$

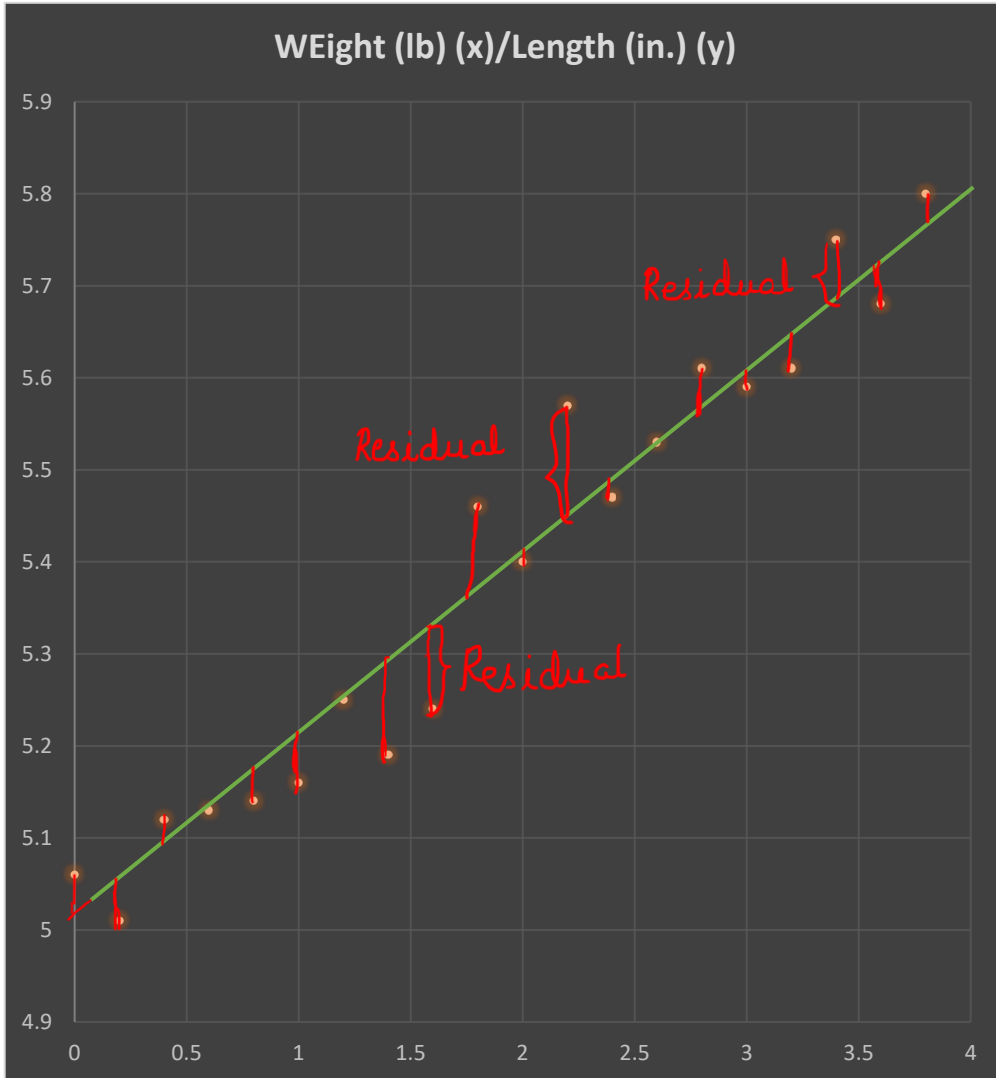
$$\begin{aligned} \therefore \text{Residual} &= y - \hat{y} \\ &= 5.16 - 5.21 \\ &= -0.05 \end{aligned}$$

2. For $x = 3.8, y = 5.80$ and $\hat{y} = 5.78$

$$\begin{aligned} \therefore \text{Residual} &= y - \hat{y} \\ &= 5.80 - 5.78 \\ &= 0.02 \end{aligned}$$

Least Square Line :

WEight (lb) (x)/Length (in.) (y)



NOTE : The least square line is defined to be the line for which the sum of squared residuals is minimum.

❖ *That is, it is the line for which $\sum_{i=1}^n e_i^2$ is minimum.*

$$\therefore \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

❖ *Using some Mathematical computations it can be shown that,*

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Least Squares Line : Summary

Scenario #1 : If there is no measurement error then the data points lie on the straight line

$y = \beta_0 + \beta_1 x$ and values of β_0 and β_1 can be obtained easily by calculating the slope and the intercept.

Scenario #2 : If there is a measurement error ε_i , then

- ❖ the exact value of β_0 and β_1 cannot be determined
- ❖ the values of β_0 and β_1 are computed by calculating the least square line.
- ❖ The least square line is given by $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

where

- $\hat{\beta}_0 \rightarrow$ the y – intercept of the least square line
 $\rightarrow$ gives an estimate of β_0 , the initial length of the spring.
- $\hat{\beta}_1 \rightarrow$ the slope of the least square line
 $\rightarrow$ gives an estimate of the actual value of the spring constant β_1 .

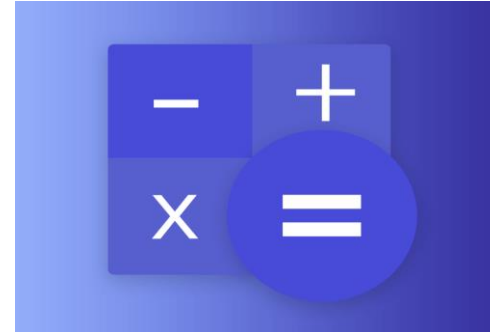
Computing formulas

Remark :

$$\diamond \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$$

$$\diamond \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

$$\diamond \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2$$



For computational purposes we use the equivalent formula that is specified in the RHS.

Try This !!!

Using the Hooke's law data given in the table

- Compute the least squares estimates of the spring constant and the unloaded length of the spring.
- Write the equation of the least squares line.
- Estimate the length of the spring under a load of 1.3 lb.
- Estimate the length of the spring under a load of 1.4 lb.
- Obtain the Residuals corresponding to all the points (x_i, y_i) .

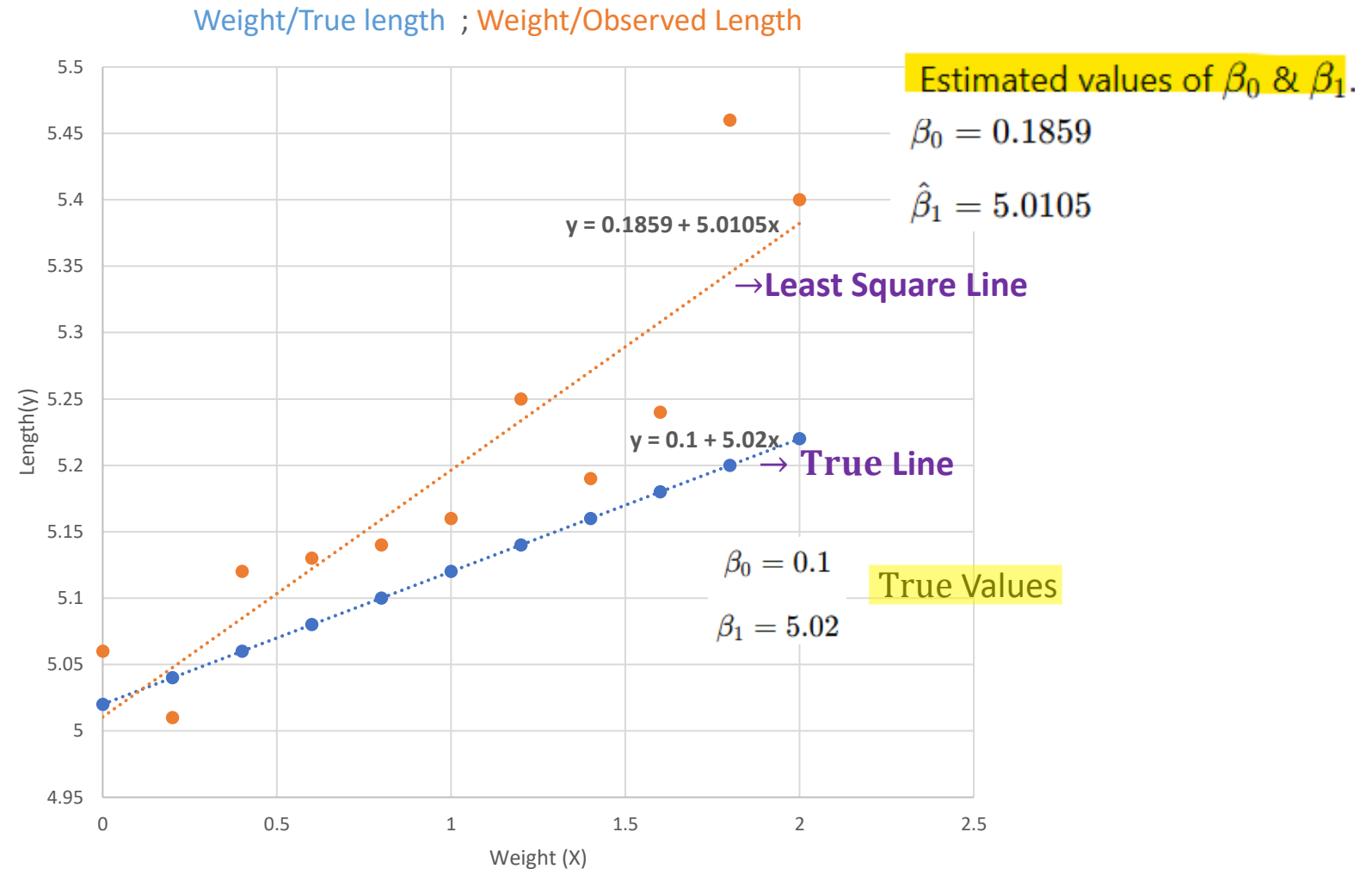
Weight (lb) (x)	Length (in.) (y)	Weight (lb) (x)	Length (in.) (y)
0.0	5.06	2.0	5.40
0.2	5.01	2.2	5.57
0.4	5.12	2.4	5.47
0.6	5.13	2.6	5.53
0.8	5.14	2.8	5.61
1.0	5.16	3.0	5.59
1.2	5.25	3.2	5.61
1.4	5.19	3.4	5.75
1.6	5.24	3.6	5.68
1.8	5.46	3.8	5.80

Some Observations :

- ❖ The Estimates are not the same as true values
- ❖ The Residuals are not the same as the Errors.
- ❖ Don't extrapolate outside the range of the data.
- ❖ Don't use the Least Squares line when the data aren't linear.

The Estimates are not the same as true values

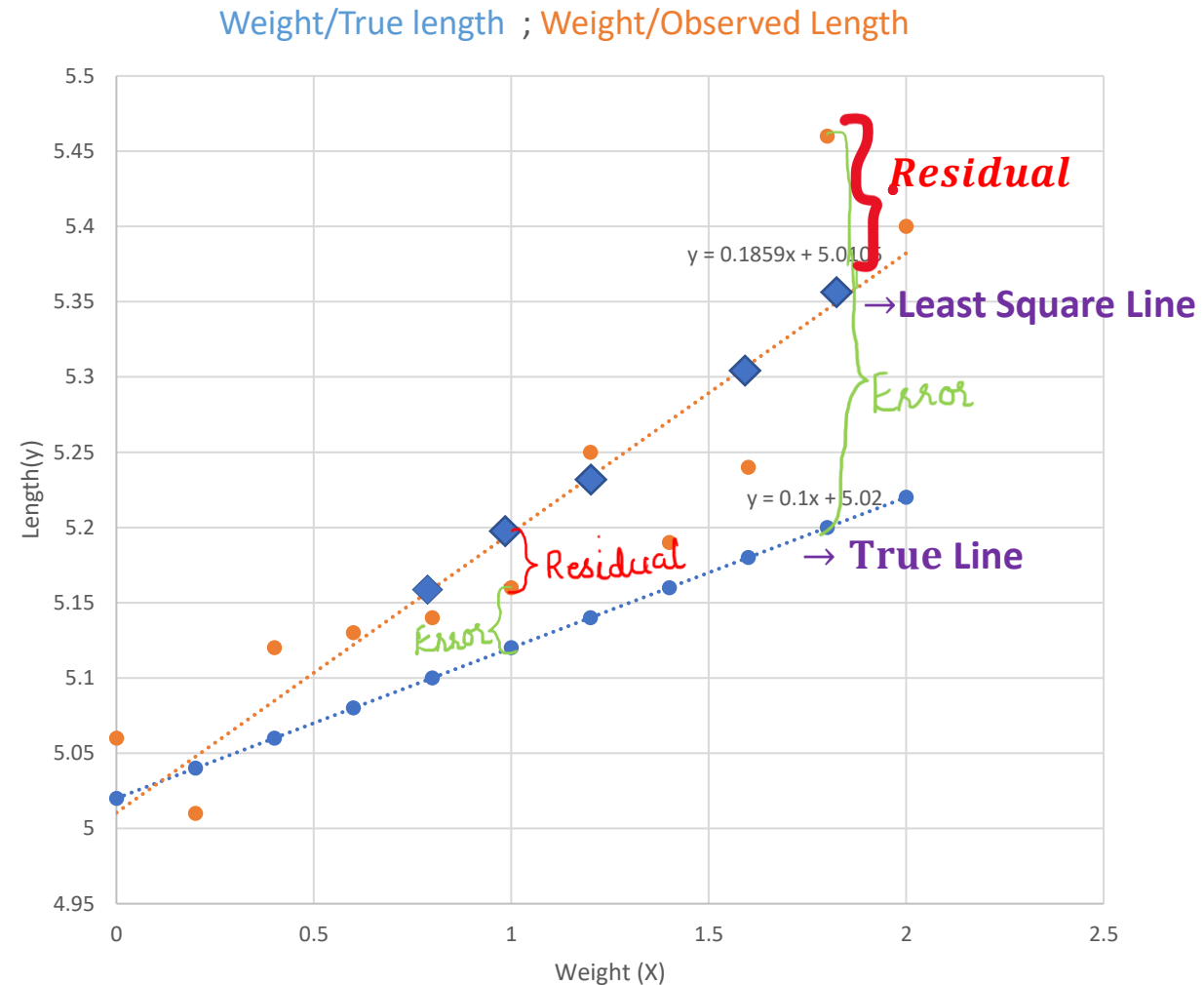
Weight (lb) (x)	True Length (in.) (y)	Length (in.) (y)
0.0	5.02	5.06
0.2	5.04	5.01
0.4	5.06	5.12
0.6	5.08	5.13
0.8	5.10	5.14
1.0	5.12	5.16
1.2	5.14	5.25
1.4	5.16	5.19
1.6	5.18	5.24
1.8	5.20	5.46
2.0	5.22	5.40



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The Residuals are not the same as Errors

Weight (lb) (x)	Length (in.) (y)	Length (in.) (y)
0.0	5.02	5.06
0.2	5.04	5.01
0.4	5.06	5.12
0.6	5.08	5.13
0.8	5.10	5.14
1.0	5.12	5.16
1.2	5.14	5.25
1.4	5.16	5.19
1.6	5.18	5.24
1.8	5.20	5.46
2.0	5.22	5.40



Don't Extrapolate outside the range of the data!!

Eg. 1: The details pertaining to the no. of hours spent by students in preparing for an entrance exam and the marks scored (on a scale of 0 – 100) is provided in the following table.

Using these values,

i. Estimate the marks scored by a student who has spent 2.35 hours.

48.6748

ii. Predict the marks that a student can score if he/she invests 20 hours.

163.5523

SL No.	No. of hours spent	Marks Scored
1	6	82
2	10	88
3	2	56
4	4	64
5	6	77
6	7	92
7	0	23
8	1	41
9	8	80
10	5	59
11	3	47

Don't Extrapolate outside the range of the data!!

Eg. 2:

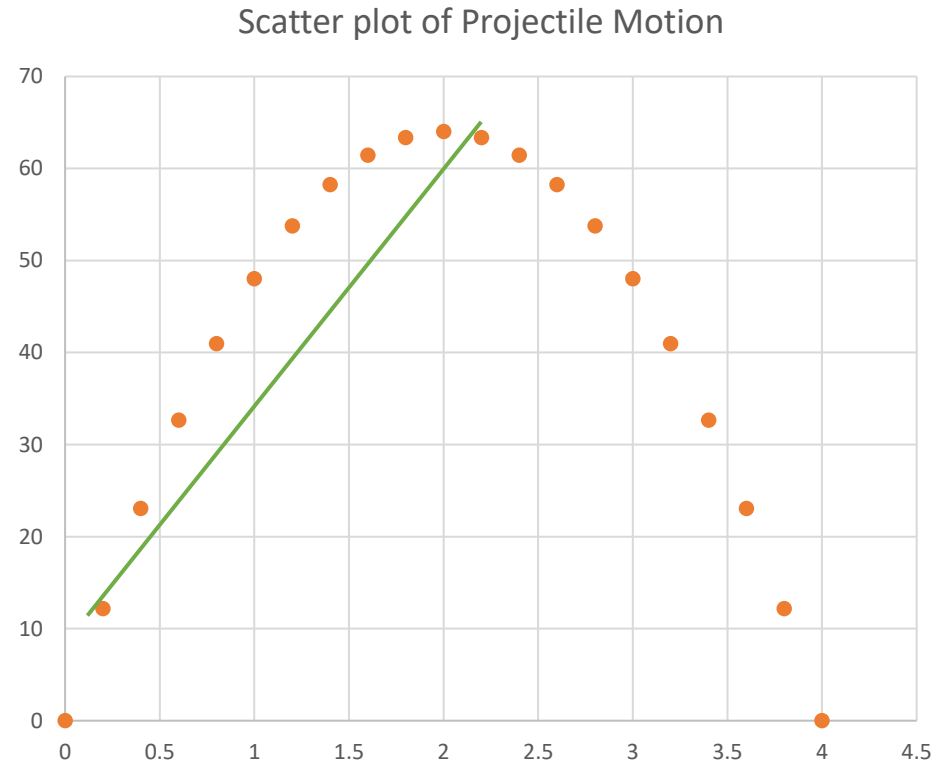
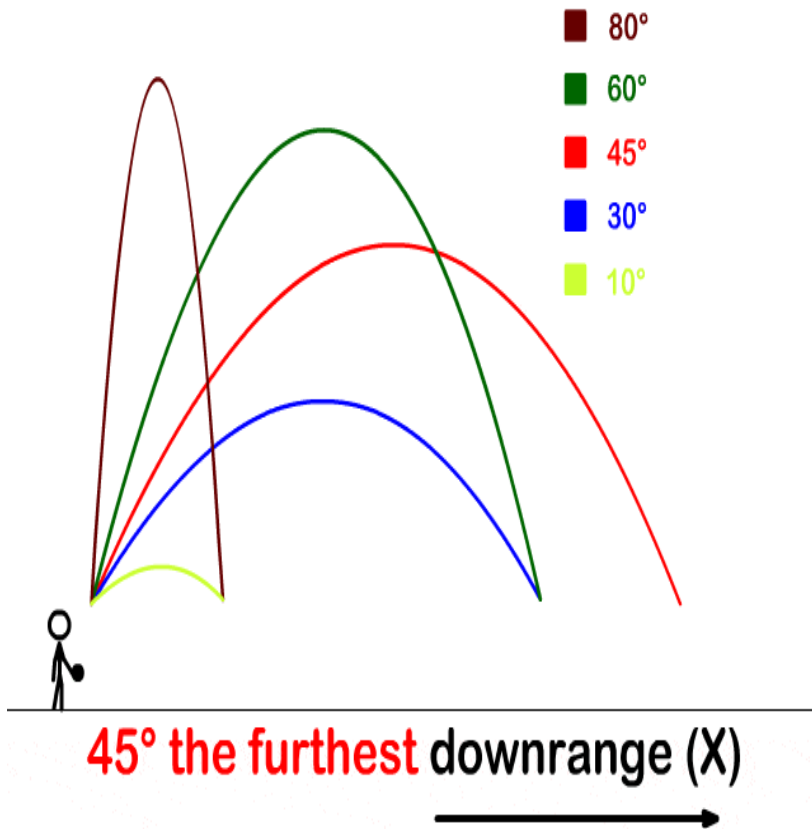
Least square line: $y=0.2046x+4.997$

For weight, $x=100\text{lb}$

Length $y=0.2046x100+4.997$
 $=25.46\text{in}$

Weight (lb) (x)	Length (in.) (y)	Weight (lb) (x)	Length (in.) (y)
0.0	5.06	2.0	5.40
0.2	5.01	2.2	5.57
0.4	5.12	2.4	5.47
0.6	5.13	2.6	5.53
0.8	5.14	2.8	5.61
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1.4	5.19	3.4	5.75
1.6	5.24	3.6	5.68
1.8	5.46	3.8	5.80

Don't use the Least Squares Line when the data aren't linear



Note : In some cases the Least – Squares line can be used for *non linear data*, but only after *variable transformation* is applied.

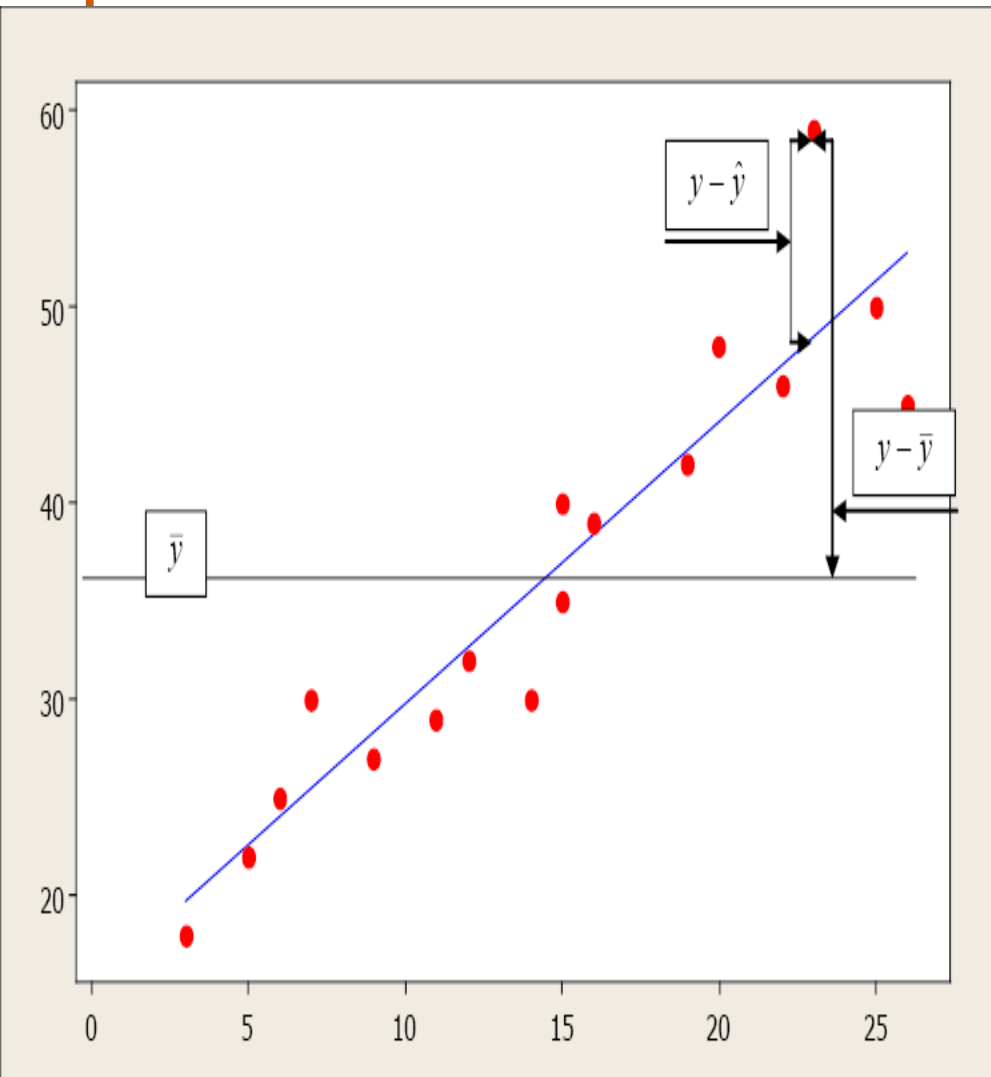
Measuring goodness of fit

- ❖ A goodness of fit statistic is a quantity that measures how well a model explains a given set of data.
- ❖ A linear model fits well if there is a strong relationship between the variables involved.
- ❖ The strength of a linear relationship can be measured by considering,

$$\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

- ❖ The above relation is also referred to as a goodness-of-fit statistic.
- ❖ The draw back of this statistic relation is that it cannot be used to compare the goodness-of-fit of two models which have different data set. (That is, data sets having different units)
- ❖ Hence we use the relation, $r^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$
which is obtained by using the correlation coefficient.
- ❖ This is also referred to as the ***co-efficient of determination***.

Visualisation of r^2



$y_i - \hat{y}_i$: distance of (x_i, y_i) from the least square line.

$y_i - \bar{y}$: distance of (x_i, y_i) from the line $y = \bar{y}$.

$(y_i - \hat{y}_i)^2$: gives the overall spread of the data around the least square line.

$(y_i - \bar{y})^2$: gives the overall spread of the data around the line $y = \bar{y}$.

- The horizontal line represents the **mean of all observed y -values**, denoted by:

$$\bar{y} = \frac{1}{n} \sum y_i$$

$$\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2 : \text{goodness of fit statistic}$$

Some special terminologies!

Total sum of squares

Error sum of squares

$$\diamond r^2 = \frac{\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$\diamond \sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y}_i)^2$: Regression sum of squares

$\diamond$ Therefore, Total sum of squares = Regression sum of squares
+ Error sum of squares

$\diamond$ And , $r^2 = \frac{\text{Regression sum of squares}}{\text{Total sum of squares}}$

$\diamond r^2$ is also referred to as the *proportion of the variance in y*

- ❖ Is a quantity that indicates how well a statistical model fits a data set. In other words, it is a statistical measure of how close the observed data are to the fitted regression line.
- ❖ It explains how much variation in the dependent variable y is characterized by a variation in the independent variable x .
- ❖ It is used to forecast or predict the possible outcomes.
- ❖ Its value lies between 0 and 1.
- ❖ The higher the value of r^2 , the better the prediction.



THANK YOU

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Checking Assumptions and Transforming Data

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Simple Linear Regression: Correlation & Regression Analysis



Uncertainties in the Least-Squares Coefficients

The least square line is given as: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

- The errors ϵ_i create uncertainty in the estimates β_0 and β_1 .
- It is intuitively clear that if ϵ_i tend to be small in magnitude, the points will be tightly clustered around the line, and the uncertainty in the least-squares estimates β_0 and β_1 will be small.
- On the other hand, if ϵ_i tend to be large in magnitude, the points will be widely scattered around the line, and the uncertainties (standard deviations) in the least-squares estimates β_0 and β_1 will be larger.

Uncertainties in the Least-Squares Coefficients

- Assume we have n data points $(x_1, y_1), \dots, (x_n, y_n)$, and we plan to fit the least squares line.
- In order for the estimates β_1 and β_0 to be useful, we need to estimate just how large their uncertainties are. In order to do this, we need to know something about the nature of the errors ϵ_i .
- We will begin by studying the simplest situation, in which four important assumptions are satisfied.

Uncertainties in the Least-Squares Coefficients

Assumptions for Errors in Linear Models:

In the simplest situation, the following assumptions are satisfied:

1. The errors $\varepsilon_1, \dots, \varepsilon_n$ are random and independent. In particular, the magnitude of any error ε_i does not influence the value of the next error ε_{i+1} .
 2. The errors $\varepsilon_1, \dots, \varepsilon_n$ all have mean 0.
 3. The errors $\varepsilon_1, \dots, \varepsilon_n$ all have the same variance, which we denote by σ^2 .
 4. The errors $\varepsilon_1, \dots, \varepsilon_n$ are normally distributed.
- When the sample size is large, the normality assumption (4) becomes less important.
 - Mild violations of the assumption of constant variance (3) do not matter too much, but severe violations should be corrected.

Uncertainties in the Least-Squares Coefficients

- Under these assumptions, the effect of the ϵ_i is largely governed by the magnitude of the variance σ^2 , since it is this variance that determines how large the errors are likely to be.
- Therefore, in order to estimate the uncertainties in β_0 and β_1 , we must first estimate the error variance σ^2 .
- Since the magnitude of the variance is reflected in the degree of spread of the points around the least-squares line, it follows that by measuring this spread, we can estimate the variance.

Specifically, the vertical distance from each data point (x_i, y_i) to the least-squares line is given by the residual e_i .

Uncertainties in the Least-Squares Coefficients

- The spread of the points around the line can be measured by the sum of the squared residuals
- The estimate of the error variance σ^2 is *the quantity* s^2 given by

$$s^2 = \frac{\sum_{i=1}^n e_i^2}{n - 2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2}$$

$$s^2 = \frac{(1 - r^2) \sum_{i=1}^n (y_i - \bar{y})^2}{n - 2}$$

Distribution

In the linear model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, under assumptions 1 through 4, the observations $y_1, \dots, y_n$ are independent random variables that follow the normal distribution. The mean and variance of y_i are given by

$$\mu_{y_i} = \beta_0 + \beta_1 x_i$$

$$\sigma_{y_i}^2 = \sigma^2$$

The slope represents the change in the mean of y associated with an increase in one unit in the value of x .

More Distributions

Under assumptions 1 – 4:

- The quantities $\hat{\beta}_0$ and $\hat{\beta}_1$ are normally distributed random variables.
- The means of $\hat{\beta}_0$ and $\hat{\beta}_1$ are the true values β_0 and β_1 , respectively.

- The standard deviations of $\hat{\beta}_0$ and $\hat{\beta}_1$ are estimated with

$$s_{\hat{\beta}_0} = s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} \quad \text{and} \quad s_{\hat{\beta}_1} = \frac{s}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

where $s = \sqrt{\frac{(1-r)^2 \sum_{i=1}^n (y_i - \bar{y})^2}{n-2}}$ is an estimate of the

error standard deviation σ .

Notes

1. Since the quantity $\sum_{i=1}^n (x_i - \bar{x})^2$ appears in the denominators of $s_{\hat{\beta}_0}$ and $s_{\hat{\beta}_1}$, it follows that the more spread out the x 's are, the smaller the uncertainties in $\hat{\beta}_0$ and $\hat{\beta}_1$ will be.
2. Use caution: if the range of x values extends beyond the range where the linear model holds, the results will not be valid.
3. The quantities $(\hat{\beta}_0 - \beta_0)/s_{\hat{\beta}_0}$ and $(\hat{\beta}_1 - \beta_1)/s_{\hat{\beta}_1}$ have Student's t distribution with $n - 2$ degrees of freedom.

- We stated some assumptions for the errors. Here we want to see if any of those assumptions are violated.
- The single best diagnostic for least-squares regression is a plot of residuals versus the fitted values, sometimes called a **residual plot**.

More of the Residual Plot

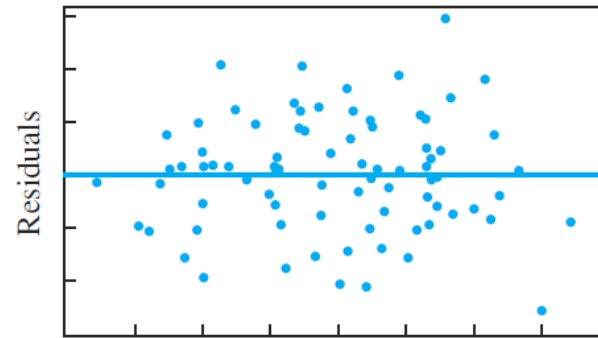
- When the linear model is valid, and assumptions 1 – 4 are satisfied, the plot will show no substantial pattern. There should be no curve to the plot, and the vertical spread of the points should not vary too much over the horizontal range of the data.
- A good-looking residual plot does not by itself prove that the linear model is appropriate. However, a residual plot with a serious defect does clearly indicate that the linear model is inappropriate.
- When the vertical spread in a scatterplot doesn't vary too much, the scatterplot is said to be **homoscedastic**. The opposite of homoscedastic is **heteroscedastic**.

A: No noticeable pattern

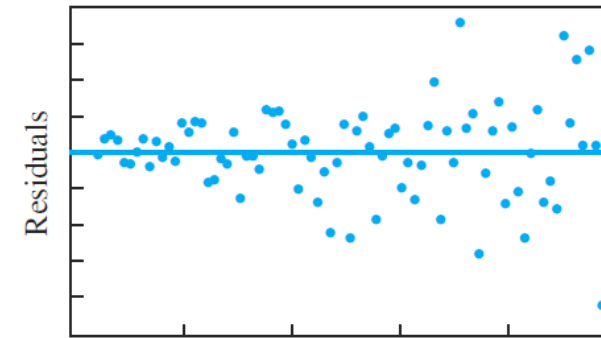
B: Heteroscedastic

C: Trend

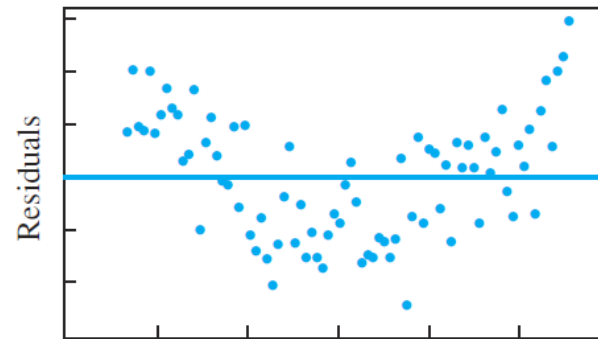
D: Outlier



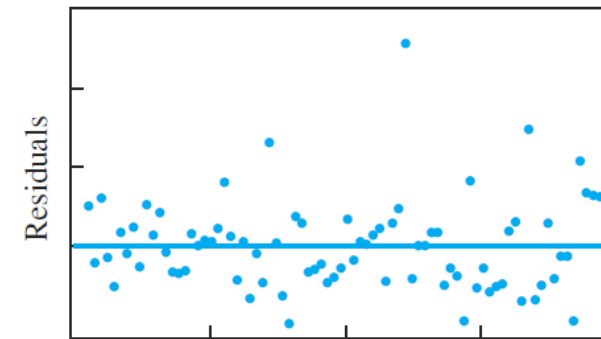
(a)



(b)



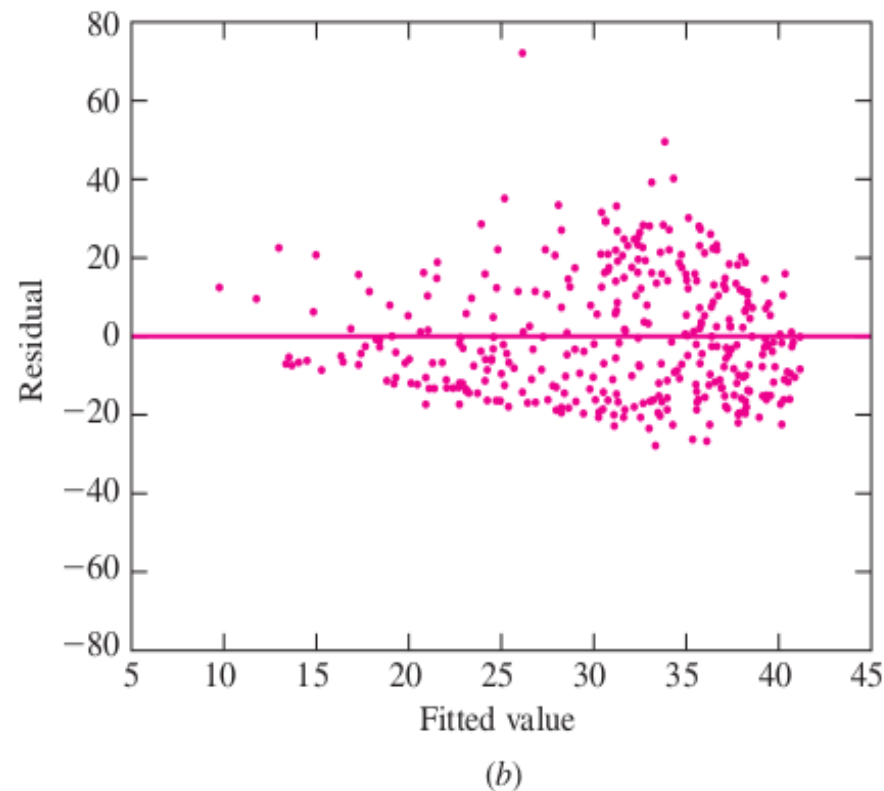
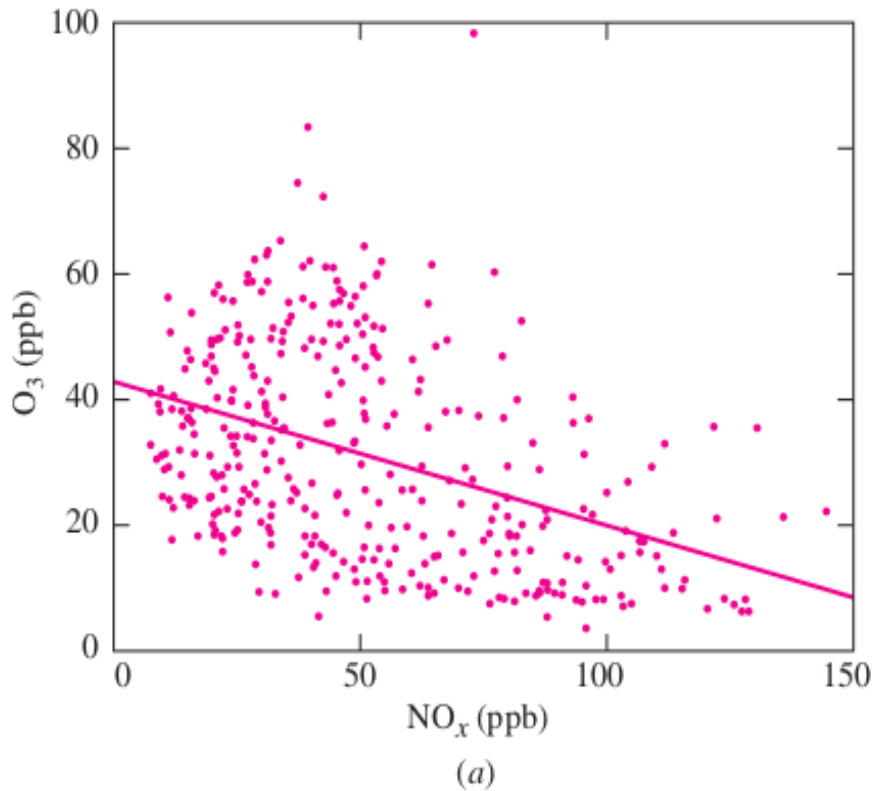
(c)



(d)

Checking Assumptions to form a Linear Model

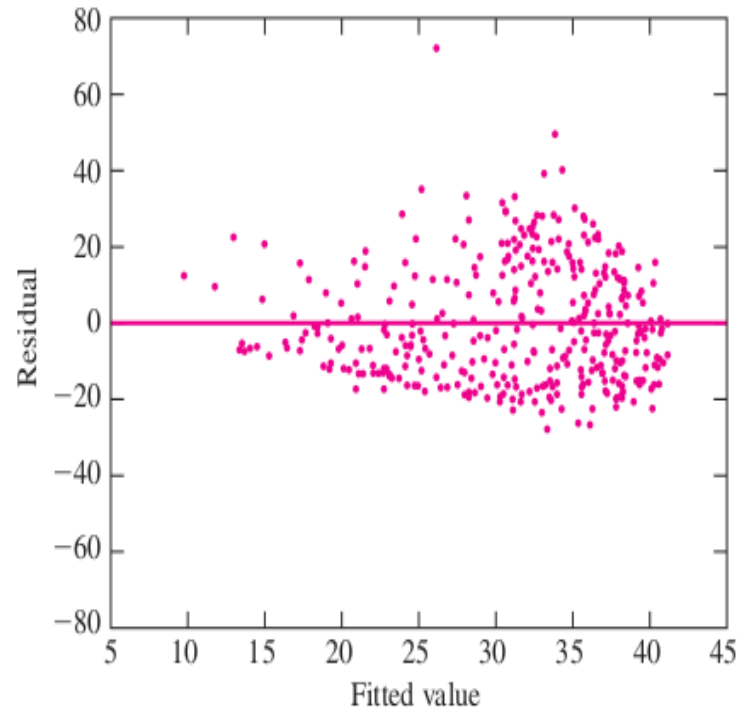
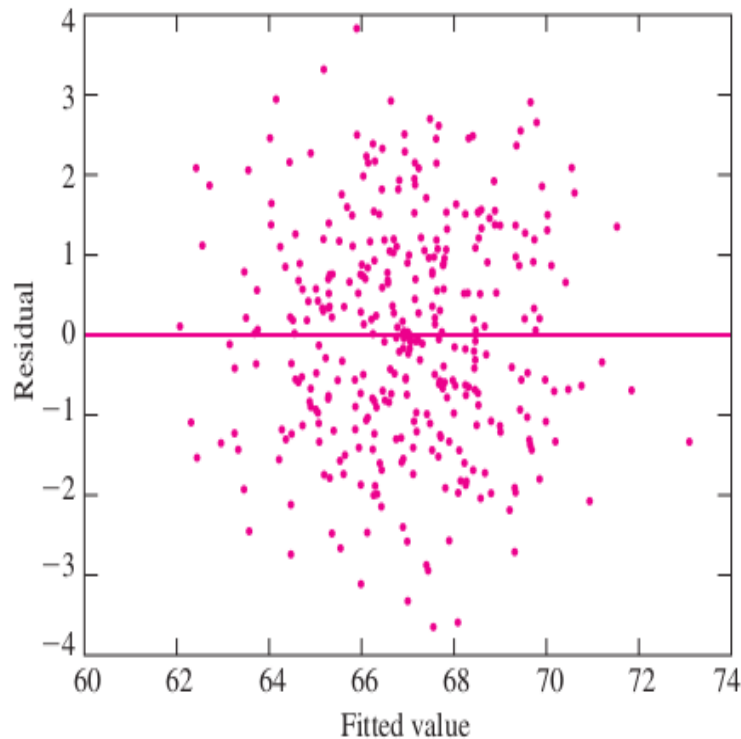
- **Example of a residual plot:** On the left is the plot of x versus the values of y , on the right the residual with the fitted values of y



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Checking Assumptions to form a Linear Model

- Below on the left the plot is homoscedastic, while on the right the spread increases with the fitted value and is thus heteroscedastic.



- If the residual plot is **homoscedastic**, and shows no substantial trend or curve, then a linear model can be found for the data plotted.
- If the residual plot is **heteroscedastic**, or shows a substantial trend or curve, then the assumptions for a linear model certainly do NOT hold! In such cases we need to transform the data or pursue other methods.

Transforming the Variables

- If we fit the linear model $y = \beta_0 + \beta_1 x + \varepsilon$ and find that the residual plot exhibits a trend or pattern, we can sometimes fix the problem by raising x , y , or both to a power.
- It may be the case that a model of the form $y^a = \beta_0 + \beta_1 x^b + \varepsilon$ fits the data well.
- Replacing a variable with a function of itself is called **transforming** the variable. Specifically, raising a variable to a power is called a **power transformation**.

Which transformation to apply?

It is possible with experience to look at a scatterplot, or a residual plot, and make an educated guess as to how to transform the variables.

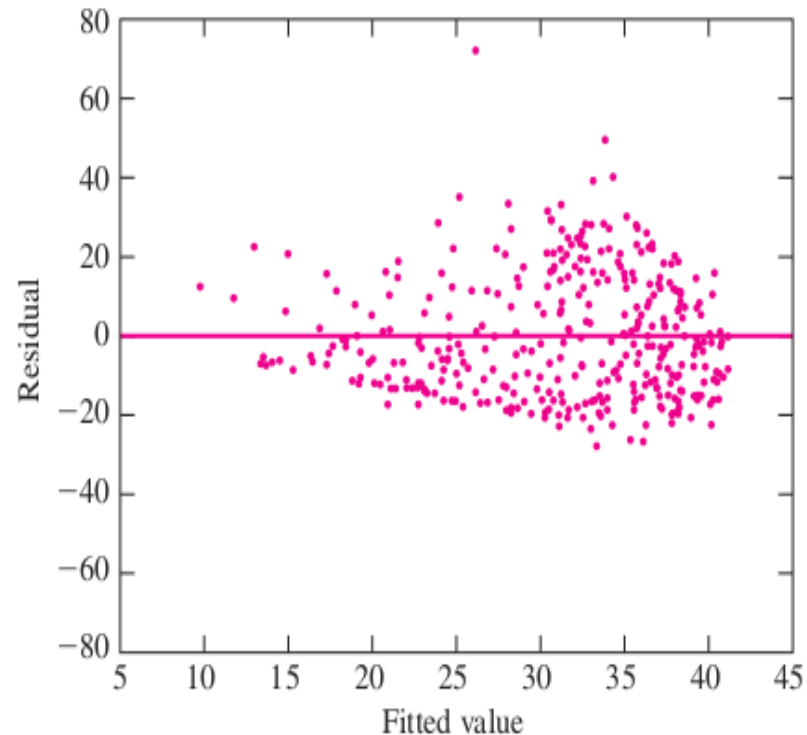
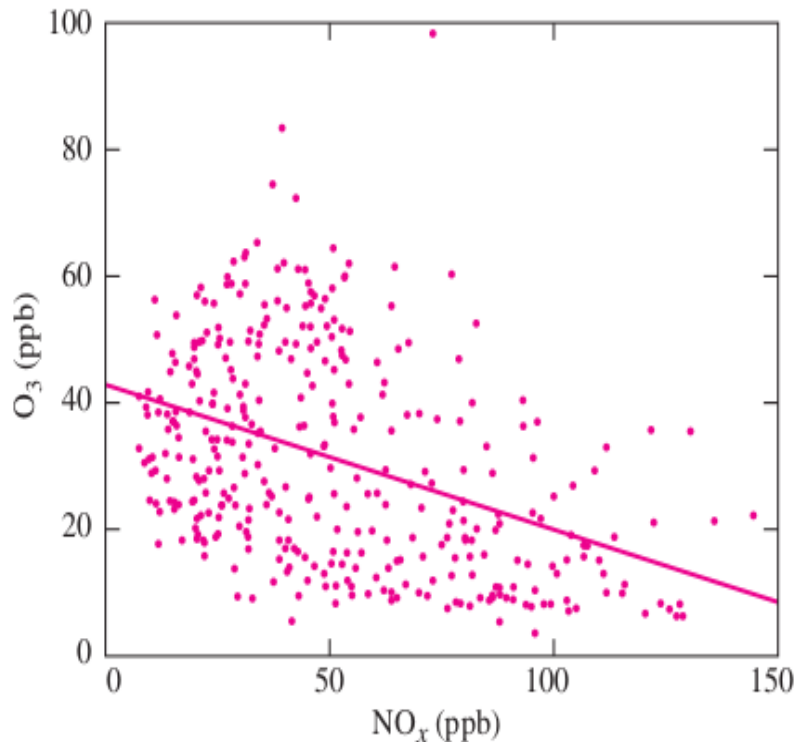
Mathematical methods are also available to determine a good transformation.

Trial and Error is fine – Try various powers on both x and y (including $\ln x$ and $\ln y$), look at the residual plots, and hope to find a homoscedastic one with no discernible pattern.

More advanced discussion in Draper and Smith (1998).

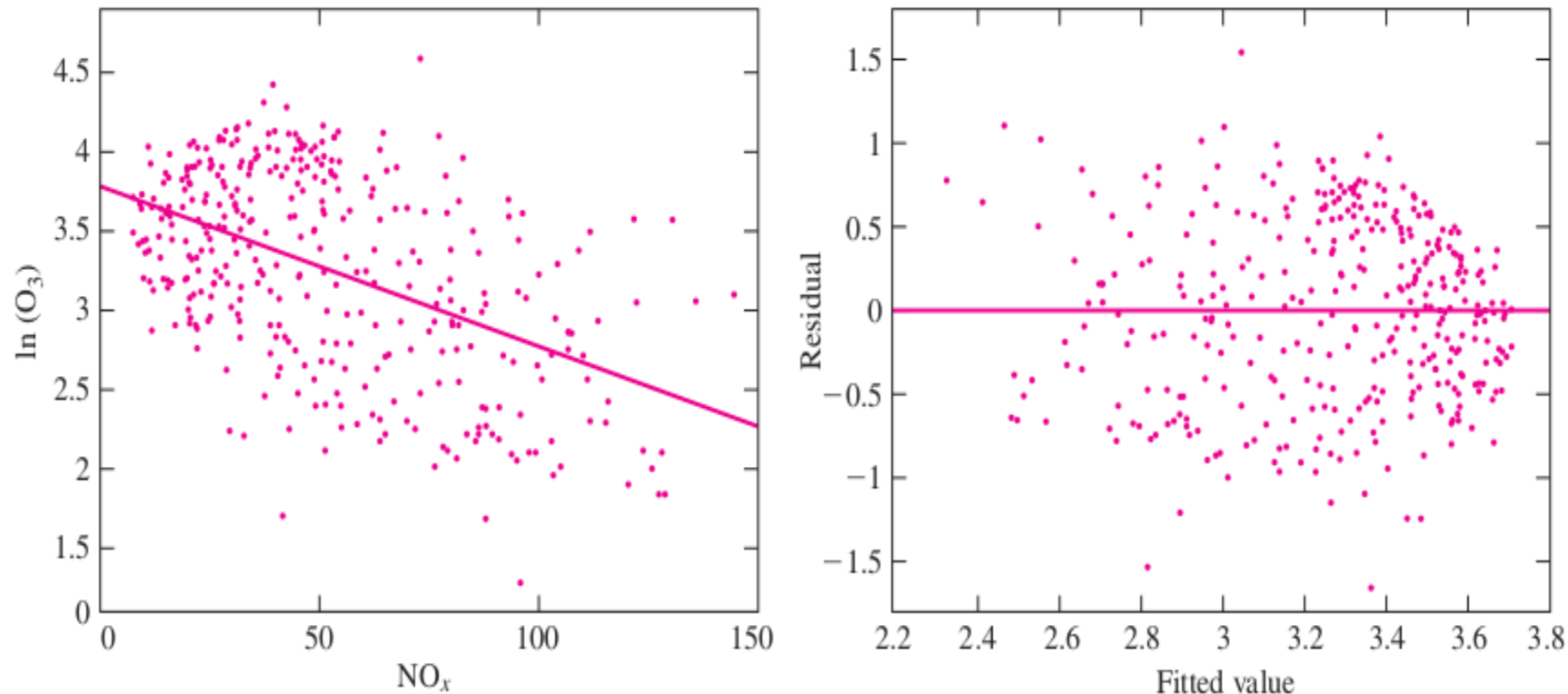
Which transformation to apply?

Recall the earlier example of a scatter plot (O_3 concentration vs NO_x concentration) whose residual plot on the right is heteroscedastic as shown below. Linear model NOT GOOD! Uh oh! Also notice the outlier with ozone concentration nearly 100.



Logarithm Transformation on One Axis

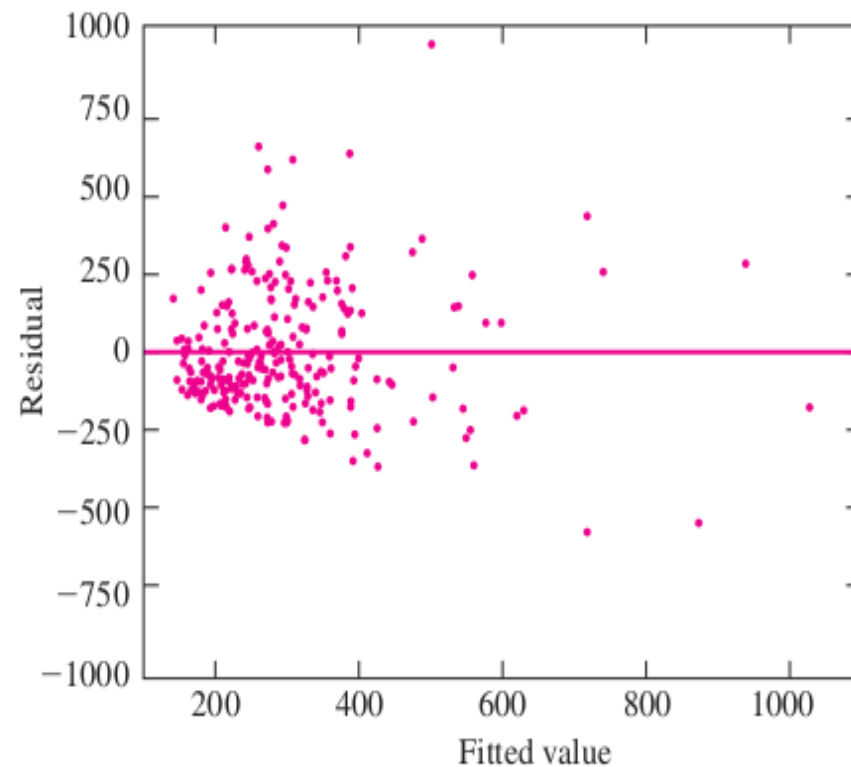
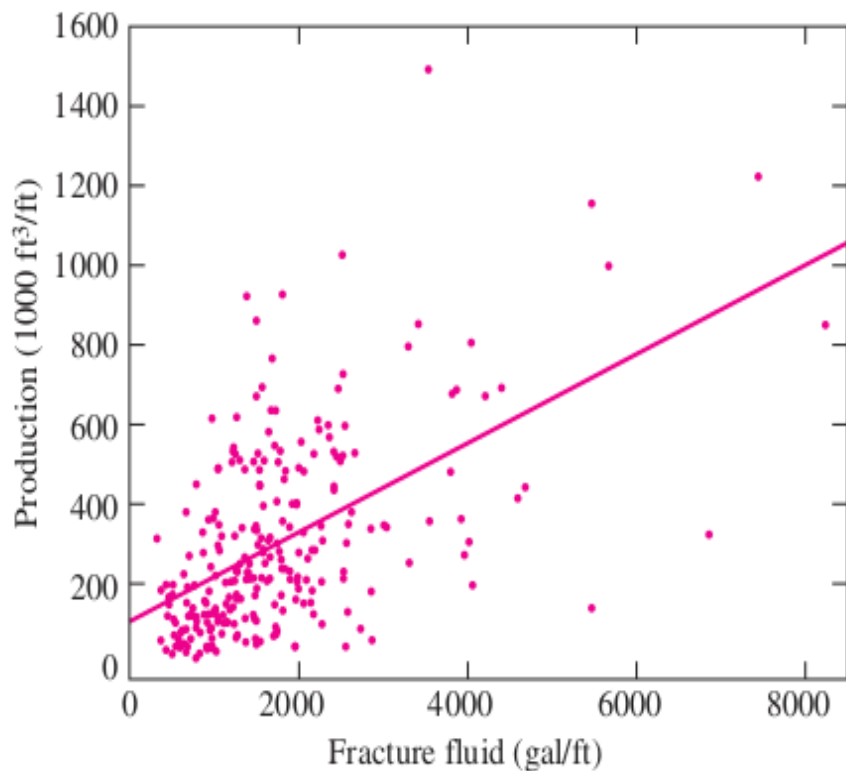
Applying the logarithm on y-axis (O_3 concentration) and obtain the following scatter plot and its residual on the right. Linear model looks GOOD! YAY! The outlier is less prominent too!



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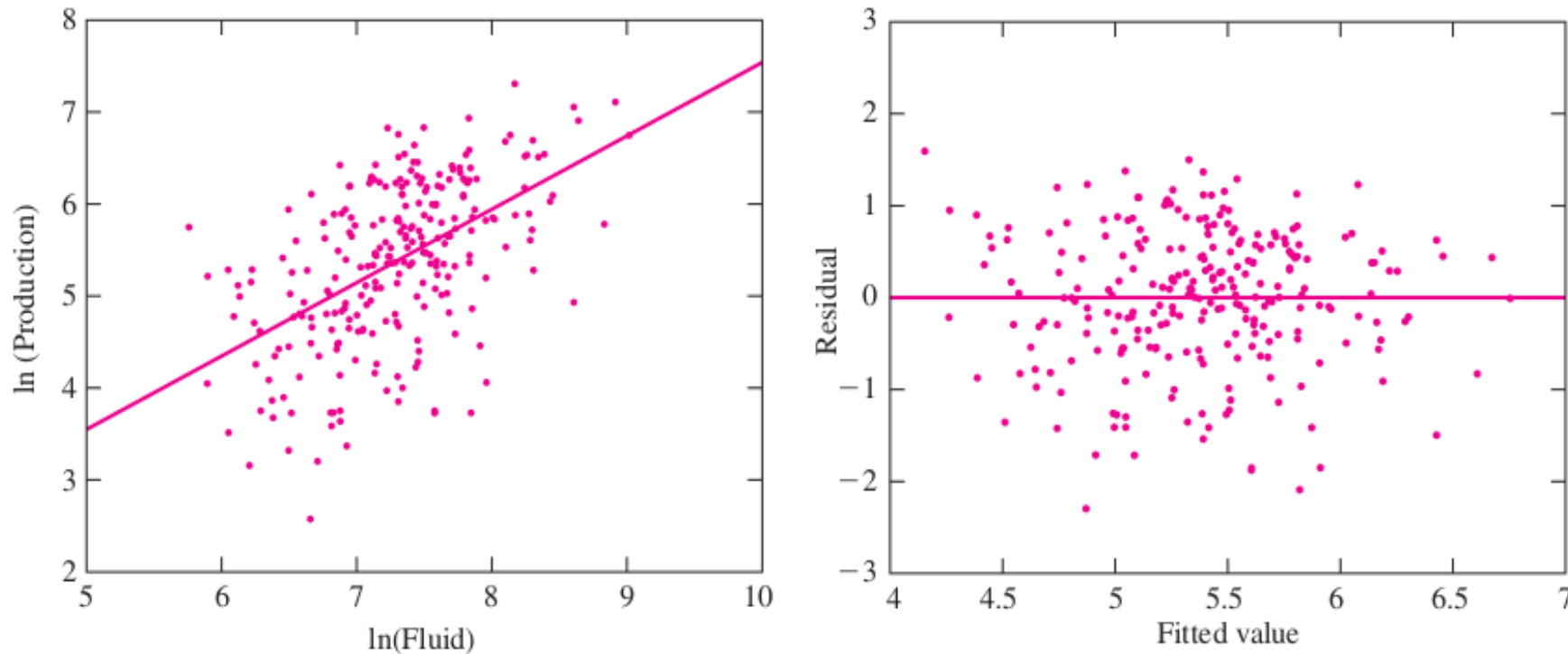
Logarithm Transformation on One Axis

Now consider an example below where The plot on the left is Production (ft^3/ft) vs Fracture fluid (gal/ft) and the residual plot is largely heteroscedastic! Not good for a linear model.



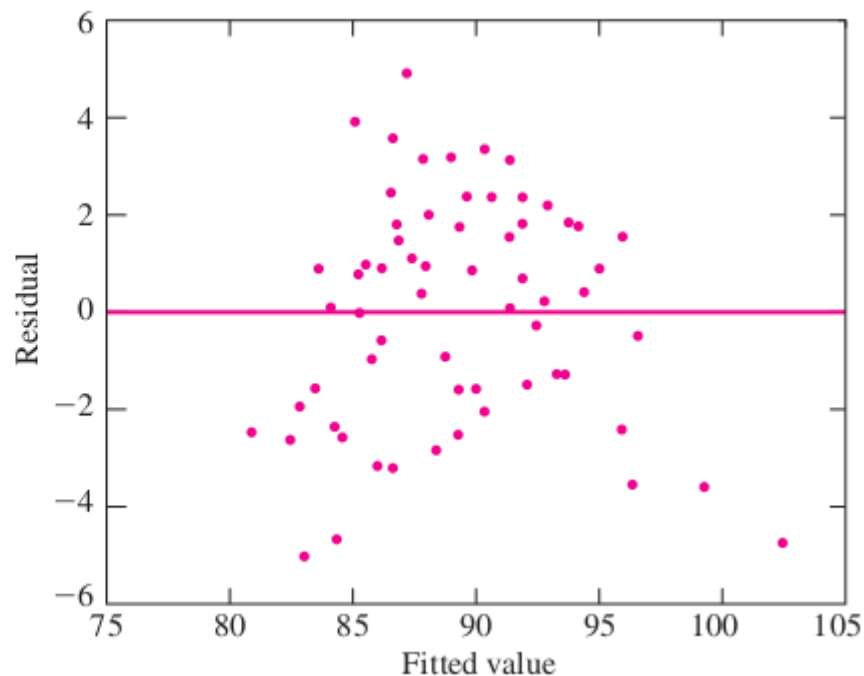
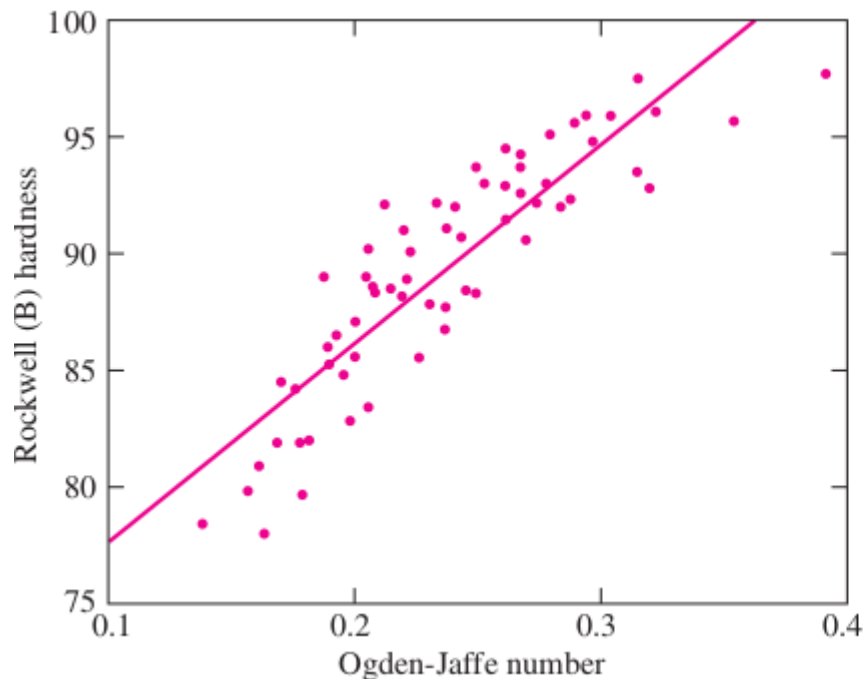
Logarithm Transformation on One Axis

Below is a plot of $\ln(\text{production})$ vs $\ln(\text{fracture fluid})$ for the same data. This time the residual plot is homoscedastic, good for linear model!



Power Transformations – The reciprocal

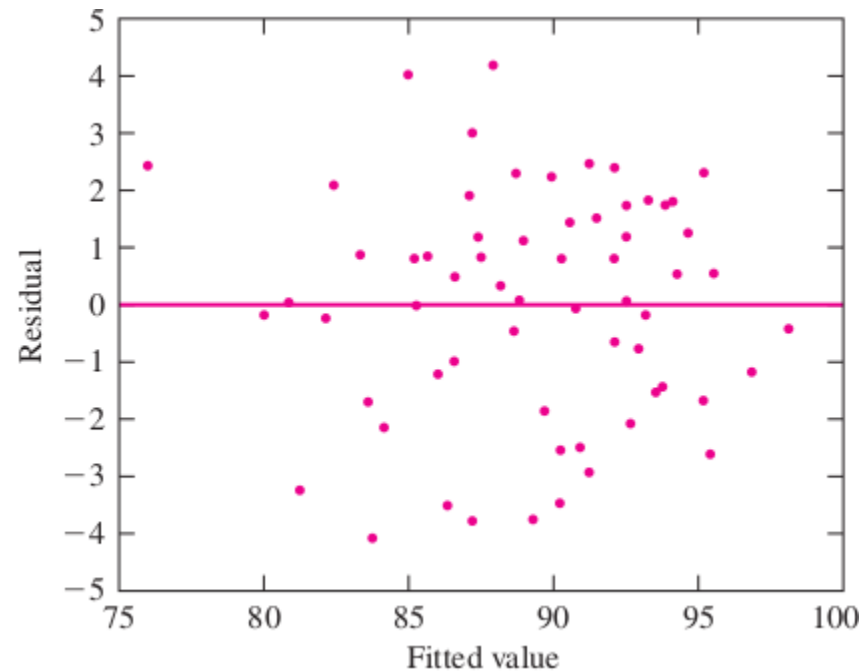
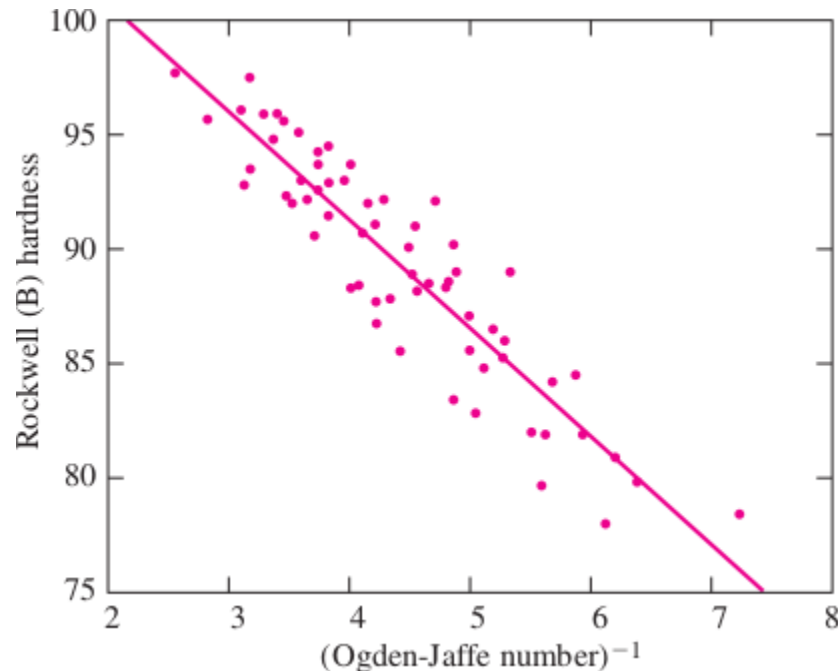
Below (left side) is a plot of Rockwell (B scale) hardness of welds versus their Ogden-Jaffe number. The residual plot (right side) shows a pattern where negative residual is observed for the extreme fitted values and positive residual for the ones in the middle. Linear model NOT OK.



Power Transformations – The reciprocal

We plot the graph of Rockwell Hardness vs $(\text{Ogden-Jaffe number})^{-1}$ for the same data (below, left side) and find that the residual plot (below, right side) is homoscedastic, having no discernible pattern.

Linear model is OK.

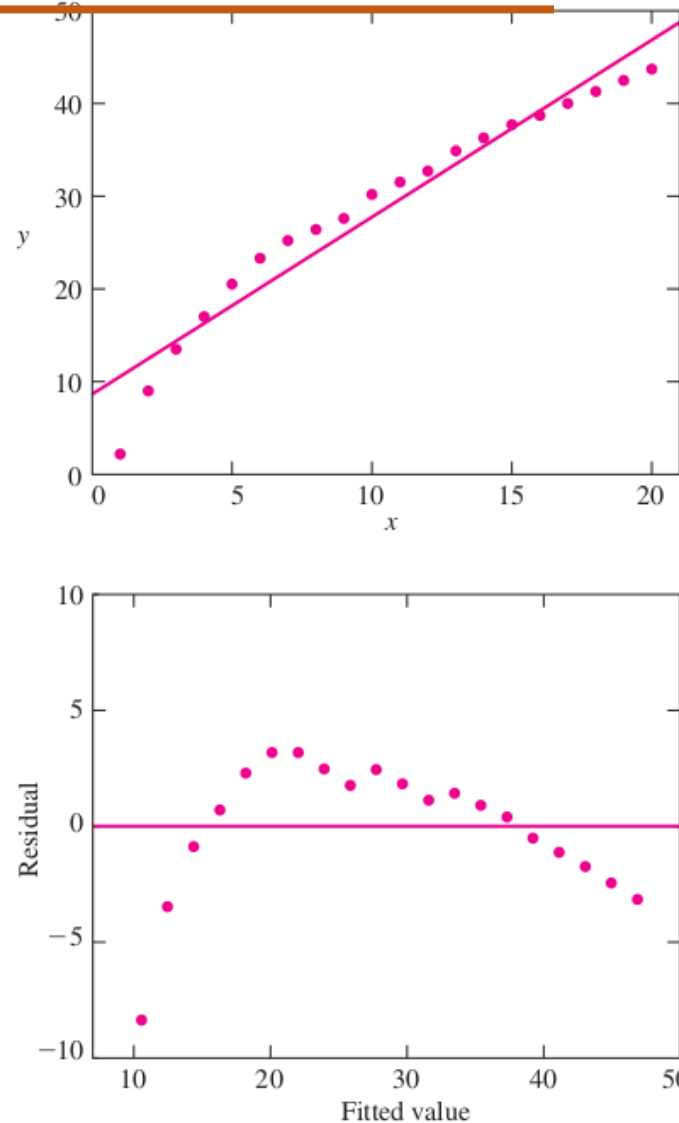


MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Power Transformations – Positive Powers

Plot y vs x and its residual plot which exhibits a discernible pattern.
Linear model is NOT OK.

x	y	x	y
1	2.2	11	31.5
2	9	12	32.7
3	13.5	13	34.9
4	17	14	36.3
5	20.5	15	37.7
6	23.3	16	38.7
7	25.2	17	40
8	26.4	18	41.3
9	27.6	19	42.5
10	30.2	20	43.7

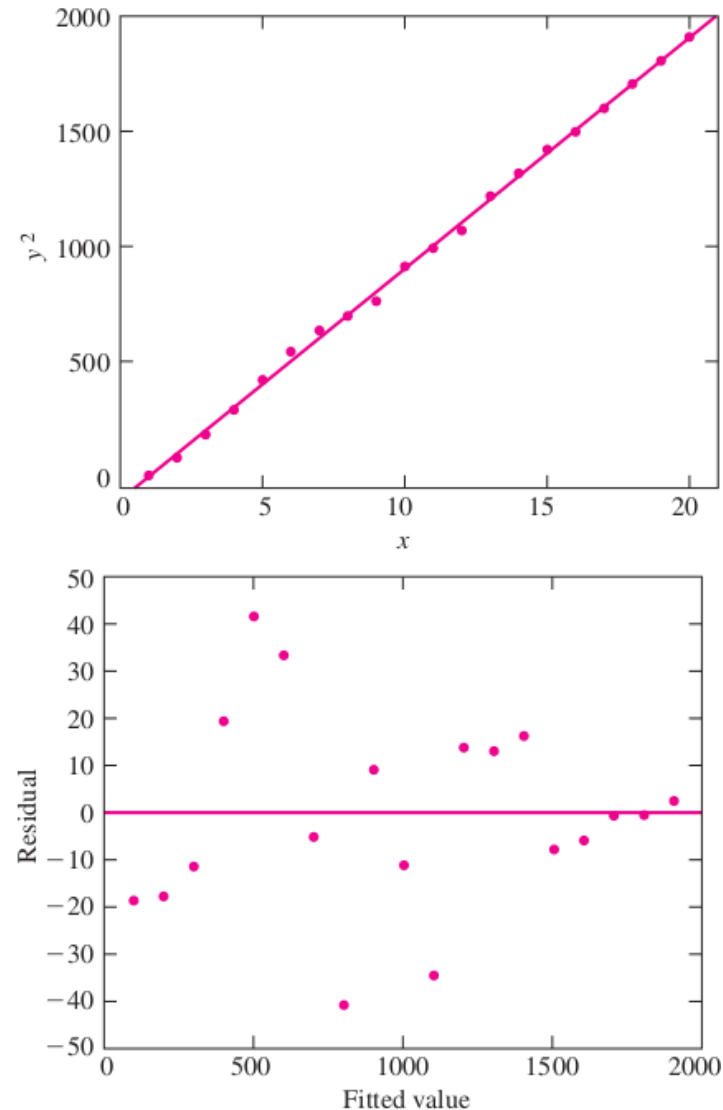


Power Transformations – Positive Powers

Plot y^2 vs x and its homoscedastic residual plot which exhibits no discernible pattern.

Linear model is OK.

x	y^2	x	y^2
1	4.84	11	992.25
2	81	12	1069.29
3	182.25	13	1218.01
4	289	14	1317.69
5	420.25	15	1421.29
6	542.89	16	1497.69
7	635.04	17	1600
8	696.96	18	1705.69
9	761.76	19	1806.25
10	912.04	20	1909.69



Transformations – Do they always work?

It is important to remember that power transformations don't always work.

Sometimes, none of the residual plots looks good, no matter what transformations are tried. In these cases, other methods should be used. One of these is multiple regression which is not covered here.

Some other methods are briefly mentioned in the next slide.

The popular methods other than transformation are:

- Weighted Least Squares
 - We assign greater weights to points in regions where the vertical spread is smaller and vice versa.
- Multiple Regression
 - We add more independent variables in order to explain the variation in the dependent variable.

How Many Points Make a Reliable Residual Plot?

When there are too few points on the residual plot, then...

- ... it may appear to have a pattern or be heteroscedastic in spite of that being just a visual effect created by one or two points.
- ... detecting outliers may become difficult

What to do if you can't interpret a residual plot reliably?

You can start by fitting a linear model but declare your result tentative; wait for more data and then a reliable decision can be made.

How Many Points Make a Reliable Residual Plot?

NOT all residual plots with few points turn out to be hard to interpret.

Some of these show a pattern which cannot be changed by relocating just one or two points.

In such a case a linear model should NOT be used!

- **Outliers** are points that are detached from the bulk of the data.
- Both the scatter plot and the residual plot should be examined for outliers.
- The first thing to do with an outlier is to determine why it is different from the rest of the points.

Outliers

- Sometimes outliers are caused by data-recording errors or equipment malfunction. In this case, the outlier can be deleted from the data set. In this case, you may present results that do not include the outlier.
- If it cannot be determined why there is an outlier, then it is not wise to delete it. Here the results presented, should be the ones from analysis with the outlier included in the data set.

Influential Point

- If there are outliers that cannot be removed from the data set, then the best thing to do is fit the whole data set and then remove the outlier and fit a line to the data set.
- If none of the outliers upon removal make a noticeable difference to the least-squares line or to the estimated standard deviation of the slope and intercept, then use the fit with the outliers included.

Influential Point

- If one or more outlier does make a difference, then the range of values for the least-squares coefficients should be reported. Avoid computing confidence and prediction intervals and performing hypothesis tests.
- An outlier that makes a considerable difference to the least-squares line when removed is called an **influential point**.
- In general, outliers with unusual x values are more likely to be influential than those with unusual y values, but every outlier should be checked.
- Some authors restrict the definition of outliers to points that have unusually large residuals.

- Transforming the variables is not the only method for analyzing data when the residual plot indicates a problem.
- There is a technique called weighted least squares regression. The effect is to make the points whose error variance is smaller have greater influence in the computation of the least-squares line.
- When the residual plot shows a trend, this sometimes indicates that more than one independent variable is needed to explain the variation in the dependent variable.

- If the relationship is nonlinear, then a method called nonlinear regression can be applied.
- If the plot of residuals versus fitted values looks good, it may be advisable to perform additional diagnostics to further check the fit of the linear model. A time series plot is used to see if time should be included in the model. A normal probability plot can be used to check the normality assumption.

- If the plot of residuals versus fitted values looks good, then further diagnostics may be used to further check the fit of the linear model.
- A time order plot of the residuals versus order in which observations were made.
- If there are trends in this plot, then x and y may be varying with time. In this case, adding a time term to the model as an additional independent variable.

- To check that the errors are normally distributed, a normal probability plot of the residuals can be made.
- If the plot looks like it follows a rough straight line, then we can conclude that the residuals are approximately normally distributed.

What does it imply if residuals show a pattern when plotted against fitted values?

- A. Homoscedasticity is present.
- B. Independence of residuals is confirmed.
- C. Mean of residuals is zero.
- D. Assumption of randomness and independence might be violated.

Solution D



THANK YOU
